MARKETSANDMARKETS™

MICROBIOME SEQUENCING SERVICES MARKET

GLOBAL FORECAST TO 2028

BY SERVICE (SAMPLE PREPARATION, SEQUENCING, LIBRARY PREPARATION),
TYPE (AMPLICON SEQUENCING, WHOLE GENOME SEQUENCING),
TECHNOLOGY (SEQUENCING BY SYNTHESIS, NANOPORE SEQUENCING), END USER

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
APAC	Asia Pacific
CIPM	Center of Innovation in Personalized Medicine
CRG	Centre for Genomic Regulation
DBT	Department of Biotechnology (GENOME India and Microbiome)
EUA	Emergency-use Authorization
FDA	Food and Drug Administration
GIP	Genome India Project
GoNL	Genome of the Netherlands
GRDI	Genomics Research and Development Initiative
HMGJ	Human MetaGenome Consortium Japan
LATAM	Latin America
MEA	Middle East & Africa
MHLW	Ministry of Health, Labour and Welfare
MRC	Microbiome Research Centre
MRD	Minimal Residual Disease
NCBI	National Center for Biotechnology Information
NGS	Next-generation Sequencing
NHGRI	National Human Genome Research Institute
R&D	Research & Development
RoAPAC	Rest of Asia Pacific
RoE	Rest of Europe
SBS	Sequencing by Synthesis
UK	United Kingdom
UKRI	UK Research and Innovation
US	United States
US FDA	United States Food and Drug Administration
USD	United States Dollar
WES	Whole-exome Sequencing

WGBS	Whole-genome Bisulfite Sequencing
WGS	Whole-genome Sequencing
WHO	World Health Organization

1 INTRODUCTION

1.1 STUDY OBJECTIVES

- To define, describe, and forecast the global microbiome sequencing services market based on service, type, technology, end user, and region
- To provide detailed information regarding the major factors influencing the market growth (such as drivers, restraints, challenges, and opportunities)
- To strategically analyze micromarkets[1] with respect to individual growth trends, prospects, and contributions to the overall microbiome sequencing services market
- To analyze the opportunities in the market for stakeholders and provide details of the competitive landscape for market leaders operating in the microbiome sequencing services market
- To forecast the revenue of the market with respect to five regional segments—North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa
- To profile the key market players and comprehensively analyze their service portfolios, market positions, and core competencies[2]
- To track and analyze competitive developments such as services launches, expansions, agreements, collaborations, partnerships, and acquisitions in the microbiome sequencing services market

1.2 MARKET DEFINITION

Microbiome sequencing is applied to decode the genetic material of all the microorganisms in a given sample which may contain bacteria, viruses, or fungi. Whole-genome sequencing (WGS) and amplicon sequencing are the two key microbiome sequencing services. Sequencing-based detection of microorganisms exhibits different advantages over traditional approaches, some of which are mentioned below:

- Detection of anaerobic bacteria
- More cost-effective microbiome characterization of multiplex microbial samples
- Identification of non-cultivable bacteria
- Estimate the relative abundance of microbes in a sample
- Profile hundreds of microorganisms in a single analysis
- Examine complex microbiomes and microbiota communities

The microbiome sequencing services market comprises services employed for sample preparation, sequencing, library preparation, and data analysis suitable for different samples, including human microbiome samples and environmental samples.

1. Micromarkets are the further segments and subsegments of the microbiome sequencing services market.
2. Core competencies of companies are captured in terms of the key developments, market shares, and key strategies adopted to sustain their positions in the market.

1.2.1 INCLUSIONS AND EXCLUSIONS

PARTICULAR	INCLUSION	EXCLUSION
Service	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing services, including sample preparation, sequencing and library preparation, and data analysis.	NA
Type	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing service types, such as amplicon sequencing, whole-genome sequencing, shotgun sequencing, and de novo sequencing.	NA
Technology	Qualitative and quantitative analysis (in terms of USD million) of technologies, such as sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and semiconductor sequencing.	NA
End User	Qualitative and quantitative analysis (in terms of USD million) of microbiome sequencing service end users such as pharmaceutical and biotechnology companies, academic and research institutes, small-scale CROs, and CDMOs.	NA

1.3 MARKET SCOPE

1.3.1 MARKETS COVERED

MICROBIOME SEQUENCING SERVICES MARKET

SERVICE
- Sample Preparation
- Sequencing and Library Preparation
- Data Analysis

TYPE
- Amplicon Sequencing
- Whole-genome Sequencing
- Shotgun Sequencing
- Other Types

TECHNOLOGY
- Sequencing by Synthesis
- Sequencing by Ligation
- Nanopore Sequencing
- Other Technologies

END USER
- Pharmaceutical and Biotechnology Companies
- Academic and Research Institutes
- Other End Users

REGION
- North America *(US and Canada)*
- Europe *(Germany, UK, France, Italy, Spain, and Rest of Europe)*
- Asia Pacific *(Japan, China, India, and Rest of Asia Pacific)*
- Latin America *(Brazil and Rest of Latin America)*
- Middle East and Africa

Note 1: Other types of microbiome sequencing services include de novo sequencing (whole-genome sequencing without any reference genome), metagenomic sequencing, and metatranscriptomic sequencing.

Note 2: Other microbiome sequencing technologies include semiconductor sequencing and Sanger sequencing.

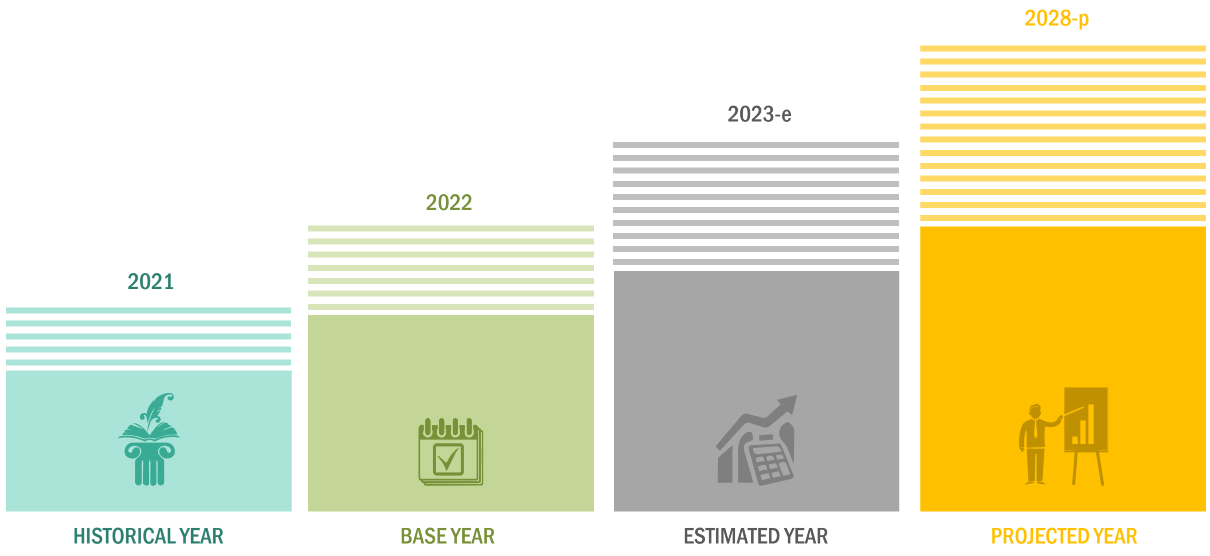
Note 3: Other end users include hospitals and clinics, small-scale CROs, and CDMOs.

Note 4: The Rest of Europe includes Switzerland, Greece, the Netherlands, Belgium, Denmark, and Ireland.

Note 5: The Rest of Asia Pacific includes South Korea, Australia, and countries in Southeast Asia.

Note 6: The Rest of Latin America includes Argentina, Peru, Chile, and Colombia.

1.3.2 YEARS CONSIDERED



Note: For market estimates, the calendar year has been considered, whereas, for company profiles, the fiscal year has been considered.

1.4 CURRENCY

- The currency used in this report is the United States Dollar (USD), with market sizes indicated in terms of USD million/billion.
- For companies reporting their revenues in USD, the revenues were taken from their respective annual reports/SEC filings.
- For companies reporting their revenues in other currencies, the average annual currency conversion rate was used for that particular year to convert the value to USD.

1.5 LIMITATIONS

- The market study does not cover the market size in terms of volume, as it is a service-oriented market.
- A limited number of industry experts participated in primary research from Africa, the Middle East, and Latin America due to specific language barriers. In such cases, the geographic market size was derived based on weightages assigned to these markets based on the qualitative insights from global industry experts, the analysis of demographic parameters, and the adoption trends for microbiome sequencing observed in these markets.
- Financial information for privately held and unlisted companies was not available in the public domain. Hence, their financial details are not provided in the company profiles section.
- Company developments not reported in the public domain are not included in this report.
- Market shares are estimated on a year-on-year basis (2020 to 2022) using historical data of companies and secondary sources.
- Several strategic approaches and recommendations from industry experts were utilized to collate the research study to overcome the limitations and maintain the effectiveness of the study.

1.6 STAKEHOLDERS

- Research Institutes
- Pharmaceutical and Biotechnology Companies
- Microbiome Sequencing Service Providers
- Metagenomic Sequencing Service Providers
- NGS Service Providers
- Venture Capitalists and Investors
- Contract Research Organizations
- Government Associations
- Healthcare Associations/Institutes
- Business Research and Consulting Service Providers

1.7 SUMMARY OF CHANGES

Refinements in the segments of the global microbiome sequencing services market:

- Market sizes are updated for the base year 2022 and forecasted from 2023 to 2028.
- The new edition of the report provides updated financial information till 2022. This will help in the easy analysis of the present status of profiled companies in terms of their financial strength, profitability, key revenue-generating region/country, and business segment focus in terms of the highest revenue-generating segment.
- Recent developments are helpful in understanding market trends and the growth strategies adopted by players in the market. The number of service launches has increased in the last three years (October 2020 to April 2023).
- Tracking the service portfolios of prominent market players helps analyze the major services in the microbiome sequencing services market. The new edition of the report provides an updated service portfolio of the companies profiled in the report.
- Key market strategies, market share analysis, and competitive leadership mapping have been updated in the competitive landscape section of the report.
- The updated version of the report includes the revenue share analysis of key market players from 2020 to 2022.
- The current edition of the report includes an assessment of the economic and financial recession and geographical conditions such as supply chain disruptions, the Russia-Ukraine war, along with the effects of inflation on the services market. The impact of the recession on year-on-year growth during the forecast period (2023 to 2028) has also been considered.

1.7.1 RECESSION IMPACT

The impact of the recession is analyzed in the current report. The factors considered while analyzing the growth of the market in the forecast period are economic and financial recession, the Russia-Ukraine war, and supply chain disruptions. These factors are anticipated to affect the market directly or indirectly in the coming years.

- The Russia-Ukraine war will have an impact on different regional markets with a direct impact on the European market.
- The decreasing demand for COVID-related drugs and vaccines is also expected to decrease drug discovery and development initiatives for the same. This, in turn, is expected to affect the revenue of key service providers in the coming years.
- Declining GDPs will affect R&D expenditures, which, in turn, will restrain the overall market to a certain extent.

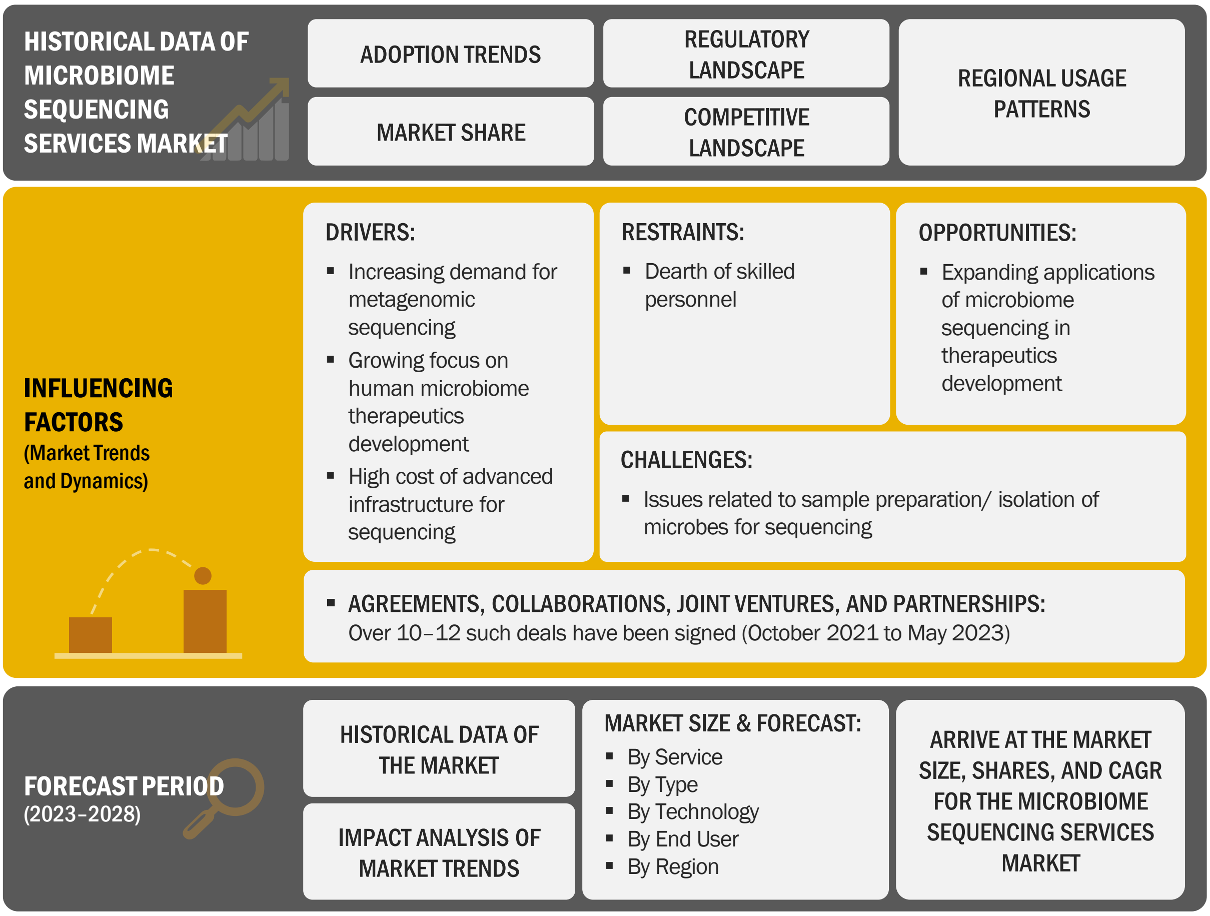
The direct and indirect impact of these factors on the growth of the microbiome sequencing services market is further analyzed in the current edition of the report.

2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

This market research study involved the extensive use of secondary sources, directories, and databases to identify and collect information useful for this technical, market-oriented, and financial study of the global microbiome sequencing services market. In-depth interviews were conducted with various primary respondents, including key industry participants, subject-matter experts (SMEs), C-level executives of key market players, and industry consultants, to obtain and verify critical qualitative and quantitative information and assess market growth prospects. The global size of the microbiome sequencing services market (estimated through various research approaches) was then triangulated with inputs from primary research to arrive at the final market size.

FIGURE 1 RESEARCH DESIGN



Sources: Secondary Research and MarketsandMarkets Analysis

2.1.1 SECONDARY DATA

The secondary sources referred to for this research study include publications from government sources, such as the National Human Genome Research Institute (NHGRI), the National Center for Biotechnology Information (NCBI), the Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation France, the Centre for Genomic Regulation (CRG) Spain, Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC) Australia, Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome) India, South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis.

Secondary sources include corporate and regulatory filings (such as annual reports, SEC filings, investor presentations, and financial statements); business magazines and research journals; press releases; and trade, business, and professional associations. Secondary data was collected and analyzed to arrive at the overall size of the global microbiome sequencing services market, which was validated through primary research.

Secondary research involved three major activities:

Background Study

- Building a basic understanding of the microbiome sequencing services market
- Analyzing the MarketsandMarkets data repository, followed by drawing and updating data tables using current information
- Identifying data gaps and key global players/service providers in each segment

Market Understanding

- Identifying stakeholders and key decision-makers in different regions
- Identifying the selection criteria of key decision-makers
- Analyzing the competitive landscape
- Analyzing major global players with their respective service portfolios
- Analyzing the strategies adopted by global players to position their services offered
- Analyzing markets at the regional/country level

Trends

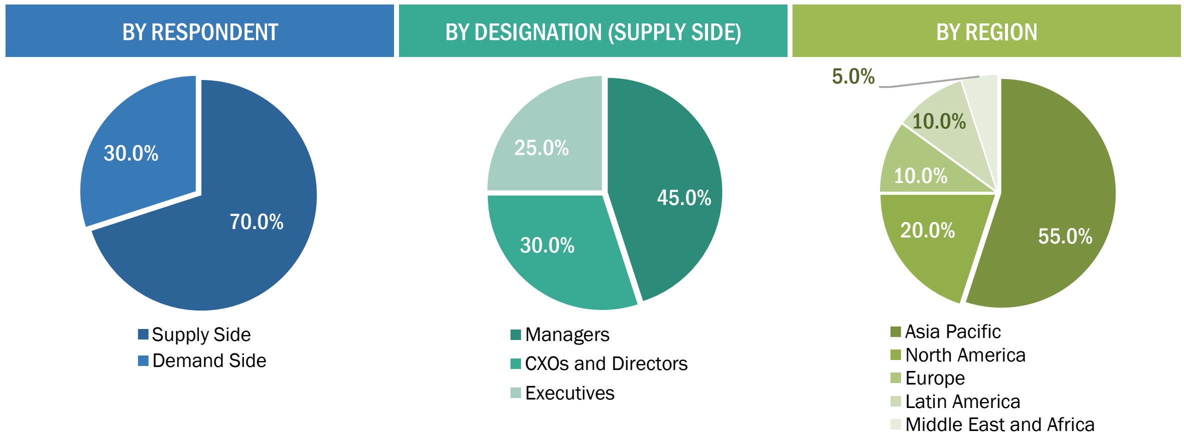
- Identifying and analyzing key products and industry trends to draw notes and include considerations in the final forecasts
- Identifying the major drivers, restraints, opportunities, and challenges in the market
- Analyzing the regulatory landscape and industry trends

2.1.2 PRIMARY DATA

Extensive primary research was conducted after acquiring basic knowledge about the global microbiome sequencing services market through secondary research. Several primary interviews were conducted with market experts from both the demand side (personnel from pharmaceutical and biopharmaceutical companies, medical device companies, and academic institutes) and supply side (C-level and D-level executives, product managers, and marketing and sales managers of key manufacturers, distributors, and channel partners, among others, from tier 1 and tier 2 companies engaged in offering services) across five major regions—North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Approximately 70% and 30% of primary interviews were conducted with supply-side and demand-side participants, respectively. This primary data was collected through questionnaires, e-mails, online surveys, personal interviews, and telephonic interviews.

FIGURE 2 MICROBIOME SEQUENCING SERVICES MARKET: BREAKDOWN OF PRIMARIES

SUPPLY-SIDE PARTICIPANTS		DEMAND-SIDE PARTICIPANTS
Microbiome Insights, Inc.	Rebiotix Inc.	Genefitletics
CosmosID	BaseClear	BIONE
Microba Life Sciences	Clarity Genomics, Inc.	Viome Life Sciences Inc.



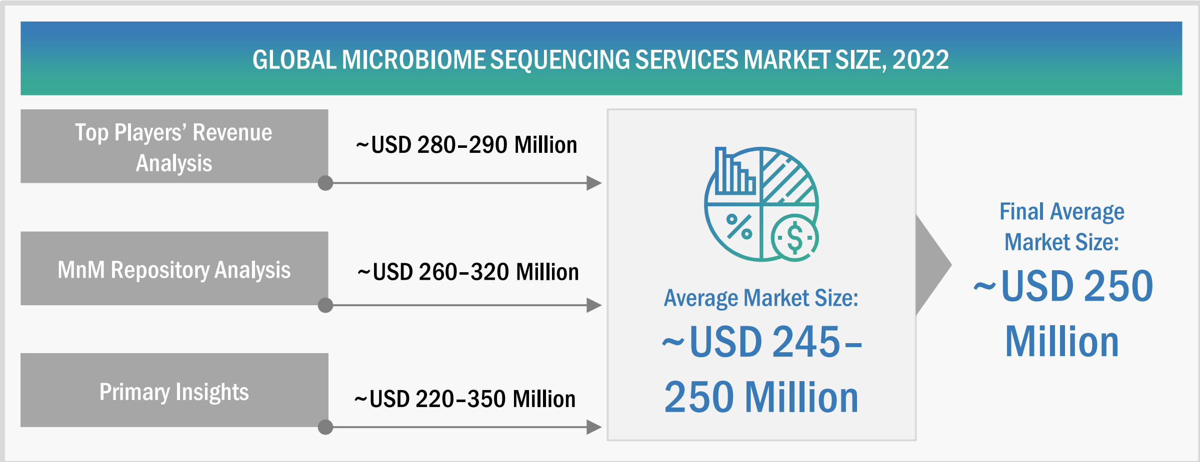
Primary research was used in this report to:

- Validate the market segmentation defined through a service portfolio assessment of the leading players in the microbiome sequencing services market
- Understand key industry trends and issues defining the strategic growth objectives of market players
- Gather both demand and supply-side validation of the key factors affecting the market growth (such as market drivers, restraints, challenges, and opportunities)
- Validate assumptions for the market sizing and forecasting model used for this study
- Understand the market position of leading players in the microbiome sequencing services market and their market shares/ranking

2.2 MARKET ESTIMATION METHODOLOGY

The report presents a detailed assessment of the microbiome sequencing services market and qualitative inputs and insights from MarketsandMarkets. The global size of the microbiome sequencing services market was estimated through multiple approaches. A detailed market estimation approach was followed to estimate and validate the value of the microbiome sequencing services market and other dependent submarkets, as mentioned below:

FIGURE 3 MICROBIOME SEQUENCING SERVICES MARKET SIZE ESTIMATION (SUPPLY-SIDE ANALYSIS), 2022

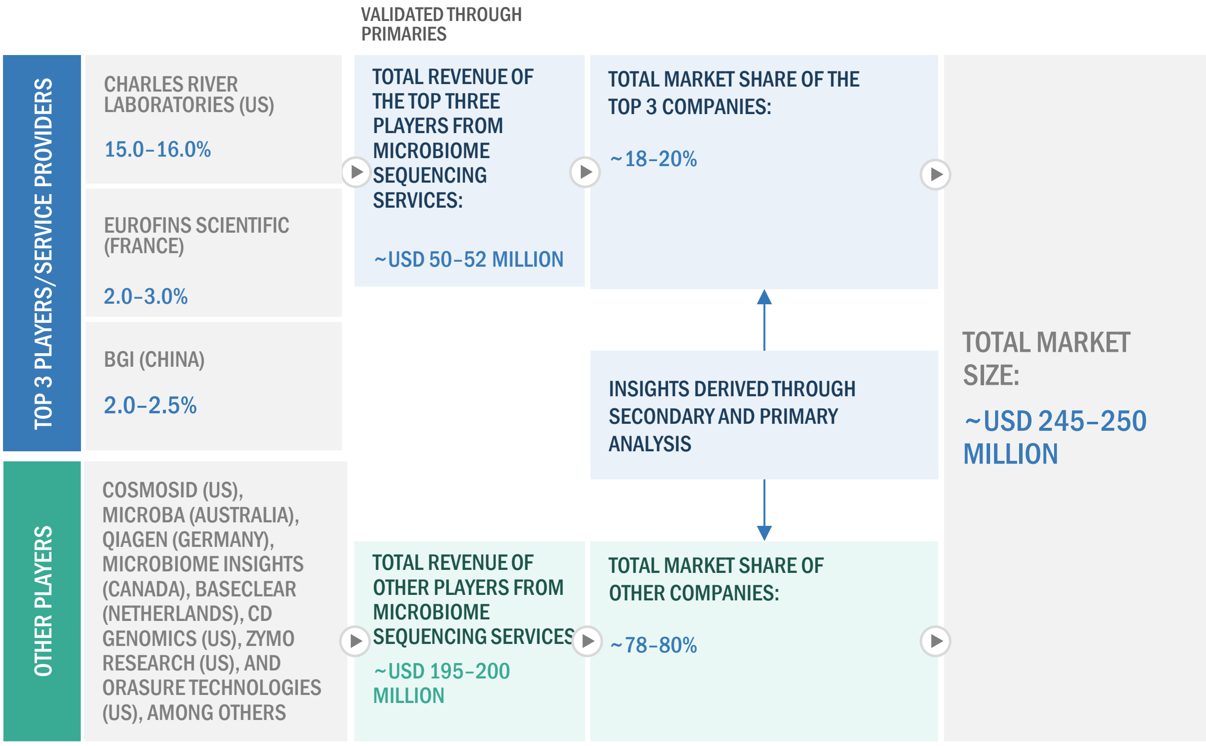


Source: MarketsandMarkets Analysis and Primary Insights

APPROACH 1: COMPANY REVENUE ANALYSIS (BOTTOM-UP APPROACH)

- Revenues for individual companies were gathered from public sources and databases.
- The microbiome sequencing services-related business shares of leading players were gathered from secondary sources to the extent available. In certain cases, the shares of microbiome sequencing services businesses were ascertained through a detailed analysis of various parameters, including service portfolios, market positioning, and primary insights.
- Individual shares or revenue estimates were validated through expert interviews.

FIGURE 4 MARKET SIZE ESTIMATION: APPROACH 1 (COMPANY REVENUE ANALYSIS-BASED ESTIMATION), 2022



Source: MarketsandMarkets Analysis and Primary Insights

For the calculation of the global market value, the segmental revenue was arrived at based on the revenue mapping of major players active in the microbiome sequencing services market. This process involved the following steps:

- A list of the major global players operating in the microbiome sequencing services market was generated.
- The annual revenues generated by major global players from the microbiome sequencing services market, or the nearest reported business unit/category, were mapped.
- The revenues of the top 3 players in the microbiome sequencing services market were identified.
- The revenue of the top 3 players was assumed to be approximately 18–20% in 2022 and added up to 100% to derive the global value of the microbiome sequencing services market.

FIGURE 5 ILLUSTRATIVE EXAMPLE OF COMPANY REVENUE ANALYSIS, 2022

COMPANY REVENUE IN THE MICROBIOME SEQUENCING SERVICES MARKET

- CALCULATING BUSINESS UNIT (BU) REVENUE OF COMPANY
- IDENTIFYING RELEVANT SERVICES OFFERED
- ESTIMATING VALUE OF THE RELEVANT SERVICES
- VALIDATING ESTIMATION THROUGH PRIMARY RESEARCH
- ARRIVING AT THE REVENUE GENERATED FROM MICROBIOME SEQUENCING SERVICES OFFERED

ILLUSTRATIVE EXAMPLE OF CRL (2022)

CRL: TOTAL REVENUE	▪ The company has 3 business segments: ▪ Research Models & Services (RMS), Drug Discovery & Safety Assessment (DSA), and Manufacturing (MFG). Its total revenue is USD 3.9 billion
CRL: RELEVANT BUSINESS REVENUE	▪ RMS Business Segment: **~USD 0.74 Billion** ▪ DSA Business Segment: **~USD 0.53 Billion** ▪ MFG Business Segment: **~USD 0.38 Billion**
CRL: REVENUE GENERATED FROM MICROBIOME SEQUENCING SERVICES	▪ All 3 business segments include microbiome sequencing service offerings. ▪ Total Revenue for Microbiome Sequencing Services: **~USD 0.04 Billion**

Source: Annual Reports, SEC Filings, Investor Presentations, Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

APPROACH 2: MNM REPOSITORY ANALYSIS

- For estimating the microbiome sequencing services market, MnM reports, such as the contract research organization (CRO) services market, next-generation sequencing (NGS) market, and metagenomic sequencing market, were considered. The global and regional market values of microbiome sequencing services and the dependent submarkets were extracted from the MnM repository and validated through secondary and primary research. The final global market size was triangulated through the average of all approaches and validated through primary interviews with industry experts.

APPROACH 3: SECONDARY RESEARCH

- The size of the microbiome sequencing services market was obtained from secondary sources.
- The shares of various microbiome sequencing services in the overall market were obtained from secondary data and validated through primary participants to arrive at the total microbiome sequencing services market.
- Primary participants further validated the numbers.

APPROACH 4: PRIMARY INTERVIEWS

- As part of the primary research process, individual respondents were interviewed for insights on market size and market growth.
- All responses were collated, and a weighted average was taken to derive a probabilistic estimate of the market size and growth rate.

2.2.1 INSIGHTS FROM PRIMARY EXPERTS

FIGURE 6 MARKET SIZE VALIDATION FROM PRIMARY SOURCES

FACTORS AFFECTING MARKET GROWTH

Product Manager	The current market is growing bigger in size as new players are entering the market with varied services for microbiome sequencing.
Group Leader	The key sequencing methods popular for microbiome profiling include amplicon sequencing and shotgun sequencing.
Research Scientist	Emerging technologies such as nanopore sequencing are set to provide lucrative opportunities for market growth.

MARKET TRENDS

R&D investments in oncology applications are positively impacting the market.

Growing adoption of new sequencing tools is expected to drive service adoption.

– Market Experts

ACCORDING TO A MAJORITY OF INDUSTRY EXPERTS, THE TOP MARKET PLAYERS ACCOUNTED FOR ~20% OF THE MICROBIOME SEQUENCING SERVICES MARKET

COMPANY	MARKET RANK (2022)
CRL (US)	1
Eurofins Scientific (France)	2
BGI (China)	3

Source: Primary Insights

The final market size was estimated based on the weighted average of values arrived at through various approaches, such as company revenue analysis, MnM repository analysis, secondary research, and primary insights. The research methodology and market numbers were further validated from primaries.

SEGMENT ASSESSMENT (BY SERVICE, TYPE, TECHNOLOGY, AND END USER)

The segmental assessment was conducted using the following approaches:

- Approach for service: Revenue contributions/splits of microbiome sequencing services were calculated from the total revenues of leading players in the market (wherever available), and respective growth trends were derived by studying the adoption patterns of various services considered in the study.
- Approach for type: Revenue contributions/splits of different segments (amplicon sequencing, shotgun sequencing, and whole-genome sequencing) to the total revenues of leading players in the market (wherever available) and respective growth trends were derived by studying the adoption patterns of various products considered in the study.
- Approach for technology: Splits for segments were derived by studying the adoption preference (based on the type of service, sample size, and turnaround time) for different technologies used in services, including sequencing by synthesis (SBS), sequencing by ligation (SBL), and nanopore sequencing. This information was validated through primary interviews.
- Approach for end users: Splits for segments were derived by studying the adoption patterns of microbiome sequencing services by different end users in the market.

- Approach for regions: The geographic assessment was conducted by studying geographic revenue contributions/splits of leading players in the market (wherever available) and respective growth trends.

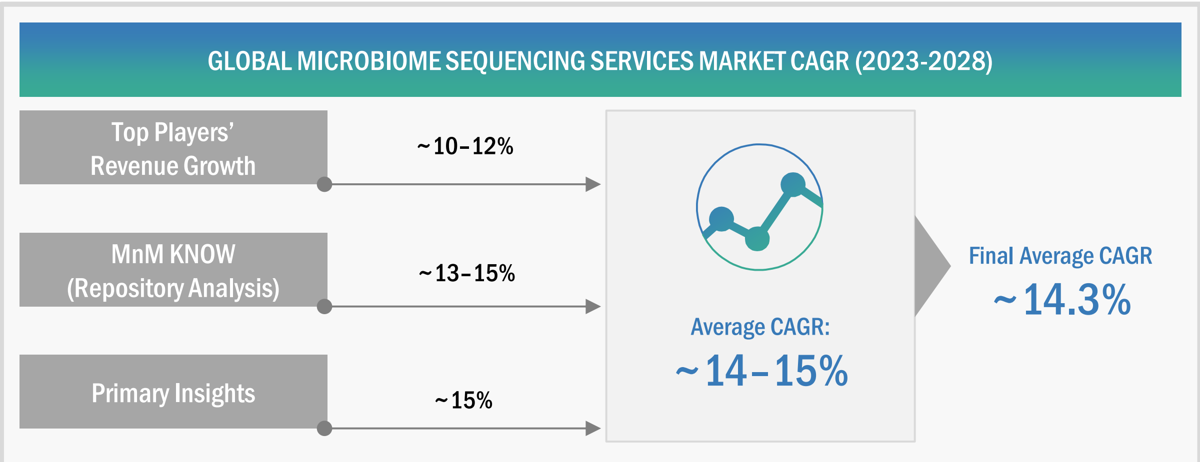
The assumptions and approaches were validated at each point through industry experts contacted during primary research. Considering the limitations of data available from secondary research, revenue estimates for individual companies (for the overall microbiome sequencing services market and geographic market assessment) were ascertained based on a detailed analysis of their respective service offerings, geographic reach/strength (direct or through distributors or suppliers), and shares of the leading players in a particular region or country.

2.3 MARKET GROWTH RATE PROJECTIONS

The growth forecast model was defined based on an assessment of the qualitative and quantitative factors affecting the market growth. These mainly included:

1. Historic revenue growth trends of the leading players in the market
    - Demand-side factor assessment, including clinical trials conducted for microbiome therapeutics, pharmaceutical R&D spending, the incidence of various diseases, the number of drug discovery and development projects across all geographies, and trial density, among other factors
    - Impact assessment of the key factors, such as drivers, restraints, challenges, and opportunities, influencing market growth
    - Impact assessment of the historic competitive trends driving market growth—service launches, strategic growth initiatives, innovation trends, and R&D initiatives
    - Impact assessment of regulatory guidelines/mandates on market growth over the forecast period. Only the defined regulatory guidelines related to the study period were considered for this assessment. The impact of any future regulatory uncertainties on market growth is not accounted for.
    - The impact of the recession on the overall market was evaluated for 2023–2028

FIGURE 7 MICROBIOME SEQUENCING SERVICES MARKET (SUPPLY-SIDE): CAGR PROJECTIONS



Source: MarketsandMarkets Analysis and Primary Insights

Considering the average of all the growth rates from the approaches mentioned above, the final CAGR for 2023–2028 was estimated to be ~14.3% for this report.

FIGURE 8 MICROBIOME SEQUENCING SERVICES MARKET: GROWTH ANALYSIS OF DEMAND-SIDE DRIVERS

DRIVERS	IMPACT LEVEL
Increasing demand for metagenomic sequencing	High
Growing focus on human microbiome therapeutics development	High
High cost of advanced infrastructure for sequencing	Medium
RESTRAINTS	**IMPACT LEVEL**
Dearth of skilled personnel	Medium
OPPORTUNITIES	**IMPACT LEVEL**
Expanding applications of microbiome sequencing in therapeutics development	Medium
CHALLENGES	**IMPACT LEVEL**
Issues related to sample preparation/isolation of microbes for sequencing	Low

IMPACT LEVEL: High Medium Low

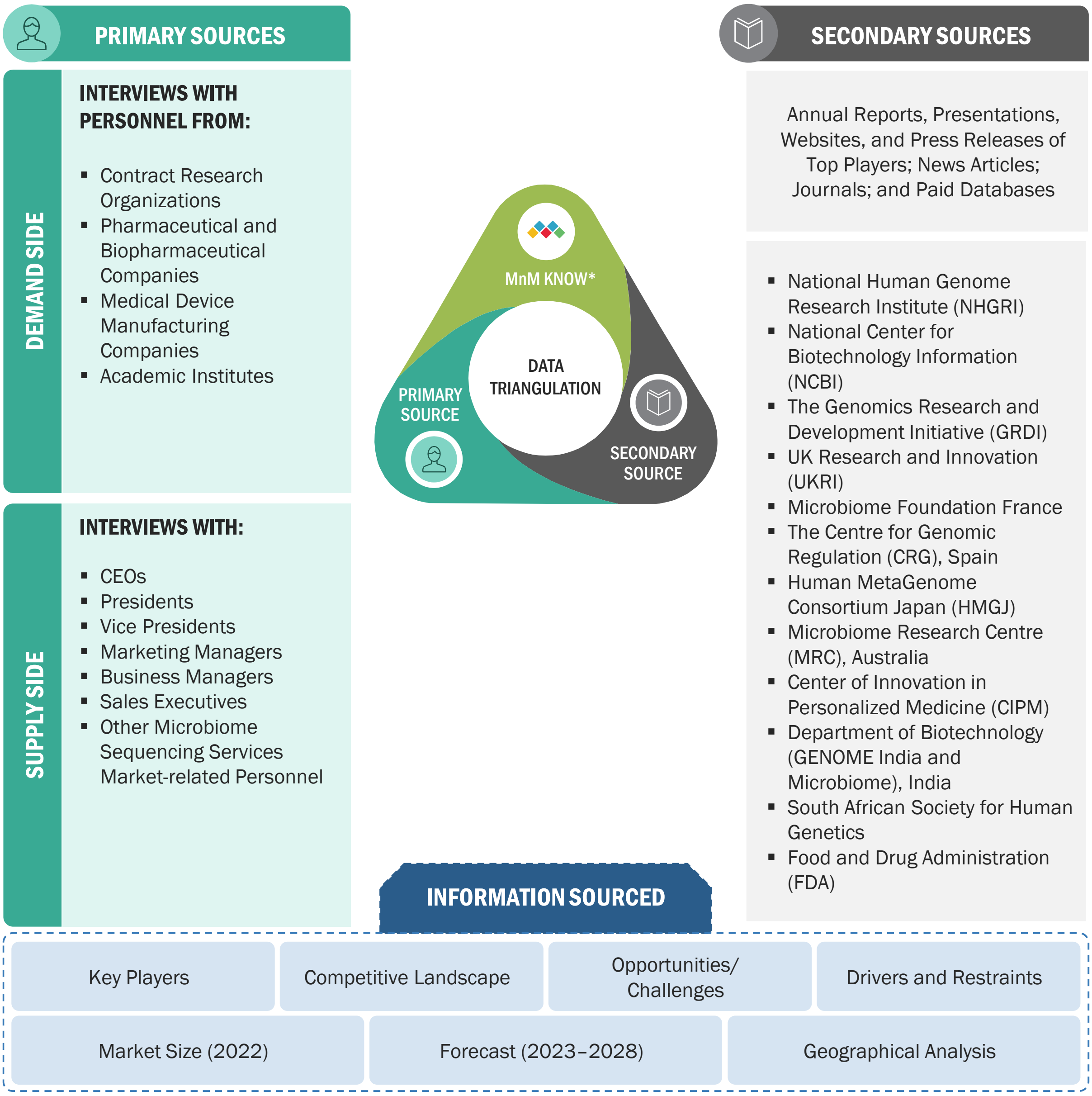
Source: Secondary Research, MarketsandMarkets Analysis, and Primary Insights

2.4 MARKET BREAKDOWN AND DATA TRIANGULATION

After arriving at the market size from the market size estimation process explained above, the total market was divided into several segments. To complete the overall market engineering process and arrive at the exact statistics for all segments, data triangulation and market breakdown procedures were employed, wherever applicable.

The following figure shows the market validation sources structure and the data triangulation methodology implemented in the market engineering process for making this report.

FIGURE 9 DATA TRIANGULATION METHODOLOGY



MnM KNOW* stands for MarketsandMarkets' 'Knowledge Asset Management' framework. In this context, it stands for existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform "Knowledge Store," subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate information gathered from primary and secondary sources.

2.5 RESEARCH ASSUMPTIONS

- Market values have been rounded off to the nearest decimal. Due to this, there may be slight differences between actual totals and the totals mentioned in market tables.
- Region-wise inflation/deflation for service pricing during 2023–2028 has not been considered.
- It has been assumed that dollar fluctuations will not affect the growth forecast to a significant extent.
- Revenues of private companies acquired from paid databases were assumed to be in line with actual company financials.
- A positive economic climate will continue in emerging countries until 2028, and the recession of 2023 will have a low impact on the microbiome sequencing services market.
- There will be no substantial changes in government regulations that will significantly impact the market.
- All market segments and subsegments listed in this report are considered to be mutually exclusive.
- Wherever the exact segmental market splits were not available, historical trends and inputs from primary respondents have been considered.

2.6 RISK ANALYSIS

DESCRIPTION	RISK ANALYSIS
The growth rate for each segment is considered based on recent market trends, key developments of companies, analysis of historical growth rates, and primary insights.	Moderate
Market sizes and shares for 2022 were estimated based on the financials available for nine months in company annual/quarterly reports and presentations for a few companies.	Low to Moderate

2.7 IMPACT OF RECESSION ON MICROBIOME SEQUENCING SERVICES MARKET

A recession is a period of temporary economic decline during which trade and industrial activities are reduced. It is generally identified by a decline in the GDP in two successive quarters. According to the World Economic Forum Forecast Report (October 2022), global GDP growth would slow from 6% in 2021 to 3.2% in 2022 and 2.7% in 2023. Alternatively, the same report suggests that global inflation would rise from 4.7% in 2021 to 8.8% in 2022 but decline to 6.5% in 2023 and to 4.1% by 2024.

The key reasons behind this slowdown are the Russia-Ukraine war (which led to a severe energy crisis in Europe, where gas prices have escalated four-fold since 2021) and frequent lockdowns in China under its 'Zero COVID policy,' which impacted global trade and led to a sharp appreciation of the US dollar.

Based on analysis by the pharmaceutical and biotechnology industry and inputs from primary experts, the pharmaceutical and biotechnology industry is expected to withstand this downturn. The demand for healthcare products and services is expected to endure, and the recession will have a low to minimal impact on overall healthcare-oriented businesses.

Healthcare, inclusive of pharma-biotech companies, is a defensive industry, which is likely to hold up relatively well to recession situations compared to other industries such as automotive, telecom, and energy, as drugs for treating health conditions are a necessity for patients. The industry is also backed by insurance and health plans, which make this sector recession-resilient.

Additionally, primary experts mentioned that investors are now looking to invest in defensive businesses (as a hedge against recessionary views), such as the pharmaceutical and biotechnology industry, and would avoid recession-sensitive businesses like consumer durables. On the other hand, small to medium-sized pharmaceutical and biotechnology companies might face uncertainties due to changes in macroeconomic conditions, such as a decrease in healthcare spending, increasing cost of raw materials, reduced margins due to rising inflation, changes to the government funding environment, and limited or significantly reduced points of access of products. Such factors might have an insignificant effect on the growth of the market in the forecast period.

As per the Healthcare Outlook 2023 report published by The Economist Intelligence Unit Limited (EIU), total healthcare spending (public and private) was expected to rise by 4.9% in 2023, propelled by higher costs and wages. However, due to high inflation and slow economic growth, EUI estimated that healthcare spending would remain similar to 2022 throughout 2023, meaning that 2023 will be the second successive year of real-term funding declines. OECD data suggests that after the global financial crisis of 2008–09, healthcare spending on preventive care and pharmaceuticals was reduced, and the same pattern is expected to be repeated. This decline in healthcare spending will be partially offset by growth in the pharmaceutical industry, the expansion of international companies into developing geographies, and investments made in 2021 and 2022.

TABLE 1 GLOBAL INFLATION RATE PROJECTIONS, 2021–2027 (% GROWTH)

	2021	2022	2023	2024	2025	2026	2027
Inflation Rate	4.7%	8.8%	6.5%	4.1%	3.2%	3.3%	3.2%

Source: IMF, World Economic Outlook Update, July 2022

TABLE 2 US HEALTH EXPENDITURE, 2019–2022 (USD MILLION)

Expenditure Amount	2019	2020	2021	2022
Total National Health Expenditure	3,757,382.0	4,144,077.0	4,255,127.0	4,442,352.6

Source: Centers for Medicare & Medicaid Services, 2022 (estimated on the basis of CMC data).

TABLE 3 US HEALTH EXPENDITURE, 2023–2027 (USD MILLION)

Expenditure Amount	2023	2024	2025	2026	2027
Total National Health Expenditure	4,657,806.7	4,904,670.4	5,169,522.6	5,449,969.3	5,746,992.6

Source: Centers for Medicare & Medicaid Services, 2022-2027 (estimated on the basis of CMC data).

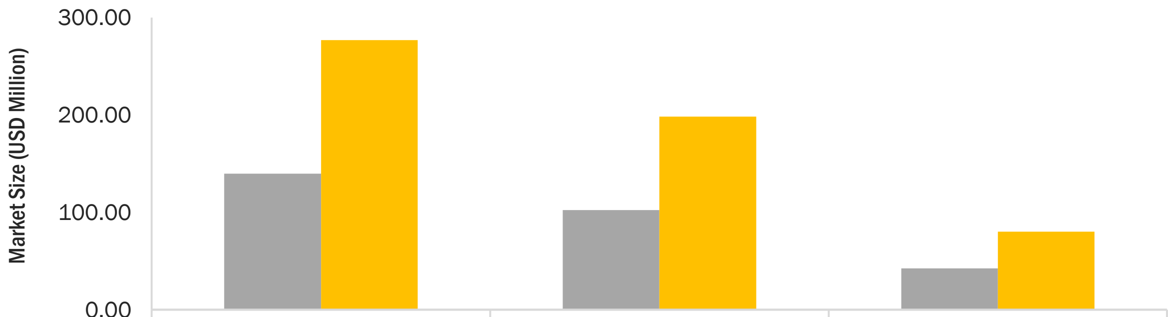
As per the Centers for Medicare & Medicaid Services data, US healthcare spending will grow at an average of 4.9% per year from 2022 to 2024 and 5.3% per year beyond that, hitting USD 6.8 trillion in 2030.

However, economists and other financial experts have stated that they do not observe any major reduction in the usage of pharma/biotech products and services in the slower-growing global economies. The demand for medicines is relatively inelastic. Moreover, many drugs are expected to go off-patent between 2025 to 2030. This is likely to boost drug innovation in the drug discovery and development field. Therefore, the microbiome sequencing services market will be one of the less-affected sectors.

3 EXECUTIVE SUMMARY

The microbiome sequencing services market is projected to grow from USD 284.31 million in 2023 to USD 555.24 million by 2028, at a CAGR of 14.3%. Growth in this market is majorly driven by the increasing demand for metagenomic sequencing, the growing focus on human microbiome therapeutics development, and the high cost of advanced infrastructure for sequencing. However, a dearth of skilled personnel is expected to hinder the growth of this market during the forecast period.

FIGURE 10 MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2023 VS. 2028 (USD MILLION)

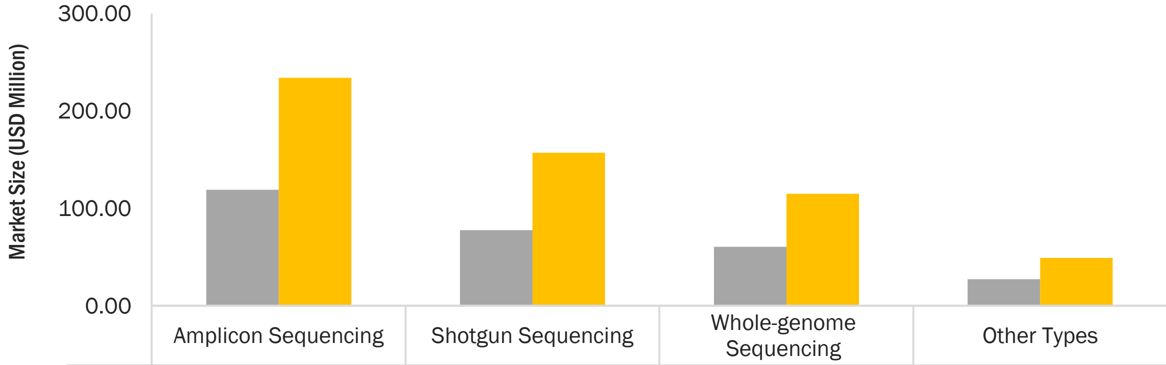


	Sequencing and Library Preparation	Sample Preparation	Data Analysis
2023	139.58	102.25	42.48
2028	276.91	198.29	80.04

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

Based on service, the microbiome sequencing services market is segmented into sample preparation, sequencing and library preparation, and data analysis services. A majority of the service providers/key market players offer all three types of services as part of the microbiome analysis workflow. Variations in sample size, associated pricing, use of different technologies, and the strength of local players are the key factors driving the revenue generation capacity of the above-mentioned segments. In 2022, the sequencing and library preparation segment accounted for the largest market share. This segment is also estimated to grow at the highest CAGR during the forecast period.

FIGURE 11 MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2023 VS. 2028 (USD MILLION)

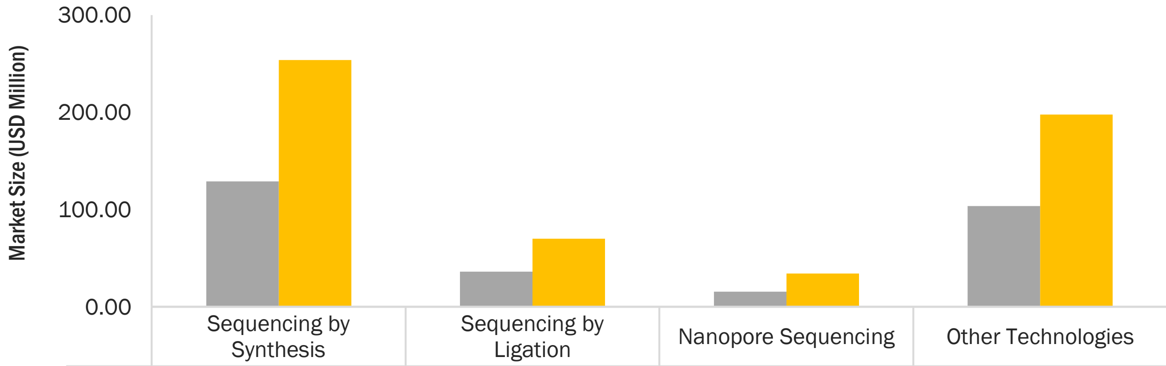


	Amplicon Sequencing	Shotgun Sequencing	Whole-genome Sequencing	Other Types
2023	118.98	77.52	60.45	27.36
2028	234.18	157.10	114.85	49.11

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

Based on type, the global microbiome sequencing services market is segmented into amplicon sequencing, whole-genome sequencing, shotgun sequencing, and other types (including de novo sequencing, metagenomic sequencing, and metatranscriptomic sequencing). The amplicon sequencing segment accounted for the largest market share in 2022. The large share of this segment can be attributed to the high adoption of 16S rRNA microbiome sequencing and targeted sequencing. Key market players/service providers have built an innovative portfolio around these services.

FIGURE 12 MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2023 VS. 2028 (USD MILLION)

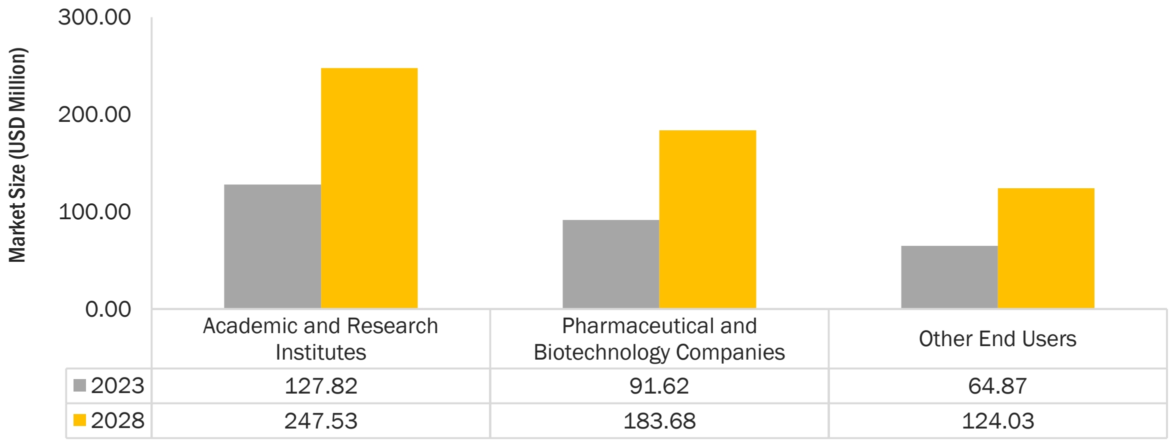


	Sequencing by Synthesis	Sequencing by Ligation	Nanopore Sequencing	Other Technologies
2023	129.01	36.24	15.42	103.64
2028	253.56	69.95	34.21	197.53

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

Based on technology, the global microbiome sequencing services market is segmented into sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and other technologies (semiconductor sequencing and Sanger sequencing, among other emerging technologies). The SBS segment accounted for the largest market share in 2022, while the nanopore sequencing segment is estimated to grow at the highest CAGR during the forecast period.

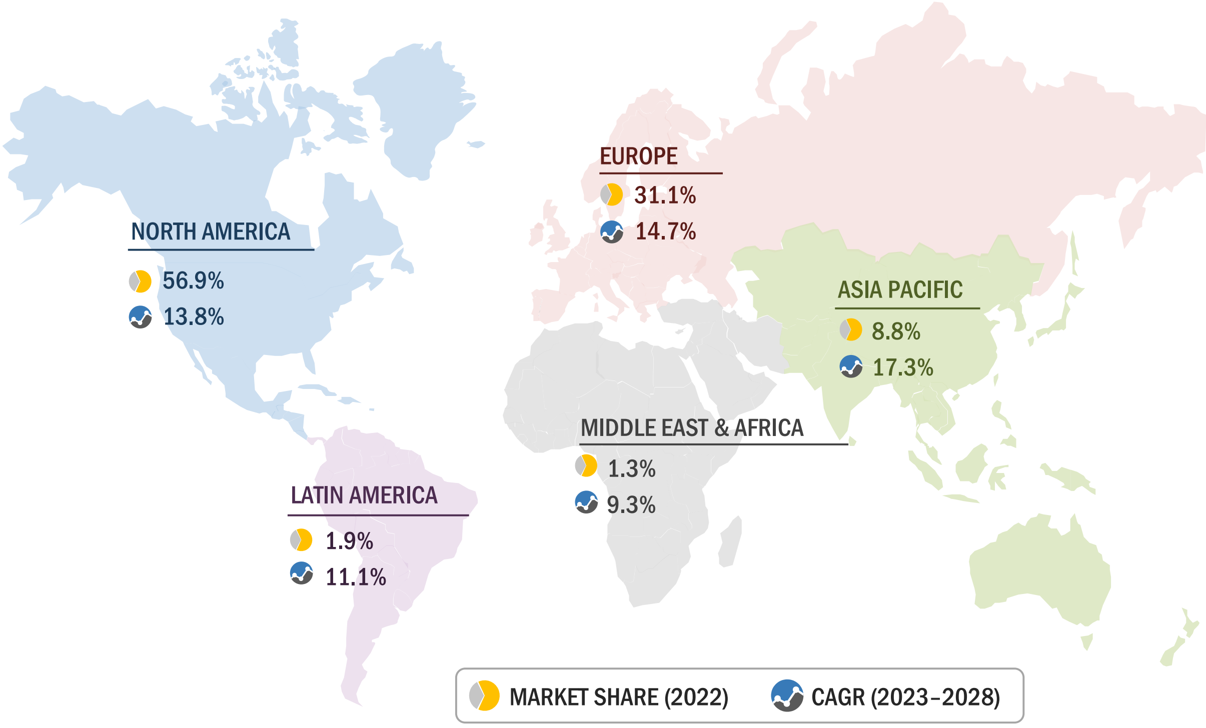
FIGURE 13 MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2023 VS. 2028 (USD MILLION)



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

Based on end users, the global microbiome sequencing services market is segmented into pharmaceutical and biotechnology companies, academic and research institutes, and other end users (including small-scale CROs, CDMOs, hospitals, and clinics). The academic and research institutes segment accounted for the largest market share in 2022. This can be attributed to the high adoption of microbiome sequencing services and clinical research in these facilities.

FIGURE 14 GEOGRAPHICAL SNAPSHOT OF MICROBIOME SEQUENCING SERVICES MARKET



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

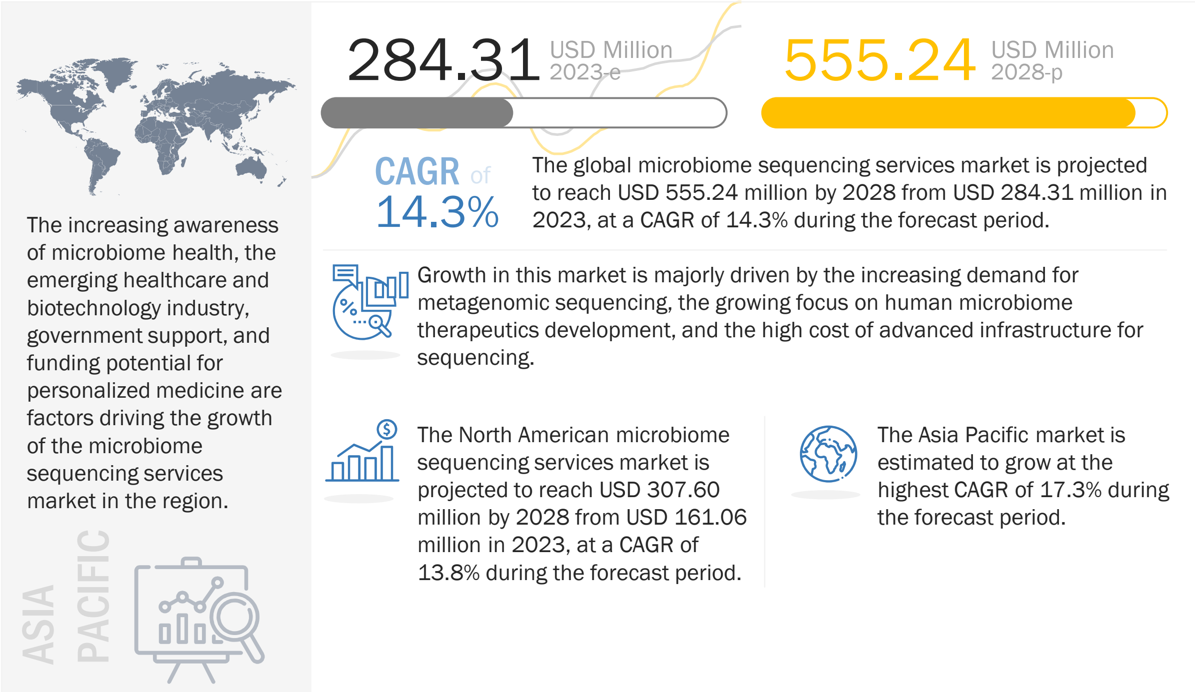
The microbiome sequencing services market is segmented into North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Factors such as established research infrastructure, technological advancements, supportive regulatory environment, robust biotechnology and healthcare industry, and the availability of funding and investment are driving the growth of the microbiome sequencing services market in North America. The microbiome sequencing services market in the Asia Pacific is expected to register the highest growth rate during the forecast period due to the increasing awareness of microbiome health, the emerging healthcare and biotechnology industry, government support, and funding potential for personalized medicine.

The prominent players in the global microbiome sequencing services market are Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea).

4 PREMIUM INSIGHTS

4.1 MICROBIOME SEQUENCING SERVICES MARKET OVERVIEW

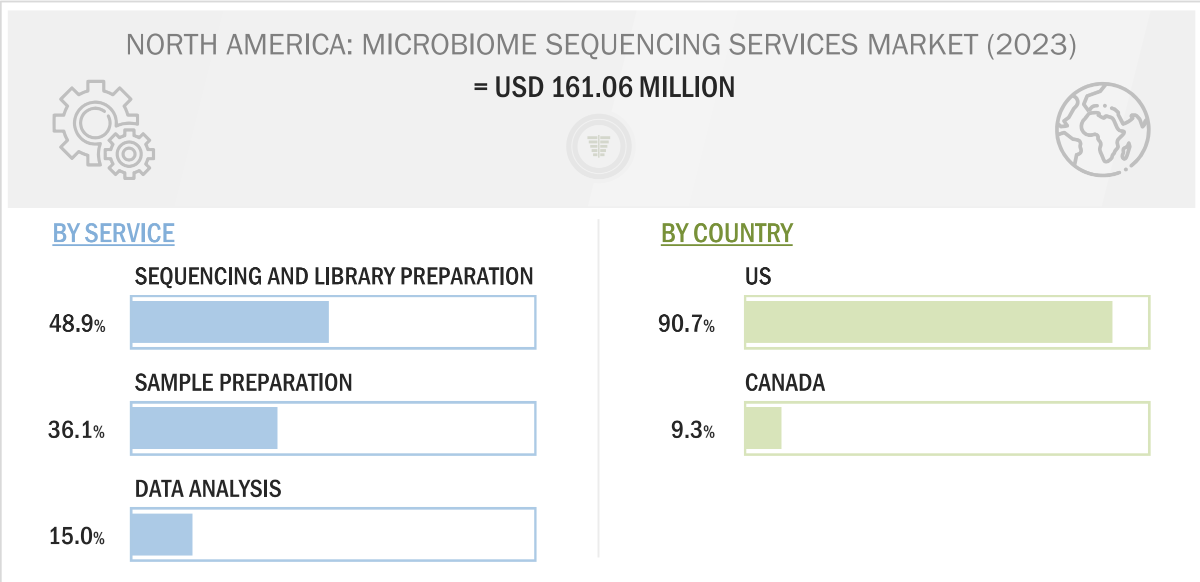
FIGURE 15 GROWING FOCUS ON HUMAN MICROBIOME THERAPEUTICS DEVELOPMENT TO DRIVE MARKET



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

4.2 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE AND COUNTRY (2023)

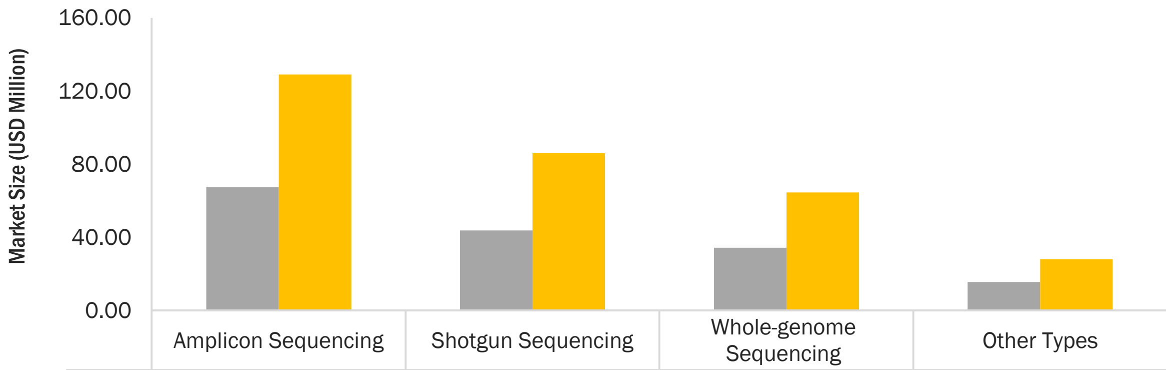
FIGURE 16 SEQUENCING AND LIBRARY PREPARATION SERVICES TO ACCOUNT FOR LARGEST SHARE OF NORTH AMERICAN MICROBIOME SEQUENCING SERVICES MARKET IN 2023



Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

4.3 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2023 VS. 2028 (USD MILLION)

FIGURE 17 SHOTGUN SEQUENCING SEGMENT TO GROW AT HIGHEST CAGR DURING FORECAST PERIOD



	Amplicon Sequencing	Shotgun Sequencing	Whole-genome Sequencing	Other Types
2023	67.44	43.79	34.29	15.54
2028	128.99	85.96	64.54	28.12

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

5 MARKET OVERVIEW

5.1 INTRODUCTION

The microbiome sequencing services market is estimated to grow at a CAGR of 14.3% during the forecast period. Growth in this market is majorly driven by the increasing demand for metagenomic sequencing, the growing focus on human microbiome therapeutics development, and the high cost of advanced infrastructure for sequencing. However, a dearth of skilled personnel is expected to hinder the growth of this market during the forecast period.

5.2 MARKET DYNAMICS

FIGURE 18 MICROBIOME SEQUENCING SERVICES MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

DRIVERS	▪ Increasing demand for metagenomic sequencing ▪ Growing focus on human microbiome therapeutics development ▪ High cost of advanced infrastructure for sequencing
RESTRAINTS	▪ Dearth of skilled personnel
OPPORTUNITIES	▪ Expanding applications of microbiome sequencing in therapeutics development
CHALLENGES	▪ Issues related to sample preparation/isolation of microbes for sequencing

Source: Secondary Literature, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 4 MICROBIOME SEQUENCING SERVICES MARKET: IMPACT ANALYSIS OF DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

FACTOR	IMPACT		
	2–3 YEARS	3–5 YEARS	5–10 YEARS
DRIVERS			
Increasing demand for metagenomic sequencing	High	High	High
Growing focus on human microbiome therapeutics development	Medium	High	Medium
High cost of advanced infrastructure for sequencing	Medium	Medium	Medium
RESTRAINTS			
Dearth of skilled personnel	High	Medium	Medium
OPPORTUNITIES			
Expanding applications of microbiome sequencing in therapeutics development	Medium	Medium	High
CHALLENGES			
Issues related to sample preparation/isolation of microbes for sequencing	Medium	Low	Low

5.2.1 DRIVERS

5.2.1.1 Increasing demand for metagenomic sequencing

Metagenomic sequencing is a target-independent approach that offers a comprehensive view of pathogenic agents and enables the detection of common and unexpected pathogens in samples. In recent years, there has been a significant increase in the demand for metagenomic sequencing in the field of microbiome research owing it its wide array of applications, such as disease diagnosis, drug discovery, and precision medicine. Recent developments in metagenomic sequencing technologies, such as improvements in sequencing accuracy, scalability, and data analysis, support the use of metagenomic sequencing. The development of long-read sequencing, single-molecule real-time sequencing, and nanopore sequencing has been particularly helpful in the metagenomic sequencing market. The demand for long-read sequencing-based metagenomics has increased in samples that are less complex, involve host-associated microbiomes, and comprise fewer microbial strains than microbiomes in marine environments, sediment, and soil. Such technologies are expected to drive the demand for microbiome sequencing services.

Over the years, the demand for precision medicine has increased. Precision medicine involves tailoring medical treatments and interventions to an individual's unique characteristics, including their microbiome. Metagenomic sequencing services are helpful in gathering essential information for personalized medicine, as well as identifying specific microbial signatures associated with disease risk and specific treatment responses. Metagenomic sequencing is also being used to study the role of microbiomes in cancer development, progression, and response to treatment. As per a study published in the Lancet in 2022, the gut microbiome of patients with metastatic melanoma can influence their response to immune checkpoint inhibitors. The study identified specific microbial species that were associated with better response rates and longer progression-free survival, suggesting that microbiome-targeted interventions could enhance the efficacy of cancer immunotherapies.

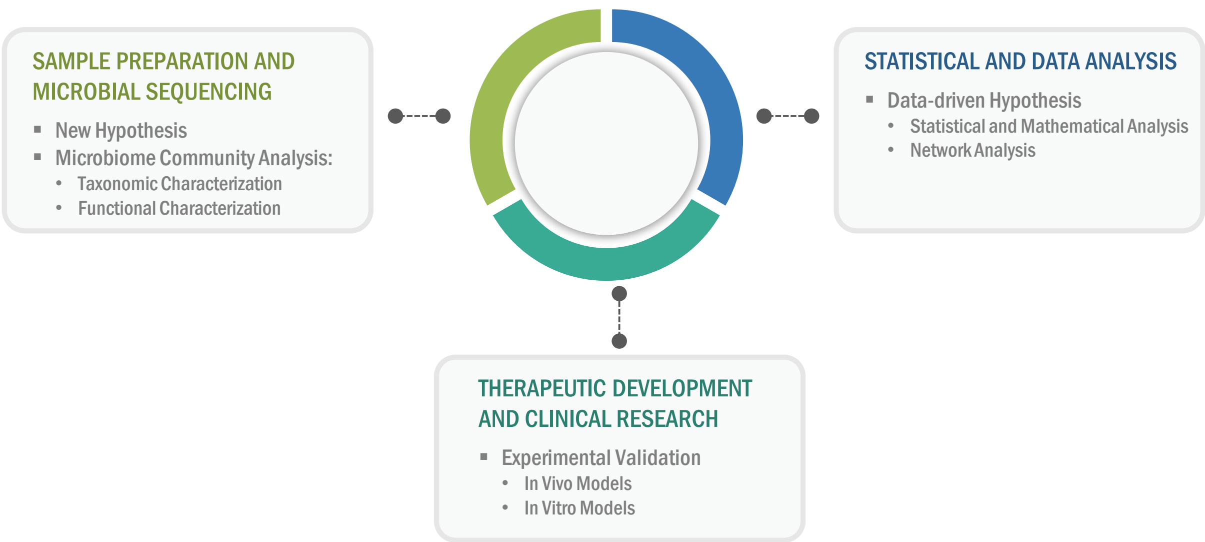
The demand for metagenomic sequencing has been driven by its applications in precision medicine, gut-brain axis research, and recent developments in sequencing technologies and data analysis methods. Metagenomic sequencing is proving to be a valuable tool for studying complex microbial communities and understanding their roles in various fields of research. The emergence of metagenomics has resolved the challenge of capturing the entire microbial community in different sample types by enabling the analysis of whole genomes without the need to culture the samples first. This is expected to boost the growth of the microbiome sequencing services market.

5.2.1.2 Growing focus on human microbiome therapeutics development

The increasing focus on human microbiome therapeutics development is driving therapeutics manufacturers to opt for sequencing, sample preparation, and data analysis services. This is further propelled by the need for a deeper understanding of the microbiome, including its composition, function, and interaction with the host, to develop microbiome-based therapeutics. Microbiome sequencing plays a critical role in providing this understanding by allowing for the identification and characterization of the microorganisms that make up the microbiome. Medium-sized and small-scale therapeutic manufacturers often prefer to outsource the initial stages of microbiome therapeutics development, such as screening, sample preparation, sequencing, and data analysis. This has offered significant momentum to large-scale CROs and service providers to develop their sequencing portfolio.

Human gastrointestinal microbiome research has the potential to offer pivotal clinical insights and facilitate the therapeutic development workflow. A research study conducted in June 2021, funded by the Australian National Health and Medical Research Council, highlighted a suitable avenue to translate microbiome research workflow, including the compositional and functional characterization of the microbiome, data-driven hypotheses generation, and the experimental validation of the hypotheses. By integrating different sequencing technologies (such as shotgun, metagenomic, and multi-omics) and data processing solutions, the ability to identify causative agents and design the appropriate microbiome therapeutics in disease treatment is expected to accelerate. The diagram below highlights the role of microbiome sequencing in such workflows.

FIGURE 19 ROLE OF MICROBIOME SEQUENCING IN THERAPEUTICS DEVELOPMENT



Source: Secondary Literature and MarketsandMarkets Analysis

Over the last few years, microbiome-based therapeutics development has witnessed significant advancements ranging from prebiotics and probiotics development to live biotherapeutics and microbiome mimetics. Advancements in bacterial culture methods from gastrointestinal (GI) samples, along with the expanding applications of metagenomic sequencing, have overcome several technical limitations in microbiome-based therapeutics development. However, a persisting challenge in this area is the development of appropriate methods to identify, refine, and test candidate therapies for microbiome medicines. Market players are progressing toward optimizing methods to identify candidate microbes, develop appropriate preclinical validation models, and progress to the personalized targeting of microbiome-based medicines through sequencing capabilities. This is expected to propel the market growth.

5.2.1.3 High cost of advanced infrastructure for sequencing

The field of microbiome sequencing requires sophisticated and expensive equipment, such as next-generation sequencing (NGS) platforms, bioinformatics tools, and data storage and analysis capabilities. Setting up and maintaining such infrastructure can be cost-prohibitive for many research and academic institutes, pharmaceutical and biotechnology companies, and other end users. Outsourcing microbiome sequencing to specialized service providers allows professionals to access advanced infrastructure and expertise without the need for substantial upfront investments. These service providers typically have state-of-the-art facilities and experienced personnel who are skilled in performing microbiome sequencing using the latest technologies and methodologies. Through outsourcing, professionals can leverage the capabilities of these specialized facilities without the burden of capital expenditures, operational costs, and maintenance expenses associated with in-house infrastructure.

The cost of NGS platforms is the highest in North America, followed by Europe. The cost in all regions is expected to increase by 2027 due to the high cost of raw materials, manufacturing, and labor. The cost of NGS platforms is low in the Middle East, Africa, and South America compared to Europe, North America, and the Asia Pacific due to the low cost of raw materials and the availability of a low-cost workforce (which reduces the overall manufacturing cost).

TABLE 5 AVERAGE SELLING PRICE OF NGS SYSTEMS, BY KEY PLAYER (2021)

COMPANY	INSTRUMENT	AVERAGE COST RANGE (USD)
Illumina	NovoSeq System	~850,000–900,000
Illumina	NextSeq System	~200,000–250,000
Illumina	MiSeq System	~80,000–85,0000
Illumina	iSeq System	~18,000–20,000
Illumina	Mini Seq System	~40,000–45,000
PacBio	Sequel System	~350,000–370,000
PacBio	Sequel IIe System	~490,000–500,000

Source: Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5.2.2 RESTRAINTS

5.2.2.1 Dearth of skilled personnel

The shortage of skilled personnel with expertise in sequencing and data analysis is a significant challenge in the field of microbiome sequencing. Microbiome research involves complex methodologies, including sample collection, DNA extraction, library preparation, sequencing, and data analysis. The proper execution of these steps requires specialized knowledge and technical expertise. However, the field of microbiome sequencing is relatively new and evolving rapidly, with a shortage of skilled personnel with the necessary expertise. Training and developing skilled personnel in microbiome sequencing can be time-consuming and resource-intensive, and there may be a lag in keeping up with the latest advancements in sequencing technologies and data analysis methods.

5.2.3 OPPORTUNITIES

5.2.3.1 Expanding applications of microbiome sequencing in therapeutics development

Microbiome sequencing is expanding its applications beyond research and is increasingly being utilized in therapeutics development, including the study of the skin microbiome and oral microbiome sequencing. These applications have the potential to advance the understanding of the role of microbiota in health and disease and inform the development of targeted interventions for improving skin health and oral health.

Skin microbiome sequencing has the potential to unravel the complex interactions between the skin microbiome and the host immune system to identify key microbial species or functional genes associated with specific skin conditions and elucidate mechanisms underlying the therapeutic effects of probiotics or other interventions. It can help inform the development of novel therapeutic strategies, including probiotics, prebiotics, and targeted antimicrobial treatments aimed at modulating the skin microbiome to improve skin health. Additionally, the speedy expansion of gut microbiome research activities has provided a foundation to explore the role of microbiota in other physiological systems, including the skin. Skin microbiome sequencing is targeted toward understanding the temporospatial distribution of cutaneous microbial communities. Research on cutaneous microbial communities is anticipated to provide crucial insights into microbiota in skin aging and pathology. Microbiome sequencing technologies are set to gain significant demand to overcome the limited information on correlative analysis, host-microbe interactions in the skin, and low taxonomic resolution. This is expected to open new avenues of business expansion for service providers operating in this market.

Oral microbiome sequencing can provide insights into the microbial etiology of oral health conditions, identify key pathogenic species or functional genes associated with oral diseases, and reveal potential therapeutic targets for intervention. This knowledge can be used to develop novel strategies for oral health management, including probiotics, antimicrobial treatments, and personalized oral care regimens tailored to an individual's oral microbiome.

5.2.4 CHALLENGES

5.2.4.1 Issues related to sample preparation/isolation of microbes for sequencing

Despite the success of using metagenomic sequence data to guide the isolation and cultivation of microbes, there are still challenges and limitations that need to be addressed for more widespread success. Anaerobic microbes may also pose technical challenges for some targeted isolation methods due to limitations with cell sorters. Determining suitable growth media and physicochemical conditions for target organisms based solely on genome sequences can be challenging, as natural environment compositions are often unknown. Contamination can also be a significant challenge in microbiome research, as it can introduce non-target microbial DNA from external sources during sample collection, processing, and sequencing. Contamination can arise from reagents, equipment, and environmental sources, leading to false-positive or false-negative results. Proper controls, including negative controls and rigorous contamination prevention measures, should be implemented to minimize contamination and ensure accurate and reliable sequencing results.

Recovering genomic data from target microbes for cultivation strategies can also be impacted by issues such as incomplete cell extraction, cell lysis, or DNA degradation, which may affect the representation of certain species in metagenome data. DNA extraction is a critical step in sample preparation for microbiome sequencing, and different DNA extraction methods can yield different results. Some DNA extraction methods may be biased towards certain microbial groups or may not effectively lyse certain tough-to-break cells, resulting in the incomplete representation of the microbial community in the sequencing data.

5.3 TECHNOLOGY ANALYSIS

Sequencing by Synthesis (SBS)

Sequencing by synthesis (SBS) is a widely used technology in microbiome sequencing services for analyzing microbial communities. SBS is a high-throughput DNA sequencing method that enables the rapid and cost-effective sequencing of millions of DNA fragments in parallel. It has revolutionized the field of microbiome research by allowing researchers to explore the diversity and composition of microbial populations in each sample with high resolution.

The basic principle of SBS involves the use of DNA polymerases, enzymes that can synthesize DNA strands, to sequentially add nucleotides to a growing DNA chain. As each nucleotide is added, it is labeled with a fluorescent dye that corresponds to the base being incorporated. The fluorescence signal is then detected and recorded, allowing the identification of the nucleotide that was added. This process is repeated multiple times, typically in cycles, to generate DNA sequences that can be assembled into a complete microbial community profile.

Sequencing by Ligation (SBL)

Sequencing by ligation (SBL) is a DNA sequencing technology that is used in microbiome sequencing services for analyzing microbial communities. SBL is a high-throughput DNA sequencing method that relies on the ligation of short DNA oligonucleotide adapters to target DNA fragments, followed by their sequential hybridization and ligation to complementary oligonucleotides, enabling the determination of the DNA sequence.

The basic principle of SBL involves the use of DNA adapters that are ligated to the DNA fragments to be sequenced. These adapters contain known sequences at their ends that serve as priming sites for the sequencing reaction. The DNA fragments, together with the adapters, are then subjected to multiple rounds of hybridization and ligation with a set of fluorescently labeled oligonucleotides that are complementary to the known adapter sequences. The ligation events are detected and recorded, allowing the determination of the DNA sequence.

Pyrosequencing

Pyrosequencing, also known as 454 sequencing, is a DNA sequencing technology that has been used in microbiome sequencing services for studying the composition and function of microbial communities. Pyrosequencing is a high-throughput NGS method that allows for the rapid and parallel sequencing of DNA fragments, providing deep insights into the genetic diversity of microbiomes.

The basic principle of pyrosequencing involves the detection of pyrophosphate (PPi) released during DNA synthesis. In this technology, DNA fragments are first amplified using PCR with primers that contain unique barcodes or indexes to enable sample multiplexing. The amplified DNA fragments are then sequenced in a massively parallel manner using a DNA polymerase enzyme and a series of nucleotide substrates that are added sequentially. As each nucleotide is added, the enzyme incorporates it into the growing DNA chain, and if the added nucleotide is complementary to the template DNA, pyrophosphate (PPi) is released as a byproduct. The released PPi is then converted into light, which is detected and quantified, allowing for the determination of the DNA sequence.

Sanger Sequencing

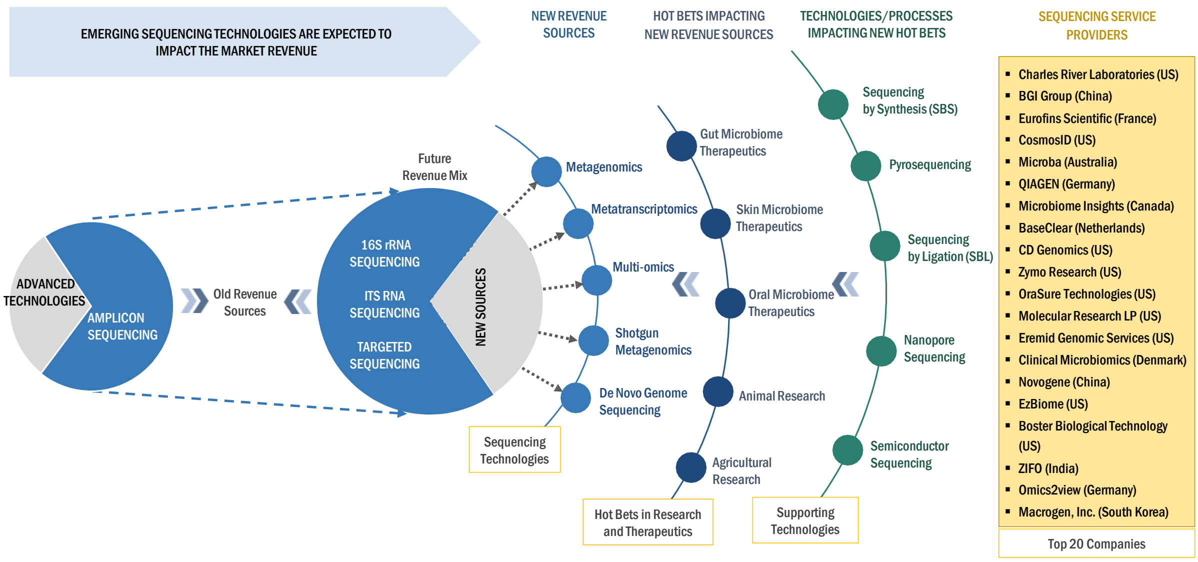
Sanger sequencing, also known as chain termination sequencing, is a traditional DNA sequencing technology used in microbiome sequencing services. While newer NGS technologies have gained popularity in recent years, Sanger sequencing still has its unique advantages and continues to be used in some microbiome research applications.

The basic principle of Sanger sequencing involves the enzymatic synthesis of DNA strands, where dideoxynucleotides (ddNTPs) are incorporated alongside regular deoxynucleotides (dNTPs) during DNA chain elongation. The incorporation of ddNTPs randomly terminates DNA synthesis, resulting in a set of DNA fragments of varying lengths that are complementary to the template DNA. The resulting DNA fragments are then separated by size using capillary electrophoresis, and the sequence is determined based on the order of the terminated ddNTPs.

5.4 TRENDS AND DISRUPTIONS IMPACTING CUSTOMERS' BUSINESSES

Robust research activities to explore the potential of the human microbiome present great potential in a number of treatments against obesity, inflammatory bowel disease, and diabetes, among other conditions. Information on microbiomes helps in the identification of the causal links between the microbiome and disease, further facilitating the development of efficient and targeted treatment strategies. Additionally, progress in state-of-the-art sequencing approaches, computational analyses, and experimental assays is supporting the growing demand for microbiome sequencing.

FIGURE 20 REVENUE SHIFT AND NEW REVENUE POCKETS FOR MICROBIOME SEQUENCING SERVICE PROVIDERS



Source: MarketsandMarkets analysis

The most commonly used sequencing technologies include 16S rRNA sequencing, ITS RNA sequencing, and whole-genome sequencing. Sequencing and experimental analyses of the microbiome continue to advance substantially. By coupling multi-omics technologies, statistical and computational analyses, and more advanced disease models, these approaches promisingly provide the opportunity to establish disease causality and subsequently inform therapeutic development.

5.5 VALUE CHAIN ANALYSIS

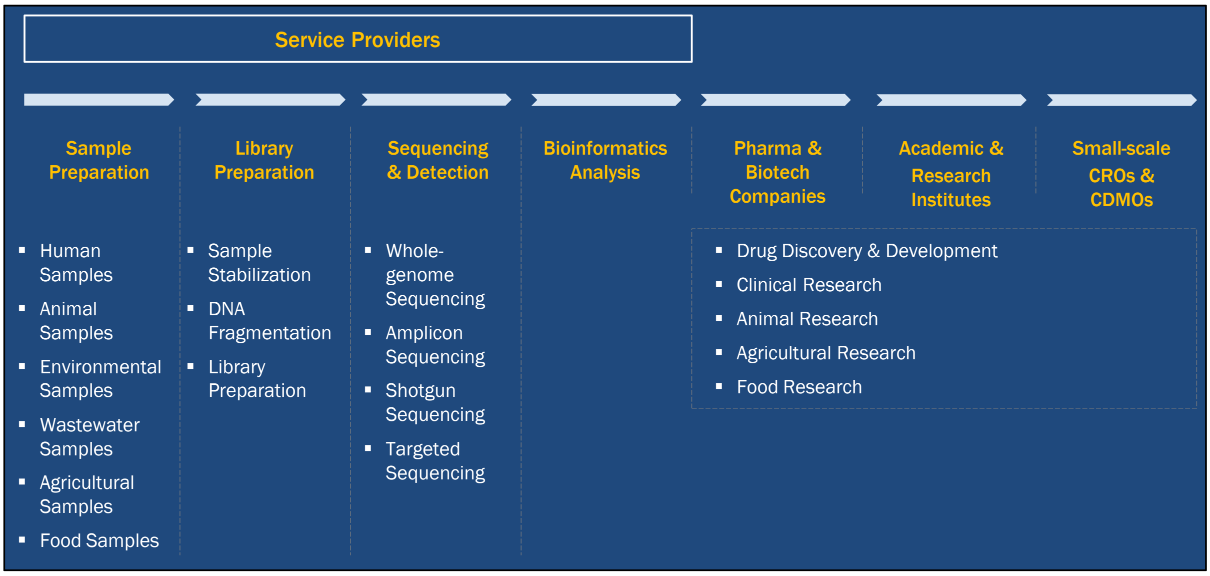
SAMPLE PREPARATION AND STUDY DESIGN SERVICES

Microbiome sequencing services start with outsourcing of study design, wherein the service provider helps in the implementation and validation of the hypothesis and the variables of interest in a sequencing project. Different variables in the aspect of study design include the number of samples required, apt sequencing technology to be used, and the requirement of follow-up studies. Service providers have expertise with respect to the collection, storage, and transport options optimized for this workflow.

SEQUENCING, LIBRARY PREPARATION, AND DATA ANALYSIS SERVICES

NGS instruments are predominantly used for the sequencing of DNA and RNA samples. Shotgun metagenomics sequencing is offered by a majority of sequencing service providers, even for deep-sequenced samples. DNA library preparation is verified for quality and yield. Genomic DNA is fragmented at specific sizes suitable for the purification/separation of desired proteins and nucleic acids. The selected fragments are then polished, A-tailed, and ligated with the full-length adapter. NGS technology is employed to sequence the sample, and the final stage involves bioinformatics analysis.

FIGURE 21 VALUE CHAIN ANALYSIS OF MICROBIOME SEQUENCING SERVICES MARKET



Source: MarketsandMarkets Analysis

5.6 ECOSYSTEM MARKET MAP

FIGURE 22 ECOSYSTEM OF MICROBIOME SEQUENCING SERVICES MARKET

	SERVICE	TYPE	END USER
CORE SEGMENT	▪ Sample Preparation ▪ Sequencing and Library Preparation ▪ Data Analysis	▪ Amplicon Sequencing ▪ Shotgun Sequencing ▪ Nanopore Sequencing ▪ De Novo Sequencing and Others	▪ Pharmaceutical and Biotechnology Companies ▪ Academic and Research Institutes ▪ Other End Users

Source: MarketsandMarkets Analysis

5.7 PORTER'S FIVE FORCES ANALYSIS

TABLE 6 MICROBIOME SEQUENCING SERVICES MARKET: PORTER'S FIVE FORCES ANALYSIS

PORTER'S FIVE FORCES	IMPACT
Intensity of Competitive Rivalry	Low
Bargaining Power of Suppliers	Moderate
Bargaining Power of Buyers	Moderate
Threat of Substitutes	Low to moderate
Threat of New Entrants	Low to moderate

Source: Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5.7.1 THREAT OF NEW ENTRANTS

The global microbiome sequencing services market is not a consolidated market. The well-established players operating in this market have narrow offerings, and a very small part of their portfolios are dedicated to microbiome sequencing. However, strong distribution channels and a global presence have made it easier for these players to maintain their market position. Moreover, existing product manufacturers have strong relationships with their customers (service providers) and are engaged in long-term contracts to supply the necessary equipment. As the industry demands high capital investments for R&D, testing, and manufacturing products, the entry barriers for new service providers in the market are estimated to be low to moderate.

5.7.2 THREAT OF SUBSTITUTES

The impact of this factor is low to moderate since there are fewer substitutes for established technologies in this market. The preference for traditional technologies for sequencing microbiomes may decrease due to the availability of advanced technology-oriented services.

5.7.3 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers is moderate in this market. This market has witnessed significant technological innovations in developing affordable and user-friendly products. Thus, suppliers have better bargaining powers if their products are highly differentiated and endowed with breakthrough technologies. Products with a high degree of technological capabilities can command a premium price in the market. The high degree of technological capabilities, coupled with a high buyer-to-seller ratio, significantly increases the bargaining power of suppliers. Furthermore, compatibility issues between instruments and consumables (which are responsible for recurrent costs) increase the bargaining power of suppliers.

5.7.4 BARGAINING POWER OF BUYERS

The bargaining power of buyers in the microbiome sequencing services market is moderate. Academic research institutes (one of the largest end users of sequencing products) are involved in bulk purchases of products and bulk sequencing of samples. This trend of bulk purchasing further increases the bargaining power of buyers in this market.

5.7.5 INTENSITY OF COMPETITIVE RIVALRY

The microbiome sequencing services market is fragmented in nature, with the top 3 players accounting for a share of ~20%. A large number of local players and small-scale CROs account for the remaining market share. These players operate in niche market segments. Industry players are expected to cash in on the various growth opportunities and gain maximum share in the market. This has created low-entry barriers for new players, and the existing players compete to expand their shares.

5.8 REGULATORY ANALYSIS

5.8.1 NORTH AMERICA

5.8.1.1 US

NGS sequencing technologies are being used predominantly in microbiome and metagenome sequencing. The increase in use is further supported by increasing advancements in genomics. Several federal agencies regulate NGS products, including the Food and Drug Administration (FDA), the Centers for Medicare and Medicaid Services (CMS), and the Federal Trade Commission (FTC).

CMS Regulations:

The CMS regulates clinical laboratories, including laboratories conducting clinical genetic testing, through its Clinical Laboratory Improvement Amendments of 1988 (CLIA) program. The CLIA established a certification that laboratories must pass to conduct clinical testing legally. The objective of the CLIA is to determine clinical testing quality, including verification of the procedures used and the qualifications of the technicians processing the tests.

FDA Regulations:

Until recent years, the FDA has practiced enforcement discretion for laboratory-developed tests (LDTs), which implies that LDTs are used in the clinic without the FDA’s assessment of their analytical and clinical validity. However, due to rapid advances in NGS, the growing need for clinical genetic testing, the growth of direct-to-consumer (DTC) genomic testing, and concerns that unregulated tests pose a public health threat, the FDA is modifying its approach.

The FDA has developed new guidance to regulate NGS genetic tests and verify their analytical and clinical validity. The agency has also drafted guidance proposing a new regulatory framework for LDTs. However, FDA guidance only represents the FDA’s current thinking on a topic and is not legally binding for the FDA or the parties it regulates. However, adhering to FDA guidance is beneficial because it can streamline the regulatory process. The draft guidance is listed below. Since they are in draft form, they are not currently being implemented.

List of FDA Draft Guidance on Genetic Tests:

- Framework for Regulatory Oversight of Laboratory Developed Tests (2014)
- FDA Notification and Medical Device Reporting for Laboratory Developed Tests (2014)
- Discussion Paper on Laboratory Developed Tests (2017)
- Next-generation Sequencing (NGS) Draft Guidance

- Use of Standards in FDA Regulatory Oversight of NGS-based In Vitro Diagnostics Used for Diagnosing Germline Diseases

In addition, the US FDA classifies medical devices into various categories. It reviews NGS instruments as Class II medical devices. The NGS manufacturer must submit a premarket notification or 510(k) to prove substantial equivalence to a legally marketed device. If there is no substantially equivalent device in the market, the manufacturer must secure premarket approval (PMA) from the FDA.

The three main classifications of medical devices by the US FDA are as follows:

TABLE 7 US FDA: MEDICAL DEVICE CLASSIFICATION

CLASS I DEVICES	CLASS II DEVICES	CLASS III DEVICES
▪ Low-risk medical devices, which are subject to the least regulatory control ▪ Exempt from premarket notification [510 (k)]	▪ Devices that pose a greater risk than Class I devices; require reasonable assurance of safety and effectiveness ▪ Require premarket notification [510 (k)], with exceptions for certain devices	▪ Devices that are used for sustaining human life and are important for preventing the impairment of human health ▪ These are the highest-risk devices and, hence, are subject to the highest level of regulatory control ▪ Require submission of clinical trial data to the US FDA and require Premarket Approval (PMA) from the FDA

TABLE 8 US: MEDICAL DEVICE REGULATORY APPROVAL PROCESS

DEVICE CLASSIFICATION	WAITING PERIOD AFTER SUBMISSION UNTIL APPROVAL IS GRANTED	VALIDITY PERIOD FOR REGISTRATION OF THE DEVICE	WHETHER REGISTRATION RENEWAL SHOULD BE STARTED IN ADVANCE	COMPLEXITY OF THE REGISTRATION PROCESS	OVERALL COST OF GAINING REGULATORY APPROVAL
Class I Device	1 Month	No Expiry Date	NA	Simple	Low
Class II Device	6–9 Months	No Expiry Date	NA	Moderate	Medium
Class III Device	18–30 Months	No Expiry Date	NA	Complex	High

5.8.1.2 Canada

Health Canada is a key regulatory authority for medical devices in the country. The Canadian Medical Devices Regulation (CMDR) is the key directive that governs the approval of medical devices (including NGS systems) in the country. In Canada, medical device manufacturers must first register and obtain a proper license from Health Canada to market their IVD devices. Health Canada has established a four-tier, risk-based classification system (Class I, II, III, and IV) for IVD products.

5.8.2 EUROPE

From May 2020, Europe has implemented new Medical Device Regulation (EU-MDR) guidelines to increase the safety of the medical devices used in the region. These guidelines require manufacturers to make significant data reporting and quality assurance changes. While new medical devices would be required to adhere to the new guidelines, existing medical devices would have to be recertified (in some cases, they would also have to be reclassified) by manufacturers.

In Europe, medical devices have been classified into four categories based on their function, risk to patients, and the manufacturer's intended use.

- Class I: Provide non-sterile or do not have a measuring function (low risk) and provide sterile and/or have a measuring function (low/medium risk); the MDR has added reusable surgical instruments to this group
- Class IIa (medium risk)
- Class IIb (medium/high risk)
- Class III (high risk)

However, with changes in MDR regulations, some of these medical devices have been reclassified. The new MDR regulations have 22 rules for classification, and the scope of medical devices has increased, including additional products.

Three Medical Device Directives mainly regulate the European market—the Medical Device Directive (93/42/EEC), the Active Implantable Medical Device Directive (90/385/EEC), and the In Vitro Diagnostic Medical Device Directive (98/79/EC). The European Medical Device Directive's Conformité Européenne (CE mark) approval is required for the distribution and marketing of any medical devices (including NGS systems) in the European (EU) region. This certification verifies that a device meets all regulatory requirements of the Medical Devices Directive (MDD). The manufacturer must be certified by the notified body to Annex II, V, or VI of the Medical Device Directives and comply with the requirements of those directives. A certified body conducts a technical review, and if all requirements are met, a CE clearance letter is issued, permitting the sale of a particular medical device in Europe. Also, manufacturers of medical devices must implement quality systems compliant with ISO 13485:2003.

5.8.3 ASIA PACIFIC

5.8.3.1 China

The National Medical Products Administration (NMPA), formerly the CFDA, is responsible for supervising and administering medical devices in the People's Republic of China (PRC). The NMPA classifies medical devices into Class I, Class II, and Class III based on the safety and effectiveness of the device.

TABLE 9 CHINA: CLASSIFICATION OF MEDICAL DEVICES

DEVICE CLASSIFICATION	POST-SUBMISSION APPROVAL (MONTHS)	LEVEL OF RISK	COMPLEXITY OF REGISTRATION	OVERALL COST OF GAINING REGULATORY APPROVAL	DEFINITION
Class I	<1	Low	Low	Moderate	Devices for which safety and effectiveness can be ensured through routine administration
Class II	16–24	High	High	Moderate-to-high	Devices for which further control is required to ensure safety and effectiveness
Class III	16–24	High	High	High	Devices that are implanted into the human body or that are used for supporting or sustaining life. They pose a significant risk to the human body and, hence, must be strictly controlled for their safety and effectiveness

NGS systems are classified as Class II devices as per NMPA rules. In China, IVD products (including NGS systems) and their components that are recognized internationally are classified as medical devices (MD), in vitro diagnostic reagents (including NGS reagents), and drugs. For an IVD system that includes an instrument and appropriate reagents, such as NGS, two product registration certificates must be acquired for market approval—medical device registration for instruments and IVD reagent registration for suited reagents.

5.8.3.2 Japan

The Ministry of Health, Labour and Welfare (MHLW) and the Pharmaceuticals and Medical Devices Agency (PMDA) govern the regulatory approval pathway to assess the safety and effectiveness of medical products in the country. Medical device manufacturers must abide by the PMDA. Obtaining PMDA approval for a new product is time-consuming (it can take approximately five years) and cumbersome. The compilation of foreign medical device manufacturers with PMDA is challenging due to the multilayered regulatory and language barriers. Hence, foreign manufacturers (with no local presence in Japan) need to appoint a Marketing Authorization Holder (MAH) to manage their device registration process and liaise with Japan's Pharmaceuticals and Medical Devices Agency.

5.8.3.3 India

According to the Central Drugs Standard Control Organisation (CDSCO), diagnostic kits are classified as notified diagnostic assays. Manufacturers must register these tests with the regulatory authorities and secure an import/manufacturing license. The remaining molecular diagnostic assays are termed non-notified diagnostic assays and need only an import license to be sold in India. The Drugs and Cosmetics Act of 1940 and the Drugs and Cosmetics Rules of 1945 are the major laws that regulate pharmaceuticals, medical devices, and diagnostics in India. The Ministry of Health & Family Welfare, through the Drug Controller General of India, is responsible for the improvement of public health by monitoring and regulating the quality, safety, and efficacy of drugs and devices through examination and investigation.

5.8.4 LATIN AMERICA

In some Latin American countries, such as Mexico and Argentina, medical device regulations are evolving, while in others, the regulations are static and, to some extent, opaque. Most Latin American countries consider premarket controls the primary feature for medical device regulation (including NGS). Medical device regulations in this region are driven by trade agreements, primarily Mercado Común del Sur (MERCOSUR), which seeks to create a common market like that in the EU. Medical goods and services can move freely among member countries. To register products in a Latin American country, a company must have an office in that country or have a local distributor (registered with the Ministry of Health) to register products on its behalf. For all Latin American registrations, a Free Sale Certificate (FSC) or a Certificate to Foreign Government (CFG) is required to confirm that a product is approved in the country of origin and can be exported without restriction.

5.8.5 MIDDLE EAST

Each member country of the Middle East has its regulatory system for medical devices in different maturity stages. Saudi Arabia has the most established regulatory framework for medical devices among Middle Eastern countries, followed by the UAE and Qatar. The regulatory systems of the UAE and Qatar are still in the early stages of development. The ministries of health in each of these countries are responsible for regulating medical devices. Most countries need overseas device manufacturers to appoint a local (or authorized) representative to facilitate the registration/regulation (in addition to distribution and sale) of their products in each country.

A marketing authorization from the Saudi Food and Drug Authority (SFDA) under Interim Regulation (1-8-1429) is required to sell medical devices in Saudi Arabia. Medical device manufacturers must obtain prior market authorization in reference markets such as Australia, Canada, the European Union, Japan, or the US. The device classification in respective reference markets will determine the classification of that device in Saudi Arabia. An authorized representative primarily manages the device registration in Saudi Arabia. The representative must be licensed with the SFDA.

5.8.6 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

TABLE 10 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS: GENOMICS, MICROBIOME RESEARCH, AND METAGENOMICS

REGION	COUNTRY	REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS
North America	US	▪ National Institute of Standards and Technology (NIST) ▪ National Human Genome Research Institute (NHGRI) ▪ Joint Genome Institute (JGI)
	Canada	▪ The Genomics Research and Development Initiative (GRDI) ▪ Genome Canada
Europe	Germany	▪ DZIF German Centre for Infection Research ▪ HELMHOLTZ-ZENTRUM FÜR INFEKTIONSFORSCHUNG GMBH
	UK	▪ UK Research and Innovation (UKRI) ▪ Microbiology Society
	France	▪ Paris Center for Microbiome Medicine (PaCeMM) FHU ▪ Microbiome Foundation France
	Italy	▪ Microbiome Research Hub Italy ▪ Council for Agricultural Research and Economics (CREA)
	Spain	▪ The Centre for Genomic Regulation (CRG) ▪ National Centre for Genomic Analysis (CNAG-CRG) ▪ The Higher Council for Scientific Research ▪ The Institute for Research in Biomedicine (IRB)
Asia	Japan	▪ Human MetaGenome Consortium Japan (HMGJ) ▪ Japan Agency for Medical Research and Development
	China	▪ Agricultural Genomics Institute at Shenzhen (AGIS) ▪ National Genomics Data Centre China
	India	▪ Department of Biotechnology (GENOME India and Microbiome) ▪ International Life Sciences Institute India (ILSI)
	South Korea	▪ National Research Foundation of Korea (NRF) ▪ Korea Biotechnology Industry Organization (KoreaBIO)
	Australia	▪ National Research Infrastructure for Australia (Australian Microbiome Initiative) ▪ Microbiome Research Centre (MRC)
Latin America	Brazil	▪ Brazilian Society of Genetics (Sociedade Brasileira de Genética in Portuguese) ▪ The São Paulo Research Foundation (Fundação de Amparo à Pesquisa do Estado de São Paulo in Portuguese)

Middle East & Africa	Saudi Arabia	▪ The Saudi Society of Medical Genetics (SSMG) ▪ King Faisal Specialist Hospital and Research Centre's (KFSH&RC) Genetics department ▪ Medical Genetics Programme ▪ Center of Innovation in Personalized Medicine (CIPM)
	South Africa	▪ South African Medical Research Council (SAMRC) ▪ South African Society for Human Genetics ▪ Cancer Association of South Africa (CANSA) ▪ Hyrax Biosciences

Source: Government Websites and MarketsandMarkets Analysis

5.9 PRICING ANALYSIS

The pricing of microbiome sequencing services can vary depending on several factors, including the specific sequencing technology used, the depth of sequencing coverage, the number of samples, the data analysis and interpretation required, and the infrastructure capacity of the service provider.

TABLE 11 AVERAGE SELLING PRICE, BY SERVICE AND TYPE

SERVICE PROVIDER	REGION/COUNTRY	TYPE	SERVICE	SAMPLE SIZE/ CUSTOMER TYPE	PRICE RANGE
Dalhousie University Integrated Microbiome Resource (IMR)	North America	Amplicon Sequencing (16S/18S/ITS Short Amplicons)	Library Preparation + Sequencing	50 to 200 Samples	USD 20–30
			Sample Preparation (DNA Extraction)	50 to 200 Samples	USD 15–20
			Standard Bioinformatics Pipeline	50 to 200 Samples	USD 500–1,000
		Metagenome Sequencing	Library Prep. + Sequencing	Academic Customers	USD 100–150
			DNA Extraction	Academic Customers	USD 15–20
			Standard Bioinformatics Pipeline	Academic Customers	USD 1,000–1,200
Genotypic Technology	Asia Pacific	Metagenomic Analysis	Library Preparation + Sequencing	Per Sample	USD 20–25

		De Novo Transcriptome Sequencing	Library Preparation + Sequencing	Per Sample	USD 500–600
		De Novo Wholeenome Sequencing	Library Preparation + Sequencing	Per Sample	USD 2,500–3,000
MR DNA	North America	Shotgun Metagenomic Sequencing	Library Preparation + Sequencing	Per Sample	USD 50–60
			Sample Preparation (DNA/RNA Extraction)	Per Sample	USD 20–70
CD Genomics	North America	Metagenomic Analysis	Library Preparation + Sequencing + Data Analysis	Per Sample	USD 100–120
UConn CORE MARS	North America	Microbiome Analysis	Data Analysis	Per Sample	USD 100–150
Dreamgenics	Europe	Metagenomic Analysis	Data Analysis	Per Sample	USD 30–40

Note: This is an indicative list of pricing specific to the particular service provider operating in a region.

Source: Company Websites and MarketsandMarkets Analysis

5.9.1 AVERAGE SELLING PRICE TREND ANALYSIS

In recent years, the cost of microbiome sequencing has generally decreased due to the rapid advancements in NGS technologies, which have made high-throughput sequencing more affordable and accessible. The decreasing cost of DNA sequencing has enabled researchers to generate larger and more comprehensive microbiome datasets, leading to a greater understanding of microbial communities and their roles in various health and disease conditions. Pricing also varies geographically, with different regions or countries having different costs associated with microbiome sequencing services due to varying market dynamics and operational costs. The average cost of microbiome sequencing services is the highest in the US, followed by Europe and the Asia Pacific.

5.10 KEY CONFERENCES AND EVENTS IN 2023–2024

TABLE 12 MICROBIOME SEQUENCING/METAGENOMIC SEQUENCING/NGS CONFERENCES, 2023–2024

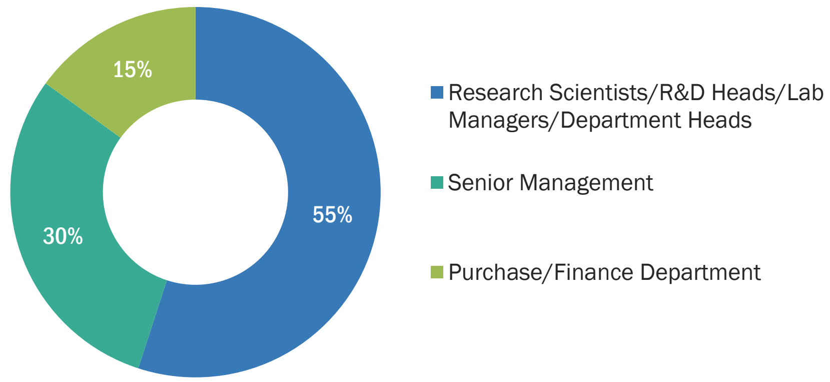
YEAR/QUARTER	NORTH AMERICA	EUROPE	ASIA PACIFIC	REST OF THE WORLD
Q1 2023	(February) International Conference on Next Generation Sequencing and Metagenomics (March) Annual Next Generation Sequencing US Congress	(January) Human Genomics and Genomic Medicine (February) International Conference on Next Generation Sequencing in Biology (March) Genomics and Molecular Biology (March) Conference on Revolutionizing Next Generation Sequencing (April) International Conference on Next Generation Sequencing Approaches in Biological Engineering	(January) International Conference on Next Generation Sequencing Approaches in Biology (February) International Conference on Next Generation Sequencing and Metagenomics	(February) International Conference on Next Generation Sequencing in Biology (April) International Conference on Next Generation Sequencing Technologies ICNGST
Q2 2023	-	-	(May) International Conference on Next Generation Sequencing in Biological Engineering	-
Q3 2023	(June) Microbiome Movement Drug Development Summit (July) International Conference on Next Generation Sequencing and Diagnostics	(June) International Conference on Microbiome Research (ICMR) (July) International Conference on Next Generation Sequencing Technology (July) International Conference on Next Generation Sequencing and Metagenomics	-	(July) International Conference on Next Generation Sequencing and Clinical Diagnostics

Q4 2023	(December) Genomics & Pharmacogenomics	(October) International Conference on Bioinformatic Approaches for Human Microbiome (ICBAHM) (November) NextGen Omics UK	-	-
Q1 2024	-	-	(March) International Conference on Microbiome Analysis (ICMA) (February) International Conference on Next Generation Sequencing and Metagenomics	-
Q2 2024	-	-	-	-
Q3 2024	-	(July) International Conference on Next Generation Sequencing	-	-
Q4 2024	-	(November) International Conference on Next Generation Sequencing	-	-

Source: Org. Websites, Expert Interviews, and MarketsandMarkets Analysis

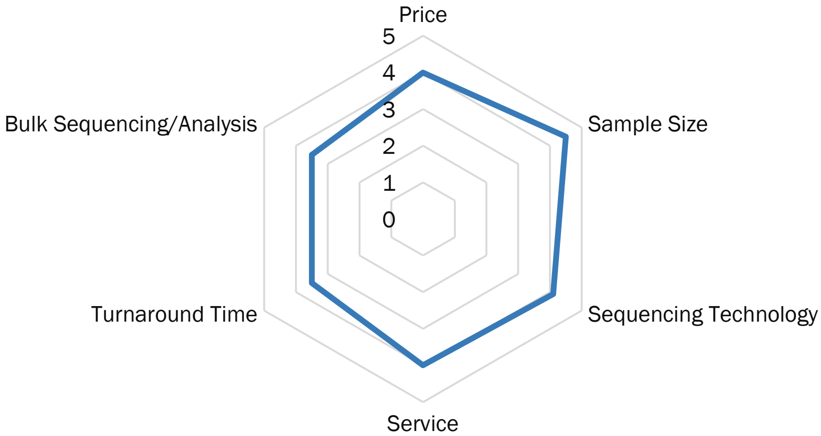
5.11 KEY STAKEHOLDERS AND BUYING CRITERIA

FIGURE 23 INFLUENCE OF STAKEHOLDERS ON BUYING PROCESS OF SEQUENCING SERVICES



Source: Expert Interviews and MarketsandMarkets Analysis

FIGURE 24 CRITERIA FOR OUTSOURCING SEQUENCING SERVICES



Note 1: Service refers to the preference for end-to-end services (sample preparation, sequencing, and data analysis) among end users.

Note 2: Bulk sequencing refers to the preference for discounted offers on bulk sequencing (sample size of more than 100) among end users.

Source: Expert Interviews and MarketsandMarkets Analysis

6 MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE

KEY FINDINGS

- The sequencing and library preparation services segment accounted for the largest share of ~48% of the microbiome sequencing services market in 2022.
- The sequencing and library preparation services segment is projected to reach USD 276.91 million by 2028 from USD 139.58 million in 2023, at a CAGR of 14.7%.
- The sample preparation services segment accounted for the second-largest share of ~36% in 2022. This segment is projected to reach USD 198.29 million by 2028.

6.1 INTRODUCTION

Based on service, the microbiome sequencing services market is segmented into sample preparation services, sequencing and library preparation services, and data analysis services. A majority of the service providers/key market players offer all three types of services as part of the microbiome analysis workflow. Variations in sample size, associated pricing, use of different technologies, and the strength of local players are the key factors driving the revenue generation capacity of the above-mentioned market segments. The sequencing and library preparation segment accounted for the largest share of ~48% in 2022, followed by the sample preparation segment. The sequencing and library preparation segment is also estimated to grow at the highest CAGR of 14.7% during the forecast period.

TABLE 13 MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	79.35	90.10	102.25	116.14	132.03	150.63	172.45	198.29	14.2%
Sequencing and Library Preparation	107.34	122.43	139.58	159.26	181.88	208.45	239.73	276.91	14.7%
Data Analysis	33.34	37.64	42.48	47.99	54.24	61.53	70.03	80.04	13.5%
Total	**220.03**	**250.17**	**284.31**	**323.39**	**368.16**	**420.61**	**482.21**	**555.24**	**14.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

6.2 SAMPLE PREPARATION

6.2.1 NEED FOR SPECIFIC SAMPLE PREPARATION PROTOCOLS BASED ON SAMPLE TYPE TO DRIVE GROWTH

This segment includes services that are applied for the collection, storage, and processing of samples subjected to microbiome sequencing. It is imperative to ensure that microbiome analysis is reproducible. Reproducibility is affected by several factors, such as sample size, storage methods, and collection methods. Microbiome sequencing helps in evaluating microbial communities across different applications, including real-time functional profiling (metatranscriptomics), functional profiling (metagenomics), and high-level community profiling (marker gene studies—16S rRNA, ITS, and 18S rRNA). Across all these applications, variation in sample collection, storage, and shipping must be avoided to maintain the reproducibility of the outcome. Sample preparation protocols differ based on sample types (oral microbiomes, gut microbiomes, and skin microbiomes), which has created robust revenue generation opportunities for service providers/players operating in the microbiome sequencing market.

TABLE 14 MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	45.42	51.34	58.01	65.60	74.25	84.33	96.11	110.02	13.7%
Europe	24.57	27.99	31.87	36.32	41.42	47.40	54.44	62.80	14.5%
Asia Pacific	6.76	7.90	9.22	10.76	12.56	14.71	17.27	20.35	17.2%
Latin America	1.54	1.70	1.88	2.08	2.30	2.55	2.83	3.16	10.9%
Middle East and Africa	1.06	1.16	1.27	1.38	1.51	1.65	1.80	1.97	9.2%
Total	**79.35**	**90.10**	**102.25**	**116.14**	**132.03**	**150.63**	**172.45**	**198.29**	**14.2%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 15 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	41.16	46.57	52.66	59.59	67.50	76.73	87.52	100.25	13.7%
Canada	4.25	4.78	5.35	6.01	6.75	7.60	8.60	9.76	12.8%
Total	**45.42**	**51.34**	**58.01**	**65.60**	**74.25**	**84.33**	**96.11**	**110.02**	**13.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 16 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	5.32	6.08	6.95	7.94	9.08	10.42	12.01	13.89	14.9%
UK	4.76	5.46	6.27	7.20	8.27	9.54	11.03	12.82	15.4%
France	3.76	4.28	4.88	5.57	6.35	7.27	8.36	9.65	14.6%
Italy	1.06	1.20	1.37	1.55	1.77	2.02	2.31	2.66	14.3%
Spain	0.81	0.92	1.05	1.19	1.35	1.53	1.75	2.01	14.0%
Rest of Europe	8.86	10.04	11.36	12.87	14.60	16.62	18.97	21.76	13.9%
Total	**24.57**	**27.99**	**31.87**	**36.32**	**41.42**	**47.40**	**54.44**	**62.80**	**14.5%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 17 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	2.86	3.34	3.90	4.56	5.32	6.23	7.32	8.63	17.2%
Japan	1.33	1.56	1.84	2.15	2.53	2.98	3.51	4.16	17.8%
India	0.17	0.20	0.23	0.27	0.31	0.36	0.43	0.50	16.9%
Rest of Asia Pacific	2.40	2.79	3.25	3.78	4.40	5.13	6.00	7.05	16.8%
Total	**6.76**	**7.90**	**9.22**	**10.76**	**12.56**	**14.71**	**17.27**	**20.35**	**17.2%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 18 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.72	0.80	0.89	0.98	1.08	1.20	1.34	1.49	11.0%
Rest of Latin America	0.82	0.90	1.00	1.10	1.21	1.35	1.49	1.66	10.8%
Total	**1.54**	**1.70**	**1.88**	**2.08**	**2.30**	**2.55**	**2.83**	**3.16**	**10.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

6.3 SEQUENCING AND LIBRARY PREPARATION

6.3.1 SEQUENCING AND LIBRARY PREPARATION SERVICES TO DOMINATE MARKET DURING FORECAST PERIOD

This segment includes sequencing and library preparation services for samples subjected to microbiome sequencing. This is a crucial step in microbiome analysis, wherein either reference sequences are created for samples or sequencing outputs are compared against the existing sequences. Different approaches are employed in microbiome sequencing and library preparation, including 16S rRNA sequencing, shotgun metagenomics, qPCR testing, and short-chain fatty acids (SCFA) analysis. Service providers use either state-of-the-art equipment and software or purchase commercial systems and software solutions. Illumina, Inc. (US) and PacBio (US) are among the popular brands adopted by a majority of service providers to offer sequencing and library preparation services. The expanding pool of service providers, particularly the ones offering NGS services, is the key contributor to the dominance of this segment.

TABLE 19 MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	61.38	69.63	78.94	89.58	101.75	115.96	132.62	152.34	14.1%
Europe	33.26	38.09	43.60	49.94	57.25	65.87	76.04	88.17	15.1%
Asia Pacific	9.17	10.79	12.68	14.91	17.53	20.67	24.44	29.01	18.0%
Latin America	2.09	2.33	2.60	2.90	3.23	3.61	4.04	4.54	11.8%
Middle East and Africa	1.45	1.60	1.76	1.93	2.13	2.34	2.58	2.85	10.2%
Total	**107.34**	**122.43**	**139.58**	**159.26**	**181.88**	**208.45**	**239.73**	**276.91**	**14.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 20 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	55.63	63.15	71.65	81.37	92.49	105.48	120.73	138.78	14.1%
Canada	5.75	6.48	7.29	8.21	9.26	10.48	11.89	13.56	13.2%
Total	**61.38**	**69.63**	**78.94**	**89.58**	**101.75**	**115.96**	**132.62**	**152.34**	**14.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 21 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	7.20	8.26	9.47	10.87	12.49	14.40	16.66	19.35	15.4%
UK	6.44	7.43	8.56	9.87	11.40	13.20	15.35	17.92	15.9%
France	5.09	5.83	6.67	7.65	8.77	10.10	11.66	13.53	15.2%
Italy	1.43	1.64	1.87	2.14	2.44	2.80	3.23	3.73	14.8%
Spain	1.10	1.26	1.43	1.64	1.87	2.14	2.46	2.84	14.7%
Rest of Europe	12.00	13.68	15.59	17.77	20.28	23.23	26.69	30.80	14.6%
Total	**33.26**	**38.09**	**43.60**	**49.94**	**57.25**	**65.87**	**76.04**	**88.17**	**15.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 22 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.88	4.56	5.36	6.30	7.41	8.74	10.33	12.26	18.0%
Japan	1.80	2.14	2.52	2.98	3.53	4.18	4.97	5.93	18.6%
India	0.23	0.27	0.32	0.37	0.44	0.51	0.61	0.72	17.8%
Rest of Asia Pacific	3.25	3.82	4.48	5.25	6.16	7.24	8.54	10.10	17.7%
Total	**9.17**	**10.79**	**12.68**	**14.91**	**17.53**	**20.67**	**24.44**	**29.01**	**18.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 23 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.98	1.10	1.22	1.36	1.52	1.70	1.91	2.14	11.9%
Rest of Latin America	1.11	1.24	1.38	1.53	1.71	1.91	2.14	2.40	11.8%
Total	**2.09**	**2.33**	**2.60**	**2.90**	**3.23**	**3.61**	**4.04**	**4.54**	**11.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

6.4 DATA ANALYSIS

6.4.1 DEVELOPMENT OF ADVANCED BIOINFORMATICS SOLUTIONS TO DRIVE GROWTH

This segment includes bioinformatics/data analysis services for samples subjected to microbiome sequencing. Researchers in the genomics field often outsource the data analysis part of their projects to bioinformatics service providers. NGS data analysis services are offered for data processing and quality control, assembly, mapping, annotation, and advanced analysis for microbiome research and population studies.

The rising awareness about NGS data analysis services, the recent launch of data analysis services by bioinformatics companies, and the increasing availability of data analysis services in developed countries are the major factors supporting the growth of this market.

TABLE 24 MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	18.95	21.38	24.11	27.21	30.73	34.83	39.61	45.25	13.4%
Europe	10.37	11.72	13.24	14.97	16.93	19.22	21.89	25.05	13.6%
Asia Pacific	2.89	3.33	3.83	4.40	5.06	5.84	6.75	7.83	15.4%
Latin America	0.66	0.72	0.78	0.85	0.92	1.00	1.09	1.19	8.7%
Middle East and Africa	0.46	0.49	0.53	0.56	0.60	0.64	0.68	0.73	6.8%
Total	**33.34**	**37.64**	**42.48**	**47.99**	**54.24**	**61.53**	**70.03**	**80.04**	**13.5%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 25 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	17.17	19.39	21.88	24.72	27.94	31.70	36.09	41.26	13.5%
Canada	1.78	1.99	2.22	2.49	2.78	3.13	3.53	3.99	12.4%
Total	**18.95**	**21.38**	**24.11**	**27.21**	**30.73**	**34.83**	**39.61**	**45.25**	**13.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 26 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	2.23	2.54	2.89	3.28	3.74	4.27	4.90	5.64	14.3%
UK	2.00	2.28	2.60	2.97	3.39	3.89	4.47	5.17	14.7%
France	1.58	1.79	2.03	2.29	2.60	2.96	3.37	3.86	13.7%
Italy	0.45	0.50	0.57	0.64	0.72	0.82	0.93	1.07	13.4%
Spain	0.34	0.39	0.43	0.49	0.55	0.62	0.70	0.79	12.8%
Rest of Europe	3.76	4.21	4.72	5.29	5.93	6.66	7.52	8.52	12.5%
Total	**10.37**	**11.72**	**13.24**	**14.97**	**16.93**	**19.22**	**21.89**	**25.05**	**13.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 27 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	1.22	1.41	1.62	1.87	2.15	2.49	2.89	3.36	15.7%
Japan	0.57	0.66	0.76	0.88	1.02	1.18	1.38	1.61	16.1%
India	0.07	0.08	0.10	0.11	0.13	0.14	0.17	0.19	15.0%
Rest of Asia Pacific	1.03	1.18	1.35	1.54	1.76	2.02	2.32	2.68	14.7%
Total	**2.89**	**3.33**	**3.83**	**4.40**	**5.06**	**5.84**	**6.75**	**7.83**	**15.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 28 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.31	0.34	0.37	0.40	0.43	0.47	0.51	0.56	8.9%
Rest of Latin America	0.35	0.38	0.41	0.45	0.48	0.53	0.57	0.62	8.5%
Total	**0.66**	**0.72**	**0.78**	**0.85**	**0.92**	**1.00**	**1.09**	**1.19**	**8.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7 MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE

KEY FINDINGS

- The amplicon sequencing segment accounted for the largest share of 41.0% of the global microbiome sequencing services market in 2022. This segment is projected to reach USD 234.18 million by 2028 from USD 118.98 million in 2023, at a CAGR of 14.5%.
- The shotgun sequencing segment is expected to grow at the highest CAGR of 15.2% from 2023 to 2028. This segment is expected to reach a value of USD 157.10 million by 2028.
- Whole-genome sequencing is the third-largest segment in terms of revenue, with a market share of approximately 21.0% in 2022.

7.1 INTRODUCTION

Based on type, the global microbiome sequencing services market is segmented into amplicon sequencing, whole-genome sequencing, shotgun sequencing, and other types (including de novo sequencing, metagenomic sequencing, and metatranscriptomic sequencing). The amplicon sequencing segment accounted for the largest market share of 41.0% in 2022. The large share of this segment can be attributed to the high adoption of 16S rRNA microbiome sequencing and targeted sequencing. Key market players/service providers have built an innovative portfolio around these services, including sample preparation and data analysis solutions.

TABLE 29 MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	91.80	104.54	118.98	135.54	154.55	176.84	203.06	234.18	14.5%
Whole-genome Sequencing	47.29	53.48	60.45	68.39	77.44	87.98	100.31	114.85	13.7%
Shotgun Sequencing	59.09	67.70	77.52	88.83	101.89	117.27	135.44	157.10	15.2%
Other Types	21.85	24.46	27.36	30.62	34.29	38.52	43.41	49.11	12.4%
Total	**220.03**	**250.17**	**284.31**	**323.39**	**368.16**	**420.61**	**482.21**	**555.24**	**14.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.2 AMPLICON SEQUENCING

7.2.1 HIGH ACCURACY OF 16S AMPLICON SEQUENCING TO DRIVE ADOPTION

This segment is inclusive of 16S rRNA sequencing, ITS sequencing, and targeted sequencing services. Amplicon sequencing is a highly targeted approach that allows researchers to analyze genetic variation in specific regions of a genome. The ultra-deep sequencing of PCR products (amplicons) allows efficient variant identification and characterization. 16S rRNA sequencing is the most common technique utilized for metagenomics studies. In this technique, the 16S rRNA genetic markers present in a bacterial genome are sequenced to study bacterial phylogeny and taxonomy from complex microbiomes or environments that are difficult or impossible to analyze. The presence of 16S rRNA genetic markers as a multigene family or operons in almost all bacteria and the large size of the 16S rRNA gene (~1,500 bp) makes it the most preferred metagenomic sequencing technique among end users.

The demand for targeted sequencing is anticipated to increase in metagenomic research activities. The targeted amplicon sequencing method involves sequencing a phylogenetically informative marker to identify organisms in community samples. The marker must be present across all samples with expected organisms to enhance taxonomic profiling through the amplification of the primer. A number of different markers are commonly used and vary by the taxonomy of interest, but the most commonly used targeted amplicon approaches include the 16S rRNA gene for bacterial samples, the 18S for eukaryotes, and ITS for fungal samples.

TABLE 30 AMPLICON SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	52.63	59.60	67.44	76.40	86.62	98.54	112.50	128.99	13.8%
Europe	28.39	32.47	37.11	42.44	48.59	55.82	64.35	74.51	15.0%
Asia Pacific	7.79	9.16	10.75	12.63	14.83	17.47	20.63	24.46	17.9%
Latin America	1.77	1.97	2.20	2.45	2.72	3.04	3.40	3.82	11.7%
Middle East and Africa	1.22	1.35	1.48	1.63	1.79	1.97	2.17	2.40	10.1%
Total	**91.80**	**104.54**	**118.98**	**135.54**	**154.55**	**176.84**	**203.06**	**234.18**	**14.5%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 31 NORTH AMERICA: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	47.70	54.05	61.22	69.40	78.74	89.64	102.42	117.52	13.9%
Canada	4.93	5.54	6.22	7.00	7.88	8.90	10.08	11.48	13.0%
Total	**52.63**	**59.60**	**67.44**	**76.40**	**86.62**	**98.54**	**112.50**	**128.99**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 32 EUROPE: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	6.16	7.06	8.08	9.26	10.62	12.22	14.11	16.37	15.2%
UK	5.50	6.34	7.29	8.40	9.68	11.20	13.00	15.15	15.7%
France	4.34	4.97	5.68	6.50	7.45	8.56	9.87	11.43	15.0%
Italy	1.22	1.40	1.59	1.82	2.08	2.38	2.74	3.16	14.7%
Spain	0.94	1.07	1.22	1.39	1.58	1.81	2.08	2.40	14.5%
Rest of Europe	10.22	11.64	13.24	15.08	17.18	19.65	22.55	25.99	14.4%
Total	**28.39**	**32.47**	**37.11**	**42.44**	**48.59**	**55.82**	**64.35**	**74.51**	**15.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 33 ASIA PACIFIC: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.30	3.88	4.55	5.34	6.27	7.39	8.72	10.34	17.8%
Japan	1.53	1.81	2.14	2.53	2.98	3.53	4.20	5.00	18.5%
India	0.19	0.23	0.27	0.31	0.37	0.43	0.51	0.60	17.7%
Rest of Asia Pacific	2.76	3.24	3.79	4.44	5.20	6.11	7.20	8.51	17.6%
Total	**7.79**	**9.16**	**10.75**	**12.63**	**14.83**	**17.47**	**20.63**	**24.46**	**17.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 34 LATIN AMERICA: AMPLICON SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.83	0.93	1.03	1.15	1.28	1.43	1.61	1.80	11.8%
Rest of Latin America	0.94	1.04	1.16	1.29	1.44	1.61	1.80	2.02	11.7%
Total	**1.77**	**1.97**	**2.20**	**2.45**	**2.72**	**3.04**	**3.40**	**3.82**	**11.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.3 WHOLE-GENOME SEQUENCING

7.3.1 ACCURATE REFERENCE GENOMES GENERATED FOR MICROBIAL IDENTIFICATION TO BOOST MARKET

Microbial whole-genome sequencing (WGS) has proved to be an important tool for mapping the genomes of novel organisms, finishing genomes of known organisms, or comparing genomes across multiple samples. This technology further helps researchers in generating accurate reference genomes for microbial identification and other comparative genomic studies. For functional studies of the microbiome, WGS metagenomic approaches are more commonly utilized. This is because WGS is affordable and particularly because of an array of computational tools to analyze microbiome sequencing data.

The advantages offered by this technique have resulted in increased adoption among scientists/researchers. As compared to other traditional genomic analysis methods, NGS-based microbial whole-genome sequencing eliminates labor-intensive cloning steps, saves time, and simplifies the workflow. This technique also helps to identify low-frequency variants and genome rearrangements that may be missed or are too expensive to identify using other methods.

TABLE 35 WHOLE-GENOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	26.93	30.39	34.29	38.72	43.75	49.62	56.47	64.54	13.5%
Europe	14.70	16.65	18.84	21.35	24.21	27.55	31.46	36.07	13.9%
Asia Pacific	4.08	4.72	5.45	6.30	7.28	8.43	9.80	11.42	15.9%
Latin America	0.93	1.02	1.11	1.21	1.32	1.45	1.59	1.74	9.4%
Middle East and Africa	0.65	0.70	0.75	0.81	0.87	0.93	1.00	1.08	7.5%
Total	**47.29**	**53.48**	**60.45**	**68.39**	**77.44**	**87.98**	**100.31**	**114.85**	**13.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 36 NORTH AMERICA: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	24.40	27.56	31.12	35.17	39.79	45.16	51.43	58.83	13.6%
Canada	2.53	2.83	3.16	3.54	3.97	4.46	5.03	5.70	12.5%
Total	**26.93**	**30.39**	**34.29**	**38.72**	**43.75**	**49.62**	**56.47**	**64.54**	**13.5%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 37 EUROPE: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	3.17	3.61	4.11	4.68	5.33	6.10	7.00	8.08	14.5%
UK	2.84	3.24	3.71	4.24	4.85	5.57	6.41	7.42	14.9%
France	2.25	2.55	2.89	3.27	3.72	4.23	4.84	5.56	14.0%
Italy	0.63	0.72	0.81	0.91	1.03	1.17	1.33	1.53	13.5%
Spain	0.49	0.55	0.62	0.70	0.79	0.89	1.01	1.15	13.2%
Rest of Europe	5.32	5.98	6.72	7.55	8.50	9.59	10.86	12.35	12.9%
Total	**14.70**	**16.65**	**18.84**	**21.35**	**24.21**	**27.55**	**31.46**	**36.07**	**13.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 38 ASIA PACIFIC: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	1.72	2.00	2.31	2.67	3.09	3.59	4.17	4.88	16.1%
Japan	0.80	0.93	1.09	1.26	1.47	1.71	2.00	2.34	16.6%
India	0.10	0.12	0.14	0.16	0.18	0.21	0.24	0.28	15.6%
Rest of Asia Pacific	1.45	1.67	1.92	2.21	2.54	2.93	3.38	3.92	15.3%
Total	**4.08**	**4.72**	**5.45**	**6.30**	**7.28**	**8.43**	**9.80**	**11.42**	**15.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 39 LATIN AMERICA: WHOLE-GENOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.44	0.48	0.52	0.57	0.62	0.68	0.75	0.83	9.5%
Rest of Latin America	0.50	0.54	0.59	0.64	0.70	0.76	0.84	0.92	9.2%
Total	**0.93**	**1.02**	**1.11**	**1.21**	**1.32**	**1.45**	**1.59**	**1.74**	**9.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.4 SHOTGUN SEQUENCING

7.4.1 ADVANCEMENTS IN DATA ANALYSIS SOLUTIONS TO SUPPORT ADOPTION

Shotgun sequencing involves breaking up large DNA sequences into multiple small pieces/sequences and then reassembling them into larger contiguous pieces representing the original large DNA sequence. Shotgun sequencing is a relatively new approach used to analyze many organisms present in a given complex sample and comprehensively sequence all genes. This technique provides researchers or end users with the necessary information to evaluate community biodiversity and its related function in the ecosystem.

The advantages offered by shotgun metagenomic sequencing over capillary sequencing, PCR-based approaches, and traditional next-generation sequencing techniques, the growing adoption of shotgun metagenomic sequencing among end users, and the growing number of metagenomic sequencing research activities across the globe are the key factors driving the growth of the shotgun metagenomic sequencing market.

Some of the advantages of shotgun metagenomic sequencing are mentioned below:

- Provides tools to study unculturable microorganisms that are difficult or impossible to analyze
- Eliminates the mapping stages, resulting in faster processing and whole-genome sequencing than clone-by-clone sequencing
- Ability to combine many samples in a single sequencing run and obtain high sequence coverage per sample
- Can detect very low abundance organisms of a microbial community, which may be missed or are too expensive to identify using other traditional methods

The challenges related to data interpretation and analysis in shotgun metagenomic sequencing and the high cost associated with the sequencing of multiple genomes are the key factors restraining the growth of this market. On the other hand, the rapid advancements and developments in informatics software are expected to improve the ease and efficiency of shotgun metagenomic analysis.

TABLE 40 SHOTGUN SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	33.80	38.48	43.79	49.86	56.83	64.99	74.58	85.96	14.4%
Europe	18.31	21.07	24.23	27.90	32.14	37.15	43.09	50.20	15.7%
Asia Pacific	5.04	5.97	7.07	8.36	9.89	11.74	13.96	16.67	18.7%
Latin America	1.15	1.29	1.45	1.63	1.83	2.06	2.32	2.63	12.6%
Middle East and Africa	0.79	0.88	0.98	1.09	1.21	1.34	1.48	1.65	11.0%
Total	**59.09**	**67.70**	**77.52**	**88.83**	**101.89**	**117.27**	**135.44**	**157.10**	**15.2%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 41 NORTH AMERICA: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	30.64	34.90	39.74	45.29	51.65	59.11	67.88	78.29	14.5%
Canada	3.17	3.58	4.05	4.57	5.18	5.88	6.70	7.67	13.6%
Total	**33.80**	**38.48**	**43.79**	**49.86**	**56.83**	**64.99**	**74.58**	**85.96**	**14.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 42 EUROPE: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	3.96	4.57	5.26	6.06	6.99	8.09	9.40	10.96	15.8%
UK	3.54	4.11	4.75	5.51	6.39	7.43	8.67	10.17	16.4%
France	2.80	3.22	3.71	4.27	4.92	5.69	6.60	7.69	15.7%
Italy	0.79	0.91	1.04	1.19	1.37	1.59	1.83	2.13	15.4%
Spain	0.61	0.70	0.80	0.91	1.05	1.21	1.40	1.62	15.3%
Rest of Europe	6.60	7.57	8.67	9.95	11.42	13.14	15.18	17.62	15.2%
Total	**18.31**	**21.07**	**24.23**	**27.90**	**32.14**	**37.15**	**43.09**	**50.20**	**15.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 43 ASIA PACIFIC: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	2.13	2.52	2.99	3.53	4.18	4.95	5.89	7.02	18.7%
Japan	0.99	1.18	1.41	1.67	1.99	2.37	2.84	3.41	19.3%
India	0.13	0.15	0.18	0.21	0.25	0.29	0.35	0.41	18.5%
Rest of Asia Pacific	1.79	2.12	2.50	2.95	3.48	4.12	4.89	5.82	18.5%
Total	**5.04**	**5.97**	**7.07**	**8.36**	**9.89**	**11.74**	**13.96**	**16.67**	**18.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 44 LATIN AMERICA: SHOTGUN SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.54	0.61	0.68	0.77	0.86	0.97	1.09	1.24	12.7%
Rest of Latin America	0.61	0.68	0.77	0.86	0.97	1.09	1.23	1.39	12.6%
Total	**1.15**	**1.29**	**1.45**	**1.63**	**1.83**	**2.06**	**2.32**	**2.63**	**12.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

7.5 OTHER TYPES

This segment is inclusive of de novo sequencing (whole-genome sequencing without any reference genome), metagenomic sequencing, and metatranscriptomic sequencing. Metatranscriptomics is a key technique in metagenomics, which allows researchers to study the function and activity of a complete set of transcripts (RNA sequences) from environmental samples.

The growing number of metagenomic research activities is the major factor driving the growth of this market segment. Over the last few years, the focus on understanding the function and relation of gut microbiome diversity, predicting resistance to specific antibiotics, understanding host-pathogen immune interactions, quantifying gene expression changes, and tracking disease progression has increased significantly.

De novo sequencing generates an initial genomic sequence of a particular organism without a reference genome. De novo sequencing services are applied in the research of animals, plants, and microorganisms, including phylogenetic studies, analysis of species diversity, genetic markers, and other genomic research. Using de novo sequencing to obtain the genomic information of microbes provides a fresh start for exploring the genetic structure and functions, studying the evolutionary origin of microbial populations, as well as developing potential applications across medicine, disease modeling, agriculture, and the environment.

TABLE 45 OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	12.39	13.88	15.54	17.41	19.53	21.97	24.80	28.12	12.6%
Europe	6.81	7.62	8.52	9.53	10.66	11.97	13.48	15.23	12.3%
Asia Pacific	1.91	2.17	2.46	2.79	3.16	3.59	4.08	4.65	13.6%
Latin America	0.44	0.47	0.50	0.53	0.57	0.61	0.65	0.69	6.6%
Middle East and Africa	0.31	0.32	0.34	0.35	0.37	0.39	0.40	0.42	4.6%
Total	**21.85**	**24.46**	**27.36**	**30.62**	**34.29**	**38.52**	**43.41**	**49.11**	**12.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 46 NORTH AMERICA: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	11.23	12.59	14.11	15.82	17.76	20.00	22.60	25.65	12.7%
Canada	1.16	1.29	1.43	1.59	1.77	1.97	2.20	2.47	11.5%
Total	**12.39**	**13.88**	**15.54**	**17.41**	**19.53**	**21.97**	**24.80**	**28.12**	**12.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 47 EUROPE: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	1.46	1.65	1.86	2.10	2.37	2.68	3.04	3.47	13.3%
UK	1.31	1.48	1.68	1.90	2.14	2.43	2.77	3.17	13.6%
France	1.04	1.17	1.31	1.46	1.64	1.84	2.08	2.35	12.5%
Italy	0.29	0.33	0.37	0.41	0.45	0.51	0.57	0.64	11.9%
Spain	0.23	0.25	0.28	0.31	0.34	0.38	0.43	0.48	11.4%
Rest of Europe	2.48	2.74	3.03	3.36	3.71	4.12	4.58	5.12	11.0%
Total	**6.81**	**7.62**	**8.52**	**9.53**	**10.66**	**11.97**	**13.48**	**15.23**	**12.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 48 ASIA PACIFIC: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	0.80	0.92	1.04	1.19	1.35	1.53	1.75	2.00	14.0%
Japan	0.37	0.43	0.49	0.56	0.64	0.73	0.83	0.95	14.3%
India	0.05	0.05	0.06	0.07	0.08	0.09	0.10	0.11	13.2%
Rest of Asia Pacific	0.68	0.77	0.87	0.97	1.10	1.23	1.39	1.57	12.7%
Total	**1.91**	**2.17**	**2.46**	**2.79**	**3.16**	**3.59**	**4.08**	**4.65**	**13.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 49 LATIN AMERICA: OTHER MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.21	0.22	0.24	0.25	0.27	0.29	0.31	0.33	6.9%
Rest of Latin America	0.23	0.25	0.27	0.28	0.30	0.32	0.34	0.36	6.4%
Total	**0.44**	**0.47**	**0.50**	**0.53**	**0.57**	**0.61**	**0.65**	**0.69**	**6.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8 MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY

KEY FINDINGS

- The sequencing by synthesis (SBS) segment accounted for the largest share of 45% of the global microbiome sequencing services market in 2022. This segment is projected to reach USD 253.56 million by 2028 from USD 129.01 million in 2023, at a CAGR of 14.5%.
- The nanopore sequencing segment is estimated to grow at the highest CAGR of 17.3% from 2023 to 2028. This segment is projected to reach USD 34.21 million by 2028.
- Other sequencing technologies (Sanger sequencing and semiconductor sequencing) form the second-largest segment in this market, with a share of ~37% in 2022.

8.1 INTRODUCTION

Based on technology, the global microbiome sequencing services market is segmented into sequencing by synthesis (SBS), sequencing by ligation (SBL), nanopore sequencing, and other technologies (semiconductor sequencing and Sanger sequencing, among other emerging technologies). The SBS segment accounted for the largest market share (~45%) in 2022, while the nanopore sequencing segment is estimated to grow at the highest CAGR (17.3%) during the forecast period.

TABLE 50 MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	99.59	113.38	129.01	146.93	167.48	191.59	219.92	253.56	14.5%
Sequencing by Ligation	28.18	31.96	36.24	41.13	46.71	53.24	60.89	69.95	14.1%
Nanopore Sequencing	11.29	13.20	15.42	18.01	21.05	24.67	29.00	34.21	17.3%
Other Technologies	80.97	91.63	103.64	117.32	132.92	151.11	172.40	197.53	13.8%
Total	**220.03**	**250.17**	**284.31**	**323.39**	**368.16**	**420.61**	**482.21**	**555.24**	**14.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8.2 SEQUENCING BY SYNTHESIS (SBS)

8.2.1 SBS TECHNOLOGY ENABLES HIGHEST PRODUCTION OF BASE PAIRS IN SHORT TIME

The SBS technology uses four fluorescently labeled nucleotides to sequence millions of clusters on the flow cell surface in parallel. During every sequencing cycle, a single-labeled deoxyribonucleoside triphosphate (dNTP) is added to the nucleic acid chain. This nucleotide label serves as a terminator for polymerization. After dNTP incorporation, the dye is first imaged to identify the base, and then it is enzymatically cleaved to allow the incorporation of another nucleotide. Since all four reversible terminator-bound dNTPs are present during each sequencing cycle, natural competition minimizes incorporation bias. The result is true base-by-base sequencing that enables the most accurate data for a broad range of applications.

The SBS technology has a wide customer base owing to its well-established precision in sequencing as well as its broad application base. It has applications in microbial genetics and metagenomics and is mainly used in whole-genome sequencing, transcriptome analysis, targeted sequencing, and genome-wide protein-nucleic acid interaction analysis.

TABLE 51 MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	57.03	64.63	73.20	82.98	94.15	107.20	122.48	140.55	13.9%
Europe	30.83	35.22	40.20	45.94	52.54	60.30	69.44	80.32	14.8%
Asia Pacific	8.47	9.93	11.63	13.62	15.96	18.75	22.09	26.12	17.6%
Latin America	1.93	2.14	2.37	2.63	2.92	3.25	3.62	4.05	11.3%
Middle East and Africa	1.33	1.46	1.60	1.75	1.91	2.10	2.30	2.52	9.5%
Total	**99.59**	**113.38**	**129.01**	**146.93**	**167.48**	**191.59**	**219.92**	**253.56**	**14.5%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 52 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	51.68	58.62	66.45	75.39	85.60	97.54	111.54	128.09	14.0%
Canada	5.34	6.01	6.75	7.59	8.55	9.66	10.94	12.46	13.0%
Total	**57.03**	**64.63**	**73.20**	**82.98**	**94.15**	**107.20**	**122.48**	**140.55**	**13.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 53 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	6.68	7.65	8.77	10.04	11.52	13.26	15.32	17.77	15.2%
UK	5.97	6.87	7.91	9.10	10.48	12.12	14.06	16.38	15.7%
France	4.72	5.39	6.15	7.03	8.05	9.24	10.64	12.31	14.9%
Italy	1.33	1.51	1.72	1.96	2.24	2.56	2.94	3.39	14.5%
Spain	1.02	1.16	1.32	1.50	1.71	1.95	2.24	2.58	14.3%
Rest of Europe	11.11	12.62	14.34	16.30	18.54	21.16	24.24	27.89	14.2%
Total	**30.83**	**35.22**	**40.20**	**45.94**	**52.54**	**60.30**	**69.44**	**80.32**	**14.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 54 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.58	4.21	4.93	5.77	6.77	7.96	9.38	11.09	17.6%
Japan	1.67	1.97	2.32	2.73	3.21	3.80	4.50	5.34	18.2%
India	0.21	0.25	0.29	0.34	0.40	0.46	0.55	0.64	17.3%
Rest of Asia Pacific	3.01	3.51	4.10	4.78	5.58	6.53	7.67	9.03	17.1%
Total	**8.47**	**9.93**	**11.63**	**13.62**	**15.96**	**18.75**	**22.09**	**26.12**	**17.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 55 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.91	1.01	1.12	1.24	1.38	1.53	1.71	1.92	11.4%
Rest of Latin America	1.02	1.13	1.26	1.39	1.54	1.71	1.91	2.13	11.2%
Total	**1.93**	**2.14**	**2.37**	**2.63**	**2.92**	**3.25**	**3.62**	**4.05**	**11.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8.3 SEQUENCING BY LIGATION (SBL)

8.3.1 INCREASING DEMAND FOR SBL IN FOOD MICROBIOME RESEARCH TO BOOST GROWTH

SBL is a DNA sequencing method that uses DNA ligase to identify the nucleotide present at a given position in a DNA sequence. SBL analyzes the mismatch in base pairs of a DNA ligase to determine the underlying sequence of a target DNA molecule.

SOLiD is an enzymatic method of sequencing that uses DNA ligase, an enzyme used widely in biotechnology, for its ability to ligate double-stranded DNA strands. Emulsion PCR is used to immobilize/amplify an ssDNA primer-binding region (known as an adapter), which has been conjugated to the target sequence on a bead. These beads are then deposited onto a glass surface with a high density of beads. This, in turn, increases the throughput of the technique.

TABLE 56 MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	16.07	18.15	20.49	23.15	26.18	29.71	33.83	38.69	13.6%
Europe	8.75	9.96	11.32	12.89	14.69	16.80	19.27	22.21	14.4%
Asia Pacific	2.42	2.83	3.29	3.84	4.48	5.23	6.13	7.22	17.0%
Latin America	0.55	0.61	0.68	0.75	0.82	0.91	1.01	1.13	10.8%
Middle East and Africa	0.38	0.42	0.46	0.50	0.54	0.59	0.65	0.71	9.2%
Total	**28.18**	**31.96**	**36.24**	**41.13**	**46.71**	**53.24**	**60.89**	**69.95**	**14.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 57 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	14.56	16.46	18.60	21.03	23.79	27.02	30.79	35.23	13.6%
Canada	1.51	1.69	1.89	2.13	2.39	2.69	3.04	3.45	12.8%
Total	**16.07**	**18.15**	**20.49**	**23.15**	**26.18**	**29.71**	**33.83**	**38.69**	**13.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 58 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	1.89	2.16	2.46	2.80	3.20	3.67	4.22	4.87	14.7%
UK	1.69	1.94	2.22	2.55	2.93	3.37	3.90	4.52	15.3%
France	1.34	1.52	1.73	1.98	2.26	2.58	2.97	3.43	14.6%
Italy	0.38	0.43	0.49	0.55	0.63	0.72	0.83	0.95	14.3%
Spain	0.29	0.33	0.37	0.42	0.48	0.55	0.62	0.72	14.0%
Rest of Europe	3.16	3.58	4.05	4.59	5.19	5.91	6.74	7.72	13.8%
Total	**8.75**	**9.96**	**11.32**	**12.89**	**14.69**	**16.80**	**19.27**	**22.21**	**14.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 59 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	1.02	1.19	1.39	1.62	1.89	2.21	2.58	3.04	16.9%
Japan	0.48	0.56	0.66	0.77	0.90	1.06	1.25	1.47	17.6%
India	0.06	0.07	0.08	0.10	0.11	0.13	0.15	0.18	16.8%
Rest of Asia Pacific	0.86	1.00	1.17	1.36	1.58	1.84	2.15	2.53	16.7%
Total	**2.42**	**2.83**	**3.29**	**3.84**	**4.48**	**5.23**	**6.13**	**7.22**	**17.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 60 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.26	0.29	0.32	0.35	0.39	0.43	0.48	0.53	10.8%
Rest of Latin America	0.29	0.32	0.36	0.39	0.44	0.48	0.53	0.60	10.7%
Total	**0.55**	**0.61**	**0.68**	**0.75**	**0.82**	**0.91**	**1.01**	**1.13**	**10.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8.4 NANOPORE SEQUENCING

8.4.1 ONLY SEQUENCING TECHNOLOGY WITH DIRECT RNA SEQUENCING CAPABILITIES

Introduced by Oxford Nanopore Technologies, nanopore sequencing technology (which is considered the fourth generation of sequencing) has emerged as a competitive, portable technology.

The technology works through the detection of unique electrical signals of different molecules as they pass across the nanopore with a semiconductor-based electronic detection system. This technology makes for a high throughput, cost-effective sequencing solution. The technology's core/heart consists of the biological nanopore, a protein pore embedded in a membrane, while the brains of the technology lie in the electronics of a semiconductor integrated circuit and proprietary chemistries. The electronic sensor embedded in the chip enables automatic membrane assembly and nanopore insertion while allowing for active control of individual sensors on the circuit. Various types of sequencing chemistries can be paired with nanopore and electronic sensor technology, which enables high-throughput and high-accuracy sequencing. However, the low accuracy of nanopore sequencing as compared to other well-established techniques is one of the major factors restraining the growth of this market segment.

In mid-2014, Oxford Nanopore Technologies launched MinION, which was the only nanopore sequencer commercially available in the market until 2015. This sequencer is portable, weighs just 100 gm, and can sequence DNA and RNA of up to ~200kb in length. Due to its portable nature, the MinION sequencer can be used for real-time sequencing. From January to July 2019, the company launched GridION, Flongle sequencer, and PromethION 24 and PromethION 48 (on the e-Commerce platform of the company to make them available globally). Also, in May 2018, the company upgraded the MinKnow software that drives the MinION, GridION, and PromethION sequencing devices.

TABLE 61 MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	6.47	7.47	8.62	9.95	11.49	13.30	15.46	18.03	15.9%
Europe	3.49	4.12	4.86	5.72	6.74	7.96	9.43	11.21	18.2%
Asia Pacific	0.96	1.18	1.44	1.76	2.14	2.62	3.20	3.92	22.2%
Latin America	0.22	0.26	0.30	0.35	0.41	0.47	0.55	0.64	16.5%
Middle East and Africa	0.15	0.18	0.20	0.24	0.27	0.31	0.36	0.41	15.2%
Total	**11.29**	**13.20**	**15.42**	**18.01**	**21.05**	**24.67**	**29.00**	**34.21**	**17.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 62 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	5.87	6.78	7.82	9.03	10.43	12.09	14.05	16.40	16.0%
Canada	0.61	0.70	0.80	0.92	1.06	1.22	1.41	1.63	15.4%
Total	**6.47**	**7.47**	**8.62**	**9.95**	**11.49**	**13.30**	**15.46**	**18.03**	**15.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 63 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	0.76	0.89	1.04	1.23	1.44	1.70	2.00	2.38	17.9%
UK	0.68	0.80	0.95	1.12	1.33	1.58	1.88	2.24	18.8%
France	0.53	0.63	0.74	0.88	1.03	1.22	1.45	1.72	18.3%
Italy	0.15	0.18	0.21	0.25	0.29	0.34	0.40	0.48	18.1%
Spain	0.12	0.14	0.16	0.19	0.22	0.26	0.31	0.37	18.0%
Rest of Europe	1.26	1.49	1.75	2.06	2.43	2.86	3.39	4.02	18.1%
Total	**3.49**	**4.12**	**4.86**	**5.72**	**6.74**	**7.96**	**9.43**	**11.21**	**18.2%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 64 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	0.41	0.50	0.61	0.74	0.90	1.09	1.33	1.63	21.9%
Japan	0.19	0.23	0.29	0.35	0.43	0.53	0.65	0.80	22.8%
India	0.02	0.03	0.04	0.04	0.05	0.07	0.08	0.10	22.2%
Rest of Asia Pacific	0.34	0.42	0.51	0.63	0.76	0.93	1.14	1.40	22.2%
Total	**0.96**	**1.18**	**1.44**	**1.76**	**2.14**	**2.62**	**3.20**	**3.92**	**22.2%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 65 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.10	0.12	0.14	0.16	0.19	0.22	0.26	0.30	16.5%
Rest of Latin America	0.12	0.14	0.16	0.19	0.22	0.25	0.29	0.34	16.6%
Total	**0.22**	**0.26**	**0.30**	**0.35**	**0.41**	**0.47**	**0.55**	**0.64**	**16.5%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8.5 OTHER TECHNOLOGIES

Thermo Fisher's ion semiconductor sequencing technology-based platforms include Ion Torrent PGM, Ion Proton, and the newly launched GeneStudio S5 Series. Ion semiconductor sequencing involves the direct translation of genetic codes of sequence (DNA bases) into digital information onto a semiconductor chip. The technology translates chemically encoded information (A, C, G, T) into digital information (0, 1) on a semiconductor chip. The company focuses on continuous enhancements in the technology to widen its scope of applications and increase its precision of sequencing.

The launch of new and advanced sequencers by Thermo Fisher and the advantages offered by ion semiconductor technology platforms are expected to drive market growth in the near future. For instance, in March 2022, the company launched the Ion Torrent Genexus Dx System. This fully validated system enables users to perform both diagnostic testing and clinical research on a single instrument. The company also provides complete NGS workflow for diagnostic solutions for comprehensive genomic profiling. The high adoption of ion semiconductor technology can be attributed to its features, including simpler workflow, time- as well as cost-efficiency, and scalability. This technology is gaining popularity in microbial profiling, thus contributing to the overall segment growth.

TABLE 66 MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	46.18	52.10	58.75	66.31	74.91	84.91	96.58	110.34	13.4%
Europe	25.14	28.51	32.32	36.67	41.63	47.44	54.24	62.28	14.0%
Asia Pacific	6.96	8.08	9.36	10.85	12.58	14.62	17.04	19.94	16.3%
Latin America	1.59	1.75	1.91	2.09	2.29	2.52	2.77	3.07	9.9%
Middle East and Africa	1.10	1.19	1.29	1.39	1.50	1.63	1.76	1.91	8.1%
Total	**80.97**	**91.63**	**103.64**	**117.32**	**132.92**	**151.11**	**172.40**	**197.53**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 67 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	41.85	47.25	53.33	60.24	68.11	77.27	87.96	100.57	13.5%
Canada	4.33	4.85	5.42	6.07	6.80	7.65	8.62	9.77	12.5%
Total	**46.18**	**52.10**	**58.75**	**66.31**	**74.91**	**84.91**	**96.58**	**110.34**	**13.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 68 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	5.43	6.18	7.04	8.02	9.15	10.47	12.02	13.86	14.5%
UK	4.86	5.56	6.35	7.27	8.32	9.56	11.03	12.76	15.0%
France	3.84	4.36	4.95	5.62	6.39	7.28	8.34	9.58	14.1%
Italy	1.08	1.23	1.39	1.57	1.78	2.02	2.30	2.64	13.7%
Spain	0.83	0.94	1.06	1.20	1.35	1.53	1.74	1.99	13.4%
Rest of Europe	9.09	10.24	11.53	12.99	14.65	16.57	18.81	21.44	13.2%
Total	**25.14**	**28.51**	**32.32**	**36.67**	**41.63**	**47.44**	**54.24**	**62.28**	**14.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 69 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	2.94	3.42	3.96	4.60	5.34	6.21	7.25	8.49	16.4%
Japan	1.37	1.60	1.86	2.17	2.53	2.96	3.47	4.08	17.0%
India	0.17	0.20	0.23	0.27	0.31	0.36	0.42	0.49	16.0%
Rest of Asia Pacific	2.48	2.86	3.30	3.81	4.40	5.09	5.90	6.88	15.8%
Total	**6.96**	**8.08**	**9.36**	**10.85**	**12.58**	**14.62**	**17.04**	**19.94**	**16.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 70 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.75	0.82	0.90	0.99	1.08	1.19	1.31	1.45	10.0%
Rest of Latin America	0.84	0.93	1.01	1.11	1.21	1.33	1.46	1.61	9.8%
Total	**1.59**	**1.75**	**1.91**	**2.09**	**2.29**	**2.52**	**2.77**	**3.07**	**9.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

9 MICROBIOME SEQUENCING SERVICES MARKET, BY END USER

KEY FINDINGS

- The academic and research institutes segment accounted for the largest share of 45.0% of the global microbiome sequencing services market in 2022. This segment is projected to reach USD 247.53 million by 2028 from USD 127.82 million in 2023, at a CAGR of 14.1%.
- The pharmaceutical and biotechnology companies segment is expected to grow at the highest CAGR of 14.9% from 2023 to 2028.
- Other end users, including small-scale CROs, CDMOs, hospitals, and clinics, collectively accounted for a share of 23.0% of the global microbiome sequencing services market in 2022.

9.1 INTRODUCTION

Based on end user, the global microbiome sequencing services market is segmented into pharmaceutical and biotechnology companies, academic and research institutes, and other end users (including small-scale CROs, CDMOs, hospitals, and clinics). The academic and research institutes segment accounted for the largest market share of 45.0% in 2022. This can be attributed to the high adoption of microbiome sequencing services and clinical research in these facilities.

TABLE 71 MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	70.17	80.20	91.62	104.76	119.89	137.69	158.69	183.68	14.9%
Academic and Research Institutes	99.25	112.66	127.82	145.15	164.97	188.15	215.34	247.53	14.1%
Other End Users	50.62	57.31	64.87	73.48	83.30	94.76	108.18	124.03	13.8%
Total	**220.03**	**250.17**	**284.31**	**323.39**	**368.16**	**420.61**	**482.21**	**555.24**	**14.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

9.2 PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES

9.2.1 ROBUST FUNDING FOR MICROBIOME-BASED THERAPEUTICS TO DRIVE MARKET

This segment is inclusive of the application of microbiome sequencing services in drug discovery, development, and clinical diagnostics. The microbiota plays an important role in the efficacy of therapeutic compounds in drug discovery. Many therapeutic compounds are processed by the catalytic functions of enzymes encoded by mammalian symbiotic bacteria. The unique nature of the human gut microbiome offers a new source of biomarkers to distinguish between patients' physiology and diseases for enhancement in personalized medicine.

Several diseases and conditions, such as diabetes, obesity, cancer, and autoimmune disorders, have been proposed to be treatable with the help of microbiome-based products. Several animal studies, preliminary human studies, uncontrolled studies, and anecdotal observations have shown positive results for these products. For instance, human trials indicating Lactobacillus GG may be useful in alleviating joint pains among rheumatoid arthritis (an autoimmune disease) patients. In addition, there are several animal studies that show that probiotics inhibit the initiation or progression of colon and bladder cancers. Furthermore, in vitro, cell culture, and animal studies have indicated that probiotics bind and prevent the absorption of aflatoxins, which have been associated with the occurrence of liver cancer in humans. This has created a significant demand for microbiome sequencing services among pharmaceutical and biotechnology companies.

The availability of funding for research is a key factor driving market growth, considering the rising pace of activity in this field. In the last five years, the NIH has extended funding of around USD 58.3 million to boost research in the human microbiome area, including funding for developing vaccines and devices. Of this, USD 49.0 million was marked for developing therapeutic products like prebiotics, probiotics, and dietary supplements and microbiome-based novel antimicrobials (Source: NIH).

Recent developments in this market include:

- The Frontiers in Microbiology Journal article titled 'Key Technologies for Progressing Discovery of Microbiome-based Medicines,' published in June 2021, reported that human gastrointestinal microbiome research can deliver critical clinical and therapeutic development. It also suggested future progress dependent on effective links between sequencing approaches, computational analyses, and experimental assays.
- In July 2021, Seres Therapeutics entered into an agreement with Nestlé Health Science to jointly commercialize SER-109, Seres' investigational oral microbiome therapeutic for recurrent Clostridioides difficile infection (CDI) in the United States and Canada.

TABLE 72 MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	40.30	45.71	51.82	58.80	66.77	76.09	87.02	99.94	14.0%
Europe	21.68	24.91	28.62	32.90	37.85	43.70	50.62	58.90	15.5%
Asia Pacific	5.92	7.03	8.33	9.86	11.68	13.87	16.52	19.74	18.8%
Latin America	1.34	1.51	1.71	1.92	2.16	2.44	2.76	3.13	12.9%
Middle East and Africa	0.93	1.04	1.15	1.28	1.43	1.59	1.77	1.97	11.3%
Total	**70.17**	**80.20**	**91.62**	**104.76**	**119.89**	**137.69**	**158.69**	**183.68**	**14.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 73 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	36.53	41.46	47.03	53.40	60.69	69.20	79.19	91.01	14.1%
Canada	3.77	4.25	4.79	5.39	6.09	6.89	7.83	8.93	13.3%
Total	**40.30**	**45.71**	**51.82**	**58.80**	**66.77**	**76.09**	**87.02**	**99.94**	**14.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 74 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	4.71	5.41	6.22	7.15	8.22	9.49	11.00	12.80	15.5%
UK	4.21	4.86	5.62	6.50	7.52	8.73	10.17	11.90	16.2%
France	3.32	3.81	4.38	5.04	5.79	6.69	7.75	9.02	15.5%
Italy	0.93	1.07	1.23	1.41	1.62	1.87	2.16	2.51	15.3%
Spain	0.72	0.82	0.94	1.08	1.24	1.42	1.65	1.91	15.2%
Rest of Europe	7.79	8.93	10.23	11.73	13.46	15.50	17.91	20.77	15.2%
Total	**21.68**	**24.91**	**28.62**	**32.90**	**37.85**	**43.70**	**50.62**	**58.90**	**15.5%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 75 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	2.51	2.98	3.52	4.16	4.93	5.84	6.95	8.30	18.7%
Japan	1.17	1.39	1.66	1.97	2.35	2.80	3.36	4.03	19.5%
India	0.15	0.18	0.21	0.25	0.29	0.35	0.41	0.49	18.7%
Rest of Asia Pacific	2.10	2.49	2.94	3.48	4.11	4.88	5.80	6.92	18.7%
Total	**5.92**	**7.03**	**8.33**	**9.86**	**11.68**	**13.87**	**16.52**	**19.74**	**18.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 76 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR PHARMACEUTICAL AND BIOTECHNOLOGY COMPANIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.63	0.71	0.80	0.90	1.02	1.15	1.30	1.47	12.9%
Rest of Latin America	0.71	0.80	0.90	1.02	1.14	1.29	1.46	1.66	12.9%
Total	**1.34**	**1.51**	**1.71**	**1.92**	**2.16**	**2.44**	**2.76**	**3.13**	**12.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

9.3 ACADEMIC AND RESEARCH INSTITUTES

9.3.1 EXPANDING APPLICATIONS OF MICROBIOME RESEARCH TO DRIVE GROWTH

This segment is inclusive of the application of microbiome sequencing services in environmental and ecological research, veterinary research, and other research activities conducted by academic and research institutes.

Metagenomics is used in microbial ecology and environmental applications to study a wide range of microbial communities present in the physical surroundings (which cannot be cultivated in laboratories). Studying microbes in the environment helps to understand their potential use in various fields, such as agriculture and biotechnology. It also provides valuable insights into other areas, such as the ecology of water bodies, debris filtered from the air, or a sample of soil, to identify the species present in the environment. It can also play a crucial role in identifying a broad range of invasive and endangered species and tracking seasonal populations and variations.

The development of NGS techniques for the large-scale analysis of microbial communities and the rising number of ecological and environmental studies conducted across the globe will support the continued demand for microbiome sequencing in this end-user segment.

Advancements in DNA sequencing methodologies and data analysis have revolutionized research in many areas of veterinary sciences, including medicine, livestock management, and disease management. The veterinary applications of metagenomics include the identification of the complex consortium of bacteria, protozoa, and fungi, among others, and their interaction. The need for rapid and accurate diagnosis of pathogens in livestock species and changes in animal husbandry—including growing herd sizes—are likely to propel the market growth in this segment.

TABLE 77 MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	56.62	64.07	72.45	82.01	92.90	105.62	120.48	138.04	13.8%
Europe	30.81	35.05	39.86	45.38	51.70	59.10	67.81	78.13	14.4%
Asia Pacific	8.52	9.93	11.55	13.44	15.64	18.25	21.36	25.09	16.8%
Latin America	1.95	2.15	2.36	2.60	2.86	3.15	3.48	3.87	10.4%
Middle East and Africa	1.35	1.47	1.59	1.73	1.87	2.03	2.21	2.41	8.6%
Total	**99.25**	**112.66**	**127.82**	**145.15**	**164.97**	**188.15**	**215.34**	**247.53**	**14.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 78 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	51.31	58.11	65.77	74.50	84.47	96.10	109.72	125.81	13.9%
Canada	5.31	5.96	6.69	7.51	8.44	9.51	10.76	12.23	12.8%
Total	**56.62**	**64.07**	**72.45**	**82.01**	**92.90**	**105.62**	**120.48**	**138.04**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 79 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	6.65	7.60	8.68	9.92	11.35	13.03	15.01	17.37	14.9%
UK	5.95	6.83	7.84	8.99	10.33	11.91	13.77	16.00	15.3%
France	4.71	5.36	6.10	6.95	7.93	9.07	10.42	12.02	14.5%
Italy	1.33	1.51	1.71	1.94	2.21	2.52	2.88	3.31	14.1%
Spain	1.02	1.16	1.31	1.48	1.68	1.91	2.18	2.50	13.8%
Rest of Europe	11.14	12.59	14.22	16.08	18.20	20.66	23.55	26.94	13.6%
Total	**30.81**	**35.05**	**39.86**	**45.38**	**51.70**	**59.10**	**67.81**	**78.13**	**14.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 80 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.60	4.20	4.89	5.69	6.63	7.75	9.07	10.67	16.9%
Japan	1.68	1.97	2.30	2.69	3.15	3.70	4.35	5.14	17.4%
India	0.21	0.25	0.29	0.33	0.39	0.45	0.53	0.62	16.5%
Rest of Asia Pacific	3.03	3.52	4.07	4.72	5.47	6.36	7.41	8.66	16.3%
Total	**8.52**	**9.93**	**11.55**	**13.44**	**15.64**	**18.25**	**21.36**	**25.09**	**16.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 81 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.91	1.01	1.11	1.22	1.35	1.49	1.65	1.83	10.5%
Rest of Latin America	1.03	1.14	1.25	1.37	1.51	1.66	1.84	2.04	10.3%
Total	**1.95**	**2.15**	**2.36**	**2.60**	**2.86**	**3.15**	**3.48**	**3.87**	**10.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

9.4 OTHER END USERS

This segment is inclusive of the application of microbiome sequencing services in clinical research, diagnostics, and collaborative R&D activities conducted by small-scale CROs, CDMOs, hospitals, and clinics. Biomarkers provided by the human gut microbiome may allow a new means to distinguish between a patient's physiology and disease conditions. It can also enable mapping microbial fingerprints and defining microbial-derived biomarkers that can help in identifying individuals at high risk of diseases, ensuring the diagnosis of existing conditions, and monitoring the efficacy of treatment.

For example, a bacterial microbiota diversity analysis—used to check for differences in the strain and number of microbes in chronic kidney disease patients and healthy people—found that there was an abundance of Bacteroidetes and Proteobacteria and fewer Firmicutes as compared to a healthy individual. Consequently, this provided a marker for the diagnosis of CKD. Furthermore, there are some diseases that occur because of the translocation of microbes from their normal habitat into some other tissues, where they can trigger diseases by setting up an inflammatory cascade. For example, it was found in a study that lupus, an autoimmune disease, was caused because of the translocation of Enterococcus gallinarum from the gut to the liver and other tissues, which triggered an autoimmune response. These are two instances of the potential use of the human microbiome as a diagnostic tool.

Several firms are focusing on investigation in this area. Enterome SA, for example, raised USD 13.8 million in venture capital funding for developing tests that use the composition of gut bacteria to diagnose inflammatory and liver diseases. It also entered into an agreement with Bristol-Myers Squibb for the discovery and development of microbiome-derived biomarkers, drug targets, and bioactive molecules to be developed as potential companion diagnostics and therapeutics for cancer. This will greatly support the prospects of the human microbiome in clinical research.

TABLE 82 MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	28.83	32.57	36.79	41.58	47.05	53.41	60.85	69.62	13.6%
Europe	15.73	17.84	20.22	22.94	26.05	29.69	33.94	38.98	14.0%
Asia Pacific	4.37	5.06	5.85	6.77	7.84	9.10	10.59	12.36	16.1%
Latin America	1.00	1.09	1.19	1.31	1.43	1.56	1.72	1.89	9.6%
Middle East and Africa	0.69	0.75	0.81	0.87	0.93	1.01	1.08	1.17	7.7%
Total	**50.62**	**57.31**	**64.87**	**73.48**	**83.30**	**94.76**	**108.18**	**124.03**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 83 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	26.12	29.54	33.39	37.78	42.78	48.61	55.42	63.47	13.7%
Canada	2.70	3.03	3.39	3.80	4.27	4.80	5.43	6.15	12.6%
Total	**28.83**	**32.57**	**36.79**	**41.58**	**47.05**	**53.41**	**60.85**	**69.62**	**13.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 84 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	3.39	3.87	4.41	5.03	5.74	6.57	7.55	8.72	14.6%
UK	3.04	3.48	3.98	4.55	5.22	6.00	6.92	8.01	15.0%
France	2.40	2.73	3.10	3.52	4.00	4.56	5.22	6.00	14.2%
Italy	0.68	0.77	0.87	0.98	1.11	1.26	1.44	1.65	13.7%
Spain	0.52	0.59	0.66	0.75	0.85	0.96	1.09	1.24	13.3%
Rest of Europe	5.70	6.41	7.21	8.12	9.15	10.34	11.73	13.36	13.1%
Total	**15.73**	**17.84**	**20.22**	**22.94**	**26.05**	**29.69**	**33.94**	**38.98**	**14.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 85 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	1.84	2.14	2.48	2.87	3.33	3.87	4.51	5.28	16.3%
Japan	0.86	1.00	1.17	1.36	1.58	1.84	2.16	2.53	16.8%
India	0.11	0.13	0.15	0.17	0.19	0.23	0.26	0.30	15.8%
Rest of Asia Pacific	1.55	1.79	2.06	2.38	2.74	3.16	3.66	4.25	15.6%
Total	**4.37**	**5.06**	**5.85**	**6.77**	**7.84**	**9.10**	**10.59**	**12.36**	**16.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 86 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER END USERS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.47	0.51	0.56	0.62	0.67	0.74	0.81	0.89	9.7%
Rest of Latin America	0.53	0.58	0.63	0.69	0.75	0.82	0.90	0.99	9.5%
Total	**1.00**	**1.09**	**1.19**	**1.31**	**1.43**	**1.56**	**1.72**	**1.89**	**9.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10 MICROBIOME SEQUENCING SERVICES MARKET, BY REGION

KEY FINDINGS

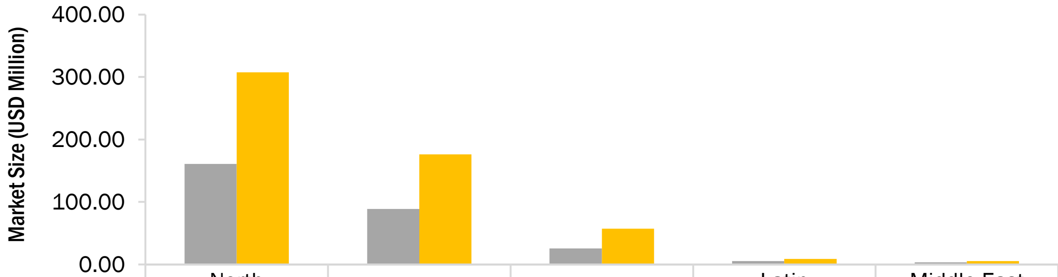
- North America accounted for the largest share of 56.9% of the microbiome sequencing services market in 2022. This market is projected to reach USD 307.60 million by 2028 from USD 161.06 million in 2023, at a CAGR of 13.8%.
- The established research infrastructure, technological advancements, and demand for precision medicine are driving the growth of the market in North America.
- Europe accounted for the second-largest share of 31.1% of the microbiome sequencing services market in 2022. This market is projected to reach USD 176.01 million by 2028, at a CAGR of 14.7% during the forecast period.
- The Asia Pacific market is estimated to grow at the highest CAGR of 17.3% during the forecast period, primarily due to the increasing awareness of microbiome health, the emerging healthcare and biotechnology industry, government support, and funding potential for personalized medicine in the region.

10.1 INTRODUCTION

The global microbiome sequencing services market has been segmented into five major regional segments, namely, North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. In 2022, North America dominated the market with the largest share of 56.9%, followed by Europe (31.1%). Factors such as the established research infrastructure, technological advancements, demand for precision medicine, supportive regulatory environment, robust biotechnology and healthcare industry, and availability of funding and investment are driving the growth of the microbiome sequencing services market in North America.

The microbiome sequencing services market in the Asia Pacific is estimated to register the highest growth rate of 17.3% during the forecast period, primarily due to the increasing awareness of microbiome health, the emerging healthcare and biotechnology industry, government support, and funding potential for personalized medicine.

FIGURE 25 MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2023 VS. 2028 (USD MILLION)



	North America	Europe	Asia Pacific	Latin America	Middle East and Africa
2023	161.06	88.70	25.73	5.26	3.55
2028	307.60	176.01	57.19	8.88	5.55
CAGR (2023–2028)	13.8%	14.7%	17.3%	11.1%	9.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 87 MICROBIOME SEQUENCING SERVICES MARKET, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	125.75	142.35	161.06	182.39	206.72	235.12	268.35	307.60	13.8%
Europe	68.21	77.80	88.70	101.22	115.60	132.49	152.38	176.01	14.7%
Asia Pacific	18.81	22.01	25.73	30.07	35.16	41.22	48.46	57.19	17.3%
Latin America	4.29	4.75	5.26	5.82	6.44	7.15	7.96	8.88	11.1%
Middle East and Africa	2.97	3.25	3.55	3.88	4.23	4.63	5.06	5.55	9.3%
Total	**220.03**	**250.17**	**284.31**	**323.39**	**368.16**	**420.61**	**482.21**	**555.24**	**14.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.2 NORTH AMERICA

North America has emerged as a leading market for microbiome sequencing services, driven by a combination of factors. One of the key drivers is the well-established healthcare and biotechnology industry in the region, with advanced infrastructure, cutting-edge research institutions, and a strong ecosystem of biotechnology companies and CROs specializing in microbiome research. The presence of top academic institutions, research hospitals, and private companies in North America with dedicated microbiome research programs and expertise has contributed to the growth of the microbiome sequencing services market. In 2022, the National Library of Medicine released a research paper that discussed the correlation of gut microbiome diversity and abundance with Gray Matter Volume (GMV) in older adults with depression.

Another factor driving North America's leadership in this market is the high level of awareness among healthcare professionals and the general population about the importance of the human microbiome in health and disease. There has been a growing recognition of the role of the microbiome in various conditions, including gastrointestinal disorders, metabolic diseases, neurodegenerative diseases, and immune-related disorders. This increased awareness has led to a rising demand for microbiome sequencing services to better understand the complex interactions between the human microbiome and host health, as well as to develop targeted interventions and therapies.

Despite the positive drivers, there are also some restraints that can impact the microbiome sequencing services market in North America. One of the major challenges is the high cost associated with certain microbiome sequencing services, which can limit the adoption of these services for some customers, especially smaller research institutions or clinics with limited budgets. Regulatory and ethical considerations related to the use of human microbiome data, such as data privacy and consent issues, can also pose challenges in the market. However, there also are significant opportunities for growth in the North American microbiome sequencing services market. Advancements in sequencing technologies, data analysis, and bioinformatics are enabling more accurate and cost-effective microbiome sequencing, which can propel the adoption of these services. Increasing research collaborations between academia, industry, and government institutions in North America can also lead to novel discoveries and applications of microbiome sequencing, thus creating new market opportunities. The rising demand for personalized medicine approaches, where understanding an individual's microbiome can inform tailored treatment plans, presents a significant growth opportunity for microbiome sequencing services.

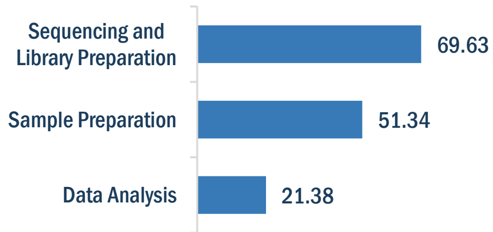
North America is home to several key companies that play a pivotal role in the microbiome sequencing services market. These companies, including Charles River Laboratories, CosmosID, and CD Genomics, are involved in the development and commercialization of microbiome sequencing technologies, platforms, and services. They are significantly contributing to the growth and innovation in the North American microbiome sequencing services market through their research, product offerings, and collaborations with academic and industry partners.

FIGURE 26 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET SNAPSHOT

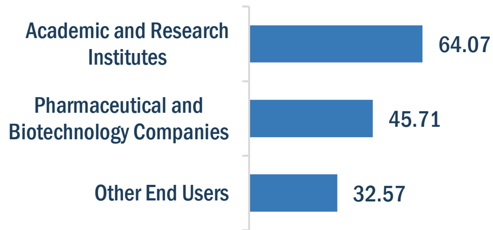
NORTH AMERICA

13.8% CAGR (2023–2028)	USD 142.35 Million Market Size in 2022	US Fastest-growing Country in the Region	56.9% Share of the Region in the Global Market in 2022

BY SERVICE, 2022 (USD MILLION)



BY END USER, 2022 (USD MILLION)



BY COUNTRY, 2022 (USD MILLION)

COUNTRY	MARKET SIZE	CAGR (2023–2028)
US	129.11	13.9%
Canada	13.24	12.9%

DRIVING FACTORS FOR GROWTH IN NORTH AMERICA

- Advanced infrastructure
- Awareness among healthcare professionals
- Growing demand for personalized medicine

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 88 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	113.97	129.11	146.19	165.68	187.93	213.91	244.33	280.29	13.9%
Canada	11.78	13.24	14.87	16.71	18.79	21.21	24.02	27.32	12.9%
Total	**125.75**	**142.35**	**161.06**	**182.39**	**206.72**	**235.12**	**268.35**	**307.60**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 89 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	45.42	51.34	58.01	65.60	74.25	84.33	96.11	110.02	13.7%
Sequencing and Library Preparation	61.38	69.63	78.94	89.58	101.75	115.96	132.62	152.34	14.1%
Data Analysis	18.95	21.38	24.11	27.21	30.73	34.83	39.61	45.25	13.4%
Total	**125.75**	**142.35**	**161.06**	**182.39**	**206.72**	**235.12**	**268.35**	**307.60**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 90 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	52.63	59.60	67.44	76.40	86.62	98.54	112.50	128.99	13.8%
Whole-genome Sequencing	26.93	30.39	34.29	38.72	43.75	49.62	56.47	64.54	13.5%
Shotgun Sequencing	33.80	38.48	43.79	49.86	56.83	64.99	74.58	85.96	14.4%
Other Types	12.39	13.88	15.54	17.41	19.53	21.97	24.80	28.12	12.6%
Total	**125.75**	**142.35**	**161.06**	**182.39**	**206.72**	**235.12**	**268.35**	**307.60**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 91 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	57.03	64.63	73.20	82.98	94.15	107.20	122.48	140.55	13.9%
Sequencing by Ligation	16.07	18.15	20.49	23.15	26.18	29.71	33.83	38.69	13.6%
Nanopore Sequencing	6.47	7.47	8.62	9.95	11.49	13.30	15.46	18.03	15.9%
Other Technologies	46.18	52.10	58.75	66.31	74.91	84.91	96.58	110.34	13.4%
Total	**125.75**	**142.35**	**161.06**	**182.39**	**206.72**	**235.12**	**268.35**	**307.60**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 92 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	40.30	45.71	51.82	58.80	66.77	76.09	87.02	99.94	14.0%
Academic and Research Institutes	56.62	64.07	72.45	82.01	92.90	105.62	120.48	138.04	13.8%
Other End Users	28.83	32.57	36.79	41.58	47.05	53.41	60.85	69.62	13.6%
Total	**125.75**	**142.35**	**161.06**	**182.39**	**206.72**	**235.12**	**268.35**	**307.60**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.2.1 US

10.2.1.1 Increasing awareness of microbiome research to drive growth

The US microbiome sequencing services market is a rapidly growing and dynamic market that is driven by several factors. One of the key drivers for market growth in the US is the increasing awareness and understanding of the role of the microbiome in human health. Research has shown that microorganisms residing in and on the human body play a crucial role in various physiological processes, including digestion, immunity, and metabolism. The increasing focus on gaining a deep understanding of this has driven the demand for microbiome sequencing services to study the composition and function of the microbiome.

NGS technologies, such as Illumina's MiSeq and HiSeq platforms, have revolutionized the field of microbiome research by enabling the high-throughput and cost-effective sequencing of microbial communities. These technologies have greatly expanded the capability to study the complex and diverse microbial populations in the human microbiome. Several key players operating in the US market are adopting strategies to develop advanced products and provide efficient metagenomic NGS solutions to their customers. For instance, in 2021, CosmosID (US) formed a partnership with Locus Biosciences (US) to provide long-term support for Locus' clinical trial initiatives. CosmosID's infrastructure for providing access to robust, compliant, and high-resolution microbiome analysis results will allow Locus to add further insights and capabilities to its precision therapeutics platform. Using shotgun metagenomics, CosmosID will help Locus understand the efficacy of its products through the highest resolution microbiome analysis, starting with its urinary tract infection clinical program.

One of the main challenges to the growth of the US market is the lack of standardized methods and protocols for microbiome sample collection, DNA extraction, library preparation, and data analysis. This can result in variability and inconsistency in results, making it challenging to compare and interpret data across different studies or laboratories.

TABLE 93 US: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	41.16	46.57	52.66	59.59	67.50	76.73	87.52	100.25	13.7%
Sequencing and Library Preparation	55.63	63.15	71.65	81.37	92.49	105.48	120.73	138.78	14.1%
Data Analysis	17.17	19.39	21.88	24.72	27.94	31.70	36.09	41.26	13.5%
Total	**113.97**	**129.11**	**146.19**	**165.68**	**187.93**	**213.91**	**244.33**	**280.29**	**13.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 94 US: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	47.70	54.05	61.22	69.40	78.74	89.64	102.42	117.52	13.9%
Whole-genome Sequencing	24.40	27.56	31.12	35.17	39.79	45.16	51.43	58.83	13.6%
Shotgun Sequencing	30.64	34.90	39.74	45.29	51.65	59.11	67.88	78.29	14.5%
Other Types	11.23	12.59	14.11	15.82	17.76	20.00	22.60	25.65	12.7%
Total	**113.97**	**129.11**	**146.19**	**165.68**	**187.93**	**213.91**	**244.33**	**280.29**	**13.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 95 US: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	51.68	58.62	66.45	75.39	85.60	97.54	111.54	128.09	14.0%
Sequencing by Ligation	14.56	16.46	18.60	21.03	23.79	27.02	30.79	35.23	13.6%
Nanopore Sequencing	5.87	6.78	7.82	9.03	10.43	12.09	14.05	16.40	16.0%
Other Technologies	41.85	47.25	53.33	60.24	68.11	77.27	87.96	100.57	13.5%
Total	**113.97**	**129.11**	**146.19**	**165.68**	**187.93**	**213.91**	**244.33**	**280.29**	**13.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 96 US: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	36.53	41.46	47.03	53.40	60.69	69.20	79.19	91.01	14.1%
Academic and Research Institutes	51.31	58.11	65.77	74.50	84.47	96.10	109.72	125.81	13.9%
Other End Users	26.12	29.54	33.39	37.78	42.78	48.61	55.42	63.47	13.7%
Total	**113.97**	**129.11**	**146.19**	**165.68**	**187.93**	**213.91**	**244.33**	**280.29**	**13.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.2.2 CANADA

10.2.2.1 Increasing prevalence of chronic diseases to boost microbiome research

The microbiome sequencing services market in Canada has been witnessing significant growth in recent years, driven by various factors. One of the key drivers is the increasing prevalence of chronic diseases, such as obesity, diabetes, and inflammatory bowel disease (IBD), in Canada. As the composition of the gut microbiome is linked to these diseases, microbiome sequencing can provide valuable insights into the underlying mechanisms and potential therapeutic strategies.

Advancements in sequencing technologies, including SBS, have made the high-throughput, cost-effective sequencing of microbial communities possible. The availability of NGS platforms has enabled the large-scale sequencing of the microbiome. The Government of Canada has undertaken several initiatives to improve the healthcare system in the country. For instance, in 2021, the government initiated the third phase of the Strategic Plan 2021–2026. This plan sets out priorities that align with the CIHR Strategic Plan 2021–2031 and the STBBI Research Initiative Strategic Plan. The plan aims to achieve excellence in infection and immunity research. In January 2019, the government announced an investment of USD 18 million over five years in microbiome research. This investment supported seven research teams from across the country that studied the microbiome to better understand its contribution to childhood asthma, cervical cancer, diabetes, youth inflammatory bowel disease, and maternal malnutrition. The investment also helped establish a Microbiome Research Core at the University of Calgary and support the teams by providing expertise in study design, sample collection, and analysis, and, eventually, a large repository of microbes.

The microbiome sequencing services market in Canada faces certain restraints, including challenges in standardization, data analysis, and interpretation. Additionally, regulatory considerations and ethical concerns related to the use of human microbiome data can impact market growth.

TABLE 97 CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	4.25	4.78	5.35	6.01	6.75	7.60	8.60	9.76	12.8%
Sequencing and Library Preparation	5.75	6.48	7.29	8.21	9.26	10.48	11.89	13.56	13.2%
Data Analysis	1.78	1.99	2.22	2.49	2.78	3.13	3.53	3.99	12.4%
Total	**11.78**	**13.24**	**14.87**	**16.71**	**18.79**	**21.21**	**24.02**	**27.32**	**12.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 98 CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	4.93	5.54	6.22	7.00	7.88	8.90	10.08	11.48	13.0%
Whole-genome Sequencing	2.53	2.83	3.16	3.54	3.97	4.46	5.03	5.70	12.5%
Shotgun Sequencing	3.17	3.58	4.05	4.57	5.18	5.88	6.70	7.67	13.6%
Other Types	1.16	1.29	1.43	1.59	1.77	1.97	2.20	2.47	11.5%
Total	**11.78**	**13.24**	**14.87**	**16.71**	**18.79**	**21.21**	**24.02**	**27.32**	**12.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 99 CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	5.34	6.01	6.75	7.59	8.55	9.66	10.94	12.46	13.0%
Sequencing by Ligation	1.51	1.69	1.89	2.13	2.39	2.69	3.04	3.45	12.8%
Nanopore Sequencing	0.61	0.70	0.80	0.92	1.06	1.22	1.41	1.63	15.4%
Other Technologies	4.33	4.85	5.42	6.07	6.80	7.65	8.62	9.77	12.5%
Total	**11.78**	**13.24**	**14.87**	**16.71**	**18.79**	**21.21**	**24.02**	**27.32**	**12.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 100 CANADA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	3.77	4.25	4.79	5.39	6.09	6.89	7.83	8.93	13.3%
Academic and Research Institutes	5.31	5.96	6.69	7.51	8.44	9.51	10.76	12.23	12.8%
Other End Users	2.70	3.03	3.39	3.80	4.27	4.80	5.43	6.15	12.6%
Total	**11.78**	**13.24**	**14.87**	**16.71**	**18.79**	**21.21**	**24.02**	**27.32**	**12.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.2.3 NORTH AMERICA: IMPACT OF RECESSION

North America is very likely to enter a recession in mid-2023, with higher-than-expected inflation, especially in the US, triggering a tightening of global financial conditions. According to the International Monetary Fund (IMF) World Economic Outlook 2022, in the US, reduced household purchasing power and tighter monetary policy will drive GDP growth down from 2.3% in 2022 to 1% in 2023. Similarly, in Canada, growth in real GDP is projected to slow from 3.2% in 2022 to 1.0% in 2023 before strengthening to 1.3% in 2024. However, people’s spending on healthcare is less likely to be affected by macroeconomic headwinds as diseases will always exist, and treatments and medicines will always be in demand, irrespective of economic turbulence. Health expenditure in North America could exceed economic growth by up to 5.9% in 2023. National health expenditure could grow at a rate of 7.1% over the next five years, from 2022 to 2027, compared with an expected economic growth rate of 4.7%.

The US government is taking action to combat the upcoming recession. In August 2022, the US government passed the Inflation Reduction Act, which features significant changes to federal prescription drug pricing policies that can reduce costs for individuals receiving care through Medicare. The bill also includes a three-year extension of expanded Affordable Care Act (ACA) healthcare benefits through 2025. Additionally, more mergers and acquisitions are expected in 2023 in the US, with a deal value of USD 225 billion to USD 275 billion across all subsectors. Companies such as Johnson & Johnson and Pfizer have entered deals with Abiomed and Biohaven Pharmaceuticals for USD 18.9 billion and USD 12.2 billion, respectively. In the pharma and life science sector, the US market is projected to exhibit higher growth in 2023. This is attributed to pharma and biotech M&A growth due to an increased focus on oncology and immunology.

10.3 EUROPE

The microbiome sequencing services market in Europe has been growing steadily in recent years, driven by several factors, such as the increasing awareness of and interest in personalized health and wellness, the growing adoption of microbiome-based therapies, and advancements in sequencing technologies and bioinformatics analysis. Consumers in Europe are becoming more aware of the role of the microbiome in health and disease and are seeking personalized solutions to optimize their microbiome composition and function. This has led to the growth of direct-to-consumer microbiome sequencing services, which offer comprehensive analysis and personalized recommendations based on an individual's microbiome composition.

Due to the role of the microbiome in various diseases and conditions, such as IBD, obesity, and autism, the focus on developing microbiome-based therapies to treat these conditions has increased in Europe in recent years. This has led to an increase in the demand for microbiome sequencing services in R&D and clinical applications.

Major companies are playing a significant role in driving innovation in this market and offering advanced technologies and services across Europe. In April 2022, Microba Life Sciences and Ginkgo Bioworks announced a partnership to identify single-strain, live bacteria product (LBP) candidates against autoimmune diseases. The collaboration aimed to build on Microba's precision approach to LBP development with an in-depth evaluation of the company's strains using Ginkgo's high-throughput and automated screening capabilities. Through this partnership, Ginkgo provided high-throughput screening for Microba's proprietary library of human microbiome-isolated strains with the goal of improving treatment for autoimmune diseases such as lupus, psoriatic arthritis, and certain autoimmune liver diseases. Microba leveraged Ginkgo's high-throughput anaerobic culturing, multi-omics data collection and analysis, functional bioassay screening, and media and fermentation optimization capabilities to generate data sets that may help characterize potential therapeutic and non-therapeutic uses of the strains. Moreover, in 2020, Microba entered into an agreement with SYNLAB Diagnostics Globales to deliver Microba’s testing services to healthcare providers in Spain, with an option to formalize distribution agreements across their affiliate organizations, which operate in 36 countries.

The use of microbiome sequencing in clinical applications is subject to regulatory oversight, and there are concerns about the accuracy and reliability of microbiome analysis and interpretation. Additionally, the collection and analysis of personal microbiome data raise concerns about data privacy and security. These factors are expected to restrain market growth in Europe.

TABLE 101 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	14.76	16.88	19.31	22.10	25.31	29.10	33.56	38.88	15.0%
UK	13.20	15.17	17.43	20.04	23.06	26.63	30.86	35.91	15.6%
France	10.43	11.90	13.58	15.51	17.72	20.32	23.39	27.04	14.8%
Italy	2.94	3.35	3.81	4.33	4.94	5.64	6.48	7.46	14.4%
Spain	2.26	2.57	2.91	3.31	3.76	4.29	4.91	5.65	14.2%
Rest of Europe	24.62	27.93	31.67	35.93	40.81	46.50	53.18	61.08	14.0%
Total	**68.21**	**77.80**	**88.70**	**101.22**	**115.60**	**132.49**	**152.38**	**176.01**	**14.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 102 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	24.57	27.99	31.87	36.32	41.42	47.40	54.44	62.80	14.5%
Sequencing and Library Preparation	33.26	38.09	43.60	49.94	57.25	65.87	76.04	88.17	15.1%
Data Analysis	10.37	11.72	13.24	14.97	16.93	19.22	21.89	25.05	13.6%
Total	**68.21**	**77.80**	**88.70**	**101.22**	**115.60**	**132.49**	**152.38**	**176.01**	**14.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 103 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	28.39	32.47	37.11	42.44	48.59	55.82	64.35	74.51	15.0%
Whole-genome Sequencing	14.70	16.65	18.84	21.35	24.21	27.55	31.46	36.07	13.9%
Shotgun Sequencing	18.31	21.07	24.23	27.90	32.14	37.15	43.09	50.20	15.7%
Other Types	6.81	7.62	8.52	9.53	10.66	11.97	13.48	15.23	12.3%
Total	**68.21**	**77.80**	**88.70**	**101.22**	**115.60**	**132.49**	**152.38**	**176.01**	**14.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 104 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	30.83	35.22	40.20	45.94	52.54	60.30	69.44	80.32	14.8%
Sequencing by Ligation	8.75	9.96	11.32	12.89	14.69	16.80	19.27	22.21	14.4%
Nanopore Sequencing	3.49	4.12	4.86	5.72	6.74	7.96	9.43	11.21	18.2%
Other Technologies	25.14	28.51	32.32	36.67	41.63	47.44	54.24	62.28	14.0%
Total	**68.21**	**77.80**	**88.70**	**101.22**	**115.60**	**132.49**	**152.38**	**176.01**	**14.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 105 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	21.68	24.91	28.62	32.90	37.85	43.70	50.62	58.90	15.5%
Academic and Research Institutes	30.81	35.05	39.86	45.38	51.70	59.10	67.81	78.13	14.4%
Other End Users	15.73	17.84	20.22	22.94	26.05	29.69	33.94	38.98	14.0%
Total	**68.21**	**77.80**	**88.70**	**101.22**	**115.60**	**132.49**	**152.38**	**176.01**	**14.7%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.1 GERMANY

10.3.1.1 Rapidly increasing government funding to support growth

Germany is the largest contributor to the European microbiome sequencing services market, with an estimated share of 21.8% in 2023. The microbiome sequencing services market in Germany is estimated to grow from USD 19.31 million in 2023 to USD 38.88 million by 2028, at a CAGR of 15.0%.

The microbiome sequencing services market in Germany is growing rapidly, driven by the increasing demand for microbiome analysis in research and clinical applications. The German government is making significant investments in the development of innovative technologies and research infrastructure, including microbiome research. This funding is driving the growth of the microbiome sequencing services market, as it allows researchers and companies to access the resources they need to advance in the field. Also, advances in sequencing technologies and data analysis tools are making microbiome analysis faster, more accurate, and more accessible.

The high cost of sequencing technologies and the lack of standardization in data analysis and interpretation are expected to restrain the growth of this market. Despite these challenges, there are significant opportunities for growth in the German microbiome sequencing services market, including the development of new technologies and applications, as well as the increasing collaborations between researchers and clinicians.

TABLE 106 GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	5.32	6.08	6.95	7.94	9.08	10.42	12.01	13.89	14.9%
Sequencing and Library Preparation	7.20	8.26	9.47	10.87	12.49	14.40	16.66	19.35	15.4%
Data Analysis	2.23	2.54	2.89	3.28	3.74	4.27	4.90	5.64	14.3%
Total	**14.76**	**16.88**	**19.31**	**22.10**	**25.31**	**29.10**	**33.56**	**38.88**	**15.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 107 GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	6.16	7.06	8.08	9.26	10.62	12.22	14.11	16.37	15.2%
Whole-genome Sequencing	3.17	3.61	4.11	4.68	5.33	6.10	7.00	8.08	14.5%
Shotgun Sequencing	3.96	4.57	5.26	6.06	6.99	8.09	9.40	10.96	15.8%
Other Types	1.46	1.65	1.86	2.10	2.37	2.68	3.04	3.47	13.3%
Total	**14.76**	**16.88**	**19.31**	**22.10**	**25.31**	**29.10**	**33.56**	**38.88**	**15.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 108 GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	6.68	7.65	8.77	10.04	11.52	13.26	15.32	17.77	15.2%
Sequencing by Ligation	1.89	2.16	2.46	2.80	3.20	3.67	4.22	4.87	14.7%
Nanopore Sequencing	0.76	0.89	1.04	1.23	1.44	1.70	2.00	2.38	17.9%
Other Technologies	5.43	6.18	7.04	8.02	9.15	10.47	12.02	13.86	14.5%
Total	**14.76**	**16.88**	**19.31**	**22.10**	**25.31**	**29.10**	**33.56**	**38.88**	**15.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 109 GERMANY: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	4.71	5.41	6.22	7.15	8.22	9.49	11.00	12.80	15.5%
Academic and Research Institutes	6.65	7.60	8.68	9.92	11.35	13.03	15.01	17.37	14.9%
Other End Users	3.39	3.87	4.41	5.03	5.74	6.57	7.55	8.72	14.6%
Total	**14.76**	**16.88**	**19.31**	**22.10**	**25.31**	**29.10**	**33.56**	**38.88**	**15.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.2 UK

10.3.2.1 Significant government investments in genomics research to propel growth

In 2022, the UK accounted for a share of 19.5% of the microbiome sequencing services market in Europe. This market is projected to reach USD 35.91 million by 2028 from USD 17.43 million in 2023, at a CAGR of 15.6% during the forecast period.

The UK government's significant investment in genomics research has helped fund and support several innovative microbiome sequencing companies and research initiatives. In addition, the country's strong academic and research base, combined with its robust healthcare system, has created a supportive environment for microbiome sequencing research and development.

The increasing prevalence of chronic diseases in the UK, such as obesity, diabetes, and cardiovascular disease, has also driven the demand for microbiome analysis to better understand these diseases and develop new treatments. In March 2022, Genetic Signatures (Australia) joined BioHub Birmingham as a launchpad for a planned further expansion into Europe, the Middle East, and Africa. BioHub is the first physical footprint within the UK and Europe for Genetic Signatures, which already provides the EasyScreen range of testing kits to countries in Europe and provides clinical diagnostics to the NHS for respiratory and gastro-intestinal diseases, as well as sexually transmitted infections.

The high cost of sequencing technologies and the lack of standardization in data analysis and interpretation are expected to restrain market growth in the UK.

TABLE 110 UK: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	4.76	5.46	6.27	7.20	8.27	9.54	11.03	12.82	15.4%
Sequencing and Library Preparation	6.44	7.43	8.56	9.87	11.40	13.20	15.35	17.92	15.9%
Data Analysis	2.00	2.28	2.60	2.97	3.39	3.89	4.47	5.17	14.7%
Total	**13.20**	**15.17**	**17.43**	**20.04**	**23.06**	**26.63**	**30.86**	**35.91**	**15.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 111 UK: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	5.50	6.34	7.29	8.40	9.68	11.20	13.00	15.15	15.7%
Whole-genome Sequencing	2.84	3.24	3.71	4.24	4.85	5.57	6.41	7.42	14.9%
Shotgun Sequencing	3.54	4.11	4.75	5.51	6.39	7.43	8.67	10.17	16.4%
Other Types	1.31	1.48	1.68	1.90	2.14	2.43	2.77	3.17	13.6%
Total	**13.20**	**15.17**	**17.43**	**20.04**	**23.06**	**26.63**	**30.86**	**35.91**	**15.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 112 UK: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	5.97	6.87	7.91	9.10	10.48	12.12	14.06	16.38	15.7%
Sequencing by Ligation	1.69	1.94	2.22	2.55	2.93	3.37	3.90	4.52	15.3%
Nanopore Sequencing	0.68	0.80	0.95	1.12	1.33	1.58	1.88	2.24	18.8%
Other Technologies	4.86	5.56	6.35	7.27	8.32	9.56	11.03	12.76	15.0%
Total	**13.20**	**15.17**	**17.43**	**20.04**	**23.06**	**26.63**	**30.86**	**35.91**	**15.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 113 UK: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	4.21	4.86	5.62	6.50	7.52	8.73	10.17	11.90	16.2%
Academic and Research Institutes	5.95	6.83	7.84	8.99	10.33	11.91	13.77	16.00	15.3%
Other End Users	3.04	3.48	3.98	4.55	5.22	6.00	6.92	8.01	15.0%
Total	**13.20**	**15.17**	**17.43**	**20.04**	**23.06**	**26.63**	**30.86**	**35.91**	**15.6%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.3 FRANCE

10.3.3.1 Growing number of microbiome sequencing startups to propel growth

The market in France is projected to reach USD 27.04 million by 2028 from USD 13.58 million in 2023, at a CAGR of 14.8% during the forecast period.

The presence of a strong healthcare system and the increasing demand for personalized medicine are the key factors driving market growth in France. Several companies in France are focusing on developing new products and services that use microbiome data to provide personalized diet and lifestyle recommendations to individuals based on their unique microbiome makeup. Microbiome sequencing startups and spin-offs in France are developing innovative technologies and services for microbiome research and analysis and creating new opportunities for market growth.

In September 2022, the Le French Gut project was launched by *the Institut national de la recherche agronomique* (INRAE). This project will contribute to the international Million Microbiome of Humans Project. INRAE's research unit, MetaGenoPolis, has initiated the project in association with AP-HP. The Le French Gut project aims to better understand healthy gut microbiota and model and predict changes in gut microbiota associated with diseases.

The lack of funding and support for research and development in this area and a lack of standardization in data analysis and interpretation are the key factors restraining market growth in France.

TABLE 114 HUMAN MICROBIOME STARTUPS IN FRANCE

COMPANY NAME	THERAPEUTIC AREA
Enterome	Develops microbiome-based diagnostics and therapeutics for chronic diseases
Eligo Bioscience	Develops microbiome-based therapies using CRISPR-Cas technology
BiomX	Develops microbiome-based therapeutics for bacterial infections and inflammatory bowel disease
TargEDys	Develops microbiome-based therapies for obesity and related disorders
LNC Therapeutics	Develops microbiome-based therapies for metabolic and inflammatory diseases
YSOPIA Bioscience	Develops microbiome-based therapies for cancer and immune disorders
Pherecydes Pharma	Develops microbiome-based phage therapies for bacterial infections
Predilife	Provides microbiome-based genetic testing and personalized health advice
InnaVirVax	Develops microbiome-based immunotherapies for chronic viral infections
TreeFrog Therapeutics	Develops microbiome-based therapies for cell therapy and regenerative medicine

TABLE 115 FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	3.76	4.28	4.88	5.57	6.35	7.27	8.36	9.65	14.6%
Sequencing and Library Preparation	5.09	5.83	6.67	7.65	8.77	10.10	11.66	13.53	15.2%
Data Analysis	1.58	1.79	2.03	2.29	2.60	2.96	3.37	3.86	13.7%
Total	**10.43**	**11.90**	**13.58**	**15.51**	**17.72**	**20.32**	**23.39**	**27.04**	**14.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 116 FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	4.34	4.97	5.68	6.50	7.45	8.56	9.87	11.43	15.0%
Whole-genome Sequencing	2.25	2.55	2.89	3.27	3.72	4.23	4.84	5.56	14.0%
Shotgun Sequencing	2.80	3.22	3.71	4.27	4.92	5.69	6.60	7.69	15.7%
Other Types	1.04	1.17	1.31	1.46	1.64	1.84	2.08	2.35	12.5%
Total	**10.43**	**11.90**	**13.58**	**15.51**	**17.72**	**20.32**	**23.39**	**27.04**	**14.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 117 FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	4.72	5.39	6.15	7.03	8.05	9.24	10.64	12.31	14.9%
Sequencing by Ligation	1.34	1.52	1.73	1.98	2.26	2.58	2.97	3.43	14.6%
Nanopore Sequencing	0.53	0.63	0.74	0.88	1.03	1.22	1.45	1.72	18.3%
Other Technologies	3.84	4.36	4.95	5.62	6.39	7.28	8.34	9.58	14.1%
Total	**10.43**	**11.90**	**13.58**	**15.51**	**17.72**	**20.32**	**23.39**	**27.04**	**14.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 118 FRANCE: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	3.32	3.81	4.38	5.04	5.79	6.69	7.75	9.02	15.5%
Academic and Research Institutes	4.71	5.36	6.10	6.95	7.93	9.07	10.42	12.02	14.5%
Other End Users	2.40	2.73	3.10	3.52	4.00	4.56	5.22	6.00	14.2%
Total	**10.43**	**11.90**	**13.58**	**15.51**	**17.72**	**20.32**	**23.39**	**27.04**	**14.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.4 ITALY

10.3.4.1 Increasing use of microbiome services in clinical research to drive growth

The market in Italy is projected to reach USD 7.46 million by 2028 from USD 3.81 million in 2023, at a CAGR of 14.4% during the forecast period.

The increasing use of microbiome services in clinical research is a key factor driving market growth in Italy. Italian researchers are using microbiome analysis to gain new insights into the role of the microbiome in a range of diseases, from IBD to cancer. This is driving the demand for high-quality microbiome sequencing services from both academic and commercial research institutions across the country. In 2022, the National Center for Biotechnology Information published a research article that explained the microbiota-gut axis in premature infants and its physio-pathological implications. In 2021, research on gut microbial dysbiosis in the pathogenesis of gastrointestinal dysmotility and metabolic disorders was conducted. It explained how constant communication between the gut microbiota-derived metabolites and human body systems regulates the physiological aspects of health and disease.

Italian companies are at the forefront of developing new microbiome-based therapies, particularly for IBD. This is driving the demand for high-quality microbiome sequencing services and creating new opportunities for collaborations and partnerships between Italian biotechnology companies and international microbiome sequencing providers.

The high cost associated with sequencing services is limiting the market growth. Although the cost of sequencing has decreased over the years, it is still relatively high and can be a significant barrier for some customers.

TABLE 119 ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.06	1.20	1.37	1.55	1.77	2.02	2.31	2.66	14.3%
Sequencing and Library Preparation	1.43	1.64	1.87	2.14	2.44	2.80	3.23	3.73	14.8%
Data Analysis	0.45	0.50	0.57	0.64	0.72	0.82	0.93	1.07	13.4%
Total	**2.94**	**3.35**	**3.81**	**4.33**	**4.94**	**5.64**	**6.48**	**7.46**	**14.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 120 ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.22	1.40	1.59	1.82	2.08	2.38	2.74	3.16	14.7%
Whole-genome Sequencing	0.63	0.72	0.81	0.91	1.03	1.17	1.33	1.53	13.5%
Shotgun Sequencing	0.79	0.91	1.04	1.19	1.37	1.59	1.83	2.13	15.4%
Other Types	0.29	0.33	0.37	0.41	0.45	0.51	0.57	0.64	11.9%
Total	**2.94**	**3.35**	**3.81**	**4.33**	**4.94**	**5.64**	**6.48**	**7.46**	**14.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 121 ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.33	1.51	1.72	1.96	2.24	2.56	2.94	3.39	14.5%
Sequencing by Ligation	0.38	0.43	0.49	0.55	0.63	0.72	0.83	0.95	14.3%
Nanopore Sequencing	0.15	0.18	0.21	0.25	0.29	0.34	0.40	0.48	18.1%
Other Technologies	1.08	1.23	1.39	1.57	1.78	2.02	2.30	2.64	13.7%
Total	**2.94**	**3.35**	**3.81**	**4.33**	**4.94**	**5.64**	**6.48**	**7.46**	**14.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 122 ITALY: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.93	1.07	1.23	1.41	1.62	1.87	2.16	2.51	15.3%
Academic and Research Institutes	1.33	1.51	1.71	1.94	2.21	2.52	2.88	3.31	14.1%
Other End Users	0.68	0.77	0.87	0.98	1.11	1.26	1.44	1.65	13.7%
Total	**2.94**	**3.35**	**3.81**	**4.33**	**4.94**	**5.64**	**6.48**	**7.46**	**14.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.5 SPAIN

10.3.5.1 Increasing prevalence of chronic diseases to drive market

The market in Spain is projected to reach USD 5.65 million by 2028 from USD 2.91 million in 2023, at a CAGR of 14.2% during the forecast period.

The increasing prevalence of chronic diseases, growing demand for personalized medicine, and rising awareness about the role of gut microbiomes in human health are the key factors driving market growth in Spain. The prevalence of chronic diseases, such as obesity, diabetes, and IBD, is increasing in Spain. These conditions are associated with dysbiosis of the gut microbiome, and understanding the composition and function of the microbiome can provide valuable insights into their pathogenesis and potential treatments. However, a lack of awareness about microbiome sequencing services among the general population and healthcare professionals is expected to limit the demand for these services.

TABLE 123 SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.81	0.92	1.05	1.19	1.35	1.53	1.75	2.01	14.0%
Sequencing and Library Preparation	1.10	1.26	1.43	1.64	1.87	2.14	2.46	2.84	14.7%
Data Analysis	0.34	0.39	0.43	0.49	0.55	0.62	0.70	0.79	12.8%
Total	**2.26**	**2.57**	**2.91**	**3.31**	**3.76**	**4.29**	**4.91**	**5.65**	**14.2%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 124 SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.94	1.07	1.22	1.39	1.58	1.81	2.08	2.40	14.5%
Whole-genome Sequencing	0.49	0.55	0.62	0.70	0.79	0.89	1.01	1.15	13.2%
Shotgun Sequencing	0.61	0.70	0.80	0.91	1.05	1.21	1.40	1.62	15.3%
Other Types	0.23	0.25	0.28	0.31	0.34	0.38	0.43	0.48	11.4%
Total	**2.26**	**2.57**	**2.91**	**3.31**	**3.76**	**4.29**	**4.91**	**5.65**	**14.2%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 125 SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.02	1.16	1.32	1.50	1.71	1.95	2.24	2.58	14.3%
Sequencing by Ligation	0.29	0.33	0.37	0.42	0.48	0.55	0.62	0.72	14.0%
Nanopore Sequencing	0.12	0.14	0.16	0.19	0.22	0.26	0.31	0.37	18.0%
Other Technologies	0.83	0.94	1.06	1.20	1.35	1.53	1.74	1.99	13.4%
Total	**2.26**	**2.57**	**2.91**	**3.31**	**3.76**	**4.29**	**4.91**	**5.65**	**14.2%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 126 SPAIN: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.72	0.82	0.94	1.08	1.24	1.42	1.65	1.91	15.2%
Academic and Research Institutes	1.02	1.16	1.31	1.48	1.68	1.91	2.18	2.50	13.8%
Other End Users	0.52	0.59	0.66	0.75	0.85	0.96	1.09	1.24	13.3%
Total	**2.26**	**2.57**	**2.91**	**3.31**	**3.76**	**4.29**	**4.91**	**5.65**	**14.2%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.6 REST OF EUROPE

The Rest of Europe region comprises Switzerland, Sweden, Poland, Austria, Belgium, Denmark, and Russia, among other European countries. Market growth in this region is primarily driven by an increase in NGS-based research, an improving clinical reimbursement structure, and government investments to support genomics research. In May 2019, Novigenix announced plans to develop a new version of Colox, its mRNA-based assay for early colorectal cancer detection that applies next-generation sequencing. The company partnered with six European collaborators, including the Swiss Federal Institute of Technology, the Lausanne University Hospital, and the Swiss Institute of Bioinformatics, to recruit 1,600 individuals for CRC screening as part of the validation study. In January 2019, the Novo Nordisk Foundation approved a framework grant of USD 157.8 million over 4.5 years to establish and operate the infrastructure of the National Genome Centre. Such developments are expected to drive market growth in this region.

TABLE 127 REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	8.86	10.04	11.36	12.87	14.60	16.62	18.97	21.76	13.9%
Sequencing and Library Preparation	12.00	13.68	15.59	17.77	20.28	23.23	26.69	30.80	14.6%
Data Analysis	3.76	4.21	4.72	5.29	5.93	6.66	7.52	8.52	12.5%
Total	**24.62**	**27.93**	**31.67**	**35.93**	**40.81**	**46.50**	**53.18**	**61.08**	**14.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 128 REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	10.22	11.64	13.24	15.08	17.18	19.65	22.55	25.99	14.4%
Whole-genome Sequencing	5.32	5.98	6.72	7.55	8.50	9.59	10.86	12.35	12.9%
Shotgun Sequencing	6.60	7.57	8.67	9.95	11.42	13.14	15.18	17.62	15.2%
Other Types	2.48	2.74	3.03	3.36	3.71	4.12	4.58	5.12	11.0%
Total	**24.62**	**27.93**	**31.67**	**35.93**	**40.81**	**46.50**	**53.18**	**61.08**	**14.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 129 REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	11.11	12.62	14.34	16.30	18.54	21.16	24.24	27.89	14.2%
Sequencing by Ligation	3.16	3.58	4.05	4.59	5.19	5.91	6.74	7.72	13.8%
Nanopore Sequencing	1.26	1.49	1.75	2.06	2.43	2.86	3.39	4.02	18.1%
Other Technologies	9.09	10.24	11.53	12.99	14.65	16.57	18.81	21.44	13.2%
Total	**24.62**	**27.93**	**31.67**	**35.93**	**40.81**	**46.50**	**53.18**	**61.08**	**14.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 130 REST OF EUROPE: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	7.79	8.93	10.23	11.73	13.46	15.50	17.91	20.77	15.2%
Academic and Research Institutes	11.14	12.59	14.22	16.08	18.20	20.66	23.55	26.94	13.6%
Other End Users	5.70	6.41	7.21	8.12	9.15	10.34	11.73	13.36	13.1%
Total	**24.62**	**27.93**	**31.67**	**35.93**	**40.81**	**46.50**	**53.18**	**61.08**	**14.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.3.7 EUROPE: IMPACT OF RECESSION

Europe was expected to witness an economic recession in the last quarter of 2022, according to the European Commission's autumn economic forecast. In Europe, inflation is expected to have a higher impact on all the major economies, such as the UK, Germany, France, Italy, and Spain. The region is expected to experience a recession till mid-2024.

Healthcare and pharmaceutical spending in European countries has been increasing consistently. The coming recession is expected to impact the healthcare and pharmaceutical industry with reduced health budgets, workers' wages, healthcare provision, pharmaceutical spending, and increased hours worked by healthcare professionals. Thus, there will be a slight impact on the healthcare market, including on sequencing technologies, with a slow market growth rate in 2023. However, the market growth rate is expected to recover in 2025-26.

This impact will be balanced by other factors, such as the expansion of business operations in Europe, an increase in clinical trials, and growing drug pipelines. International pharmaceutical companies are working toward expanding their operations in Europe. For instance, in December 2022, Pfizer announced an investment of USD 1.2 billion for its manufacturing site in Puurs, Belgium. This expansion will occur over the next three years, with some projects already started while others are scheduled to begin in early 2023. Such developments will help European healthcare markets to withstand the effects of an economic downturn.

10.4 ASIA PACIFIC

The Asia Pacific microbiome sequencing services market is estimated to reach USD 57.19 million by 2028 from USD 25.73 million in 2023, at a CAGR of 17.3% during the forecast period. In 2022, China accounted for the largest share of 42.3% of the microbiome sequencing services market in the Asia Pacific, followed by Japan with a share of 19.8%.

The Asia Pacific region is expected to be the fastest-growing market for microbiome sequencing services due to the increasing awareness of the importance of the microbiome in health and disease, rising demand for personalized medicine, and advancements in sequencing technologies. Another key driver for the Asia Pacific microbiome sequencing services market is the region's large and rapidly aging population. As people age, they become more susceptible to a range of diseases and conditions that may be influenced by the microbiome. This has led to an increased focus on preventative healthcare and personalized medicine, which is driving the demand for microbiome sequencing services. The development of high-throughput sequencing technologies has made it possible to sequence the entire microbiome in a single sample, enabling researchers to more accurately and comprehensively characterize the microbial communities that are present in samples.

Challenges in terms of data protection and privacy, as well as the high cost of microbiome sequencing services, are expected to restrain the growth of the Asia Pacific market. In addition, there may be cultural and linguistic barriers to adoption in certain regions, as well as concerns about the accuracy and clinical utility of microbiome sequencing data.

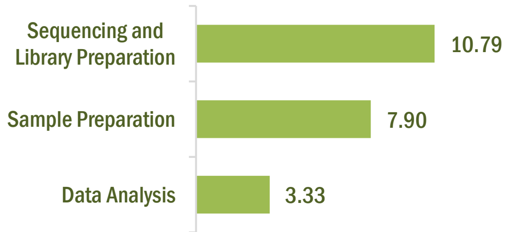
FIGURE 27 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET SNAPSHOT

ASIA PACIFIC

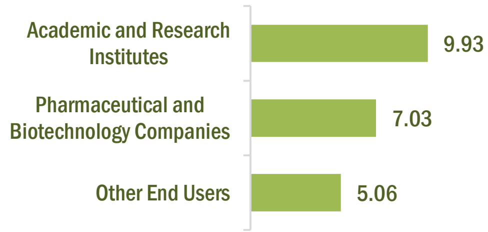
8.8%
Share of the Region in the Global Market in 2022

Japan
to register the highest CAGR of 18.0% during the forecast period

BY SERVICE, 2022 (USD MILLION)



BY END USER, 2022 (USD MILLION)



BY COUNTRY, 2022 (USD MILLION)

COUNTRY/ REGION	MARKET SIZE	CAGR (2023–2028)
China	9.31	17.4%
Japan	4.36	18.0%
India	0.55	17.1%
RoAPAC	7.79	16.9%

DRIVING FACTORS FOR GROWTH IN ASIA PACIFIC

- Increasing awareness of the importance of the microbiome in health and diseases
- Rising demand for personalized medicine
- Advancements in sequencing technologies

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 131 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	7.96	9.31	10.89	12.73	14.89	17.46	20.54	24.24	17.4%
Japan	3.70	4.36	5.12	6.02	7.08	8.34	9.86	11.70	18.0%
India	0.47	0.55	0.64	0.75	0.87	1.02	1.20	1.41	17.1%
Rest of Asia Pacific	6.68	7.79	9.08	10.57	12.32	14.39	16.86	19.83	16.9%
Total	**18.81**	**22.01**	**25.73**	**30.07**	**35.16**	**41.22**	**48.46**	**57.19**	**17.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 132 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	6.76	7.90	9.22	10.76	12.56	14.71	17.27	20.35	17.2%
Sequencing and Library Preparation	9.17	10.79	12.68	14.91	17.53	20.67	24.44	29.01	18.0%
Data Analysis	2.89	3.33	3.83	4.40	5.06	5.84	6.75	7.83	15.4%
Total	**18.81**	**22.01**	**25.73**	**30.07**	**35.16**	**41.22**	**48.46**	**57.19**	**17.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 133 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	7.79	9.16	10.75	12.63	14.83	17.47	20.63	24.46	17.9%
Whole-genome Sequencing	4.08	4.72	5.45	6.30	7.28	8.43	9.80	11.42	15.9%
Shotgun Sequencing	5.04	5.97	7.07	8.36	9.89	11.74	13.96	16.67	18.7%
Other Types	1.91	2.17	2.46	2.79	3.16	3.59	4.08	4.65	13.6%
Total	**18.81**	**22.01**	**25.73**	**30.07**	**35.16**	**41.22**	**48.46**	**57.19**	**17.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 134 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	8.47	9.93	11.63	13.62	15.96	18.75	22.09	26.12	17.6%
Sequencing by Ligation	2.42	2.83	3.29	3.84	4.48	5.23	6.13	7.22	17.0%
Nanopore Sequencing	0.96	1.18	1.44	1.76	2.14	2.62	3.20	3.92	22.2%
Other Technologies	6.96	8.08	9.36	10.85	12.58	14.62	17.04	19.94	16.3%
Total	**18.81**	**22.01**	**25.73**	**30.07**	**35.16**	**41.22**	**48.46**	**57.19**	**17.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 135 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	5.92	7.03	8.33	9.86	11.68	13.87	16.52	19.74	18.8%
Academic and Research Institutes	8.52	9.93	11.55	13.44	15.64	18.25	21.36	25.09	16.8%
Other End Users	4.37	5.06	5.85	6.77	7.84	9.10	10.59	12.36	16.1%
Total	**18.81**	**22.01**	**25.73**	**30.07**	**35.16**	**41.22**	**48.46**	**57.19**	**17.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.1 CHINA

10.4.1.1 Increasing investments in R&D to drive growth

The microbiome sequencing services market in China has seen significant growth in recent years. The rapidly expanding healthcare industry in China has led to increased investment in the research and development of new microbiome sequencing technologies. China is home to one of the world's fastest-growing biotechnology markets. In 2020, Chinese genetic testing companies in the primary market raised over RMB 19.3 billion (USD 3 billion) in financing—an annual growth rate of 153% compared to 2019.

The rising incidence of chronic diseases in China, such as cancer and diabetes, has fueled the demand for more accurate and efficient diagnostic tools. Additionally, the Chinese government has placed a strong emphasis on promoting the development of the biotechnology and life science industries, which has created opportunities for companies in the microbiome sequencing services market to receive funding and support for R&D efforts. This has also encouraged collaborations between academic institutions and private companies in China, which has facilitated the development of new technologies and therapies. For instance, Biomica Ltd. announced the signing of a definitive agreement for a USD 20 million financing round to be led by Shanghai Healthcare Capital (SHC). The financing is subject to customary closing conditions, including clearance by Chinese regulatory authorities. The financing round will enable Biomica to forge ahead, developing its pipeline of microbiome-based therapeutics. Moreover, Siolta Therapeutics announced that it raised USD 30 million in a series B investment round to support the clinical development of its lead product, STMC-103H. The series B funding involved participation from top-tier investors and venture firms, including series A investors Khosla Ventures and Marc Benioff, as well as new investors such as Seventure (Health for Life Capital Fund), SymBiosis, and Global Brain (Kirin Health Innovation Fund/GB-VII). Siolta's live biotherapeutic product (LBP) platform is designed to develop microbiome-based medicines and diagnostics for the prevention and treatment of chronic diseases in targeted populations.

One of the main restraints for the microbiome sequencing services market in China is the lack of regulatory oversight and standardization in the industry. This has created uncertainty for investors and may hinder the growth of the market in the long term. Additionally, the high cost of technology and limited reimbursement options may limit the accessibility of microbiome sequencing services among patients.

Despite these challenges, there still are several opportunities for the growth of the microbiome sequencing services market in China. For instance, the growing demand for personalized medicine and precision healthcare has increased the focus on individualized microbiome analysis, which has the potential to drive the demand for microbiome sequencing services. The rapid growth of the Chinese middle class has led to an increased demand for high-quality healthcare services, which could further accelerate the growth of the microbiome sequencing services market.

TABLE 136 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	2.86	3.34	3.90	4.56	5.32	6.23	7.32	8.63	17.2%
Sequencing and Library Preparation	3.88	4.56	5.36	6.30	7.41	8.74	10.33	12.26	18.0%
Data Analysis	1.22	1.41	1.62	1.87	2.15	2.49	2.89	3.36	15.7%
Total	**7.96**	**9.31**	**10.89**	**12.73**	**14.89**	**17.46**	**20.54**	**24.24**	**17.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 137 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	3.30	3.88	4.55	5.34	6.27	7.39	8.72	10.34	17.8%
Whole-genome Sequencing	1.72	2.00	2.31	2.67	3.09	3.59	4.17	4.88	16.1%
Shotgun Sequencing	2.13	2.52	2.99	3.53	4.18	4.95	5.89	7.02	18.7%
Other Types	0.80	0.92	1.04	1.19	1.35	1.53	1.75	2.00	14.0%
Total	**7.96**	**9.31**	**10.89**	**12.73**	**14.89**	**17.46**	**20.54**	**24.24**	**17.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 138 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	3.58	4.21	4.93	5.77	6.77	7.96	9.38	11.09	17.6%
Sequencing by Ligation	1.02	1.19	1.39	1.62	1.89	2.21	2.58	3.04	16.9%
Nanopore Sequencing	0.41	0.50	0.61	0.74	0.90	1.09	1.33	1.63	21.9%
Other Technologies	2.94	3.42	3.96	4.60	5.34	6.21	7.25	8.49	16.4%
Total	**7.96**	**9.31**	**10.89**	**12.73**	**14.89**	**17.46**	**20.54**	**24.24**	**17.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 139 CHINA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	2.51	2.98	3.52	4.16	4.93	5.84	6.95	8.30	18.7%
Academic and Research Institutes	3.60	4.20	4.89	5.69	6.63	7.75	9.07	10.67	16.9%
Other End Users	1.84	2.14	2.48	2.87	3.33	3.87	4.51	5.28	16.3%
Total	**7.96**	**9.31**	**10.89**	**12.73**	**14.89**	**17.46**	**20.54**	**24.24**	**17.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.2 JAPAN

10.4.2.1 Significant growth in geriatric population to present opportunities for market growth

The microbiome sequencing services market in Japan has been growing rapidly in recent years, driven by several factors. One key driver is the increasing prevalence of chronic diseases, such as cancer, diabetes, and IBD, which have been linked to dysbiosis or imbalances in the gut microbiome. This has led to a growing interest in the potential of microbiome analysis for personalized medicine and disease diagnosis and treatment. Another driver is the development of microbiome-based therapeutics, such as fecal microbiota transplantation (FMT), which have been approved in Japan for the treatment of recurrent Clostridioides difficile infections.

In addition to these drivers, Japan's aging population presents a unique opportunity for microbiome sequencing services. Japan's population is aging and shrinking fast. With a median age of 48.4 years, Japan's population is the oldest in the world. The government of Japan projects that there will be almost one elderly person for each person of working age by 2060. Over 29.18% of the total population is 65 years or older. As the population ages, the incidence of age-related diseases such as dementia and Parkinson's disease is increasing, and there is a growing interest in the potential of the gut microbiome to play a role in these conditions. Japan's highly developed healthcare system and advanced research infrastructure also make it an attractive market for microbiome sequencing services.

However, there are several restraints to the growth of the microbiome sequencing services market in Japan. One is the lack of public awareness and understanding of the microbiome and its potential applications in healthcare. This has led to the slow adoption of microbiome analysis services by both patients and healthcare providers. In addition, regulatory hurdles and a lack of reimbursement for microbiome analysis services have hindered the growth of this market.

Despite these challenges, the increasing availability of NGS technologies and bioinformatics tools, which have greatly improved the speed and accuracy of microbiome analysis, are expected to offer potential growth opportunities for market growth in Japan.

TABLE 140 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.33	1.56	1.84	2.15	2.53	2.98	3.51	4.16	17.8%
Sequencing and Library Preparation	1.80	2.14	2.52	2.98	3.53	4.18	4.97	5.93	18.6%
Data Analysis	0.57	0.66	0.76	0.88	1.02	1.18	1.38	1.61	16.1%
Total	**3.70**	**4.36**	**5.12**	**6.02**	**7.08**	**8.34**	**9.86**	**11.70**	**18.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 141 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.53	1.81	2.14	2.53	2.98	3.53	4.20	5.00	18.5%
Whole-genome Sequencing	0.80	0.93	1.09	1.26	1.47	1.71	2.00	2.34	16.6%
Shotgun Sequencing	0.99	1.18	1.41	1.67	1.99	2.37	2.84	3.41	19.3%
Other Types	0.37	0.43	0.49	0.56	0.64	0.73	0.83	0.95	14.3%
Total	**3.70**	**4.36**	**5.12**	**6.02**	**7.08**	**8.34**	**9.86**	**11.70**	**18.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 142 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.67	1.97	2.32	2.73	3.21	3.80	4.50	5.34	18.2%
Sequencing by Ligation	0.48	0.56	0.66	0.77	0.90	1.06	1.25	1.47	17.6%
Nanopore Sequencing	0.19	0.23	0.29	0.35	0.43	0.53	0.65	0.80	22.8%
Other Technologies	1.37	1.60	1.86	2.17	2.53	2.96	3.47	4.08	17.0%
Total	**3.70**	**4.36**	**5.12**	**6.02**	**7.08**	**8.34**	**9.86**	**11.70**	**18.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 143 JAPAN: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	1.17	1.39	1.66	1.97	2.35	2.80	3.36	4.03	19.5%
Academic and Research Institutes	1.68	1.97	2.30	2.69	3.15	3.70	4.35	5.14	17.4%
Other End Users	0.86	1.00	1.17	1.36	1.58	1.84	2.16	2.53	16.8%
Total	**3.70**	**4.36**	**5.12**	**6.02**	**7.08**	**8.34**	**9.86**	**11.70**	**18.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.3 INDIA

10.4.3.1 Increasing prevalence of chronic diseases to support growth

The microbiome sequencing services market in India is experiencing significant growth due to several drivers. One of the major drivers is the increasing prevalence of chronic diseases in the country. According to the World Health Organization, in 2019, non-communicable diseases such as cardiovascular diseases, diabetes, and cancer accounted for 66% of all deaths in India. The microbiome has been shown to play a crucial role in the development and progression of these diseases, making microbiome sequencing services an important tool in disease prevention and management.

Another driver is the growing demand for personalized medicine. With advancements in microbiome research, there is a growing understanding of the unique microbiome profiles of individuals and how they can impact their health. Microbiome sequencing services can provide personalized insights into an individual’s microbiome composition and help in developing personalized treatment plans.

Government initiatives and funding programs are also driving the growth of the microbiome sequencing services market in India. The Department of Biotechnology (DBT) has launched several programs to promote microbiome research in the country, such as the Human Microbiome Project and the Microbial

Culture Collection program. These initiatives have created opportunities for the development of new microbiome sequencing services and technologies in India.

One of the major restraints for market growth in India is the lack of awareness and understanding of the microbiome among the public and healthcare professionals. This is limiting the adoption of microbiome sequencing services and their incorporation into clinical practice.

TABLE 144 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.17	0.20	0.23	0.27	0.31	0.36	0.43	0.50	16.9%
Sequencing and Library Preparation	0.23	0.27	0.32	0.37	0.44	0.51	0.61	0.72	17.8%
Data Analysis	0.07	0.08	0.10	0.11	0.13	0.14	0.17	0.19	15.0%
Total	**0.47**	**0.55**	**0.64**	**0.75**	**0.87**	**1.02**	**1.20**	**1.41**	**17.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 145 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.19	0.23	0.27	0.31	0.37	0.43	0.51	0.60	17.7%
Whole-genome Sequencing	0.10	0.12	0.14	0.16	0.18	0.21	0.24	0.28	15.6%
Shotgun Sequencing	0.13	0.15	0.18	0.21	0.25	0.29	0.35	0.41	18.5%
Other Types	0.05	0.05	0.06	0.07	0.08	0.09	0.10	0.11	13.2%
Total	**0.47**	**0.55**	**0.64**	**0.75**	**0.87**	**1.02**	**1.20**	**1.41**	**17.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 146 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	0.21	0.25	0.29	0.34	0.40	0.46	0.55	0.64	17.3%
Sequencing by Ligation	0.06	0.07	0.08	0.10	0.11	0.13	0.15	0.18	16.8%
Nanopore Sequencing	0.02	0.03	0.04	0.04	0.05	0.07	0.08	0.10	22.2%
Other Technologies	0.17	0.20	0.23	0.27	0.31	0.36	0.42	0.49	16.0%
Total	**0.47**	**0.55**	**0.64**	**0.75**	**0.87**	**1.02**	**1.20**	**1.41**	**17.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 147 INDIA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.15	0.18	0.21	0.25	0.29	0.35	0.41	0.49	18.7%
Academic and Research Institutes	0.21	0.25	0.29	0.33	0.39	0.45	0.53	0.62	16.5%
Other End Users	0.11	0.13	0.15	0.17	0.19	0.23	0.26	0.30	15.8%
Total	**0.47**	**0.55**	**0.64**	**0.75**	**0.87**	**1.02**	**1.20**	**1.41**	**17.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.4 REST OF ASIA PACIFIC

The RoAPAC includes Australia, Taiwan, New Zealand, Thailand, South Korea, Singapore, and Malaysia, among other Asia Pacific countries. Factors such as public and private initiatives to support the adoption and development of microbiome technologies, increasing R&D expenditure, and increasing awareness of the diagnostic applications of microbiome sequencing are driving the growth of the RoAPAC market.

In September 2019, the South Korean government proposed plans to increase the budget of government subsidies in the biotech sector by 16% and finance a project to secure the genome data of 1 million people. The government aims to increase its budget spending to help the pharmaceutical sector develop innovative new drugs and medical devices. As per this agenda, the government will invest USD 951.8 million in R&D projects in the sector in 2020, with an increase of 16% in 2019. Over the past few years, a few events focusing on spreading awareness about the benefits of microbiome sequencing and the latest advancements in this field have been organized in the RoAPAC region.

In June 2022, Singapore-based precision gut microbiome company AMILI received USD 10.5 million in a Series A funding round led by technology and life sciences investor Vulcan Capital. New investors also joined the round, including Pruksa Group, TVM Capital Healthcare, Emtek Group, Capital Code, Pureland Group, Blue7, and Enterprise Singapore's investment arm SEEDS Capital.

TABLE 148 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	2.40	2.79	3.25	3.78	4.40	5.13	6.00	7.05	16.8%
Sequencing and Library Preparation	3.25	3.82	4.48	5.25	6.16	7.24	8.54	10.10	17.7%
Data Analysis	1.03	1.18	1.35	1.54	1.76	2.02	2.32	2.68	14.7%
Total	**6.68**	**7.79**	**9.08**	**10.57**	**12.32**	**14.39**	**16.86**	**19.83**	**16.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 149 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	2.76	3.24	3.79	4.44	5.20	6.11	7.20	8.51	17.6%
Whole-genome Sequencing	1.45	1.67	1.92	2.21	2.54	2.93	3.38	3.92	15.3%
Shotgun Sequencing	1.79	2.12	2.50	2.95	3.48	4.12	4.89	5.82	18.5%
Other Types	0.68	0.77	0.87	0.97	1.10	1.23	1.39	1.57	12.7%
Total	**6.68**	**7.79**	**9.08**	**10.57**	**12.32**	**14.39**	**16.86**	**19.83**	**16.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 150 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	3.01	3.51	4.10	4.78	5.58	6.53	7.67	9.03	17.1%
Sequencing by Ligation	0.86	1.00	1.17	1.36	1.58	1.84	2.15	2.53	16.7%
Nanopore Sequencing	0.34	0.42	0.51	0.63	0.76	0.93	1.14	1.40	22.2%
Other Technologies	2.48	2.86	3.30	3.81	4.40	5.09	5.90	6.88	15.8%
Total	**6.68**	**7.79**	**9.08**	**10.57**	**12.32**	**14.39**	**16.86**	**19.83**	**16.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 151 REST OF ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	2.10	2.49	2.94	3.48	4.11	4.88	5.80	6.92	18.7%
Academic and Research Institutes	3.03	3.52	4.07	4.72	5.47	6.36	7.41	8.66	16.3%
Other End Users	1.55	1.79	2.06	2.38	2.74	3.16	3.66	4.25	15.6%
Total	**6.68**	**7.79**	**9.08**	**10.57**	**12.32**	**14.39**	**16.86**	**19.83**	**16.9%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.4.5 ASIA PACIFIC: IMPACT OF RECESSION

The impact of the recession on the global market would affect the Asia Pacific region as well. The Association of Southeast Asian Nations (ASEAN) projected a steady decline in the economic growth of Asia Pacific countries owing to the slower growth of their trading partners, namely, Europe, the US, and China. The overall GDP decline in the coming years is expected to affect the healthcare expenditure, spending, and wages in the industry with reduced import and export activities.

However, in Asia, countries focus on increasing their R&D in the pharmaceutical and biotechnology industry. India has seen a progression in terms of its R&D funding and rising government support. According to a report by India Brand Equity Foundation (IBEF), India's domestic pharmaceutical market was estimated at USD 42 billion in 2021. It is likely to reach USD 65 billion by 2024 and may further expand to reach USD 120–130 billion by 2030. Sun Pharma, a leading pharmaceutical company in India, announced that it intends to invest USD 600–650 million in R&D. On the other hand, Chinese pharmaceutical companies invested nearly 10% of their sales revenue in R&D in 2021.

For the overall Asia Pacific region, the impact of the recession is expected to be low during the forecast period. Japan’s economy is expected to be more stable than other Asian countries. However, price inflation leading to lower consumption would lead to a slight decline in Japan’s economic growth in 2023.

10.5 LATIN AMERICA

The Latin American microbiome sequencing services market is expected to reach USD 8.88 million by 2028 from USD 5.26 million in 2023, at a CAGR of 11.1% from 2023 to 2028.

The Latin American microbiome sequencing services market has been growing steadily in recent years. The region is witnessing an increased prevalence of chronic diseases, such as cancer, IBD, and diabetes, which have been linked to imbalances in the gut microbiome. As a result, there is a growing demand for microbiome sequencing services to better understand and manage these conditions. Additionally, the region's large and diverse population provides a unique opportunity for microbiome research and discovery.

The lack of awareness and understanding of the potential benefits of microbiome sequencing among healthcare providers and the general population is a key factor restraining the growth of this regional market. This has resulted in a relatively small market size and limited investment in the field. Additionally, the high cost of sequencing and the lack of standardization and regulation of microbiome testing can be a barrier to entry for smaller companies.

TABLE 152 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	2.01	2.23	2.47	2.74	3.04	3.37	3.76	4.20	11.1%
Rest of Latin America	2.28	2.52	2.79	3.08	3.41	3.78	4.20	4.69	11.0%
Total	**4.29**	**4.75**	**5.26**	**5.82**	**6.44**	**7.15**	**7.96**	**8.88**	**11.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 153 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.54	1.70	1.88	2.08	2.30	2.55	2.83	3.16	10.9%
Sequencing and Library Preparation	2.09	2.33	2.60	2.90	3.23	3.61	4.04	4.54	11.8%
Data Analysis	0.66	0.72	0.78	0.85	0.92	1.00	1.09	1.19	8.7%
Total	**4.29**	**4.75**	**5.26**	**5.82**	**6.44**	**7.15**	**7.96**	**8.88**	**11.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 154 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.77	1.97	2.20	2.45	2.72	3.04	3.40	3.82	11.7%
Whole-genome Sequencing	0.93	1.02	1.11	1.21	1.32	1.45	1.59	1.74	9.4%
Shotgun Sequencing	1.15	1.29	1.45	1.63	1.83	2.06	2.32	2.63	12.6%
Other Types	0.44	0.47	0.50	0.53	0.57	0.61	0.65	0.69	6.6%
Total	**4.29**	**4.75**	**5.26**	**5.82**	**6.44**	**7.15**	**7.96**	**8.88**	**11.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 155 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.93	2.14	2.37	2.63	2.92	3.25	3.62	4.05	11.3%
Sequencing by Ligation	0.55	0.61	0.68	0.75	0.82	0.91	1.01	1.13	10.8%
Nanopore Sequencing	0.22	0.26	0.30	0.35	0.41	0.47	0.55	0.64	16.5%
Other Technologies	1.59	1.75	1.91	2.09	2.29	2.52	2.77	3.07	9.9%
Total	**4.29**	**4.75**	**5.26**	**5.82**	**6.44**	**7.15**	**7.96**	**8.88**	**11.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 156 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	1.34	1.51	1.71	1.92	2.16	2.44	2.76	3.13	12.9%
Academic and Research Institutes	1.95	2.15	2.36	2.60	2.86	3.15	3.48	3.87	10.4%
Other End Users	1.00	1.09	1.19	1.31	1.43	1.56	1.72	1.89	9.6%
Total	**4.29**	**4.75**	**5.26**	**5.82**	**6.44**	**7.15**	**7.96**	**8.88**	**11.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.5.1 BRAZIL

10.5.1.1 Growing adoption of next-generation sequencing technologies to propel growth

Brazil has been increasingly adopting NGS technologies for microbiome sequencing, which offer higher throughput, greater accuracy, and lower cost compared to traditional sequencing methods. This has led to the development of new microbiome sequencing services and the expansion of existing ones in the country.

The increasing prevalence of chronic diseases such as diabetes, obesity, and IBD is another major driver for market growth in Brazil, as it has created a high demand for microbiome sequencing services to better understand the role of the microbiome in disease development and progression.

TABLE 157 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.72	0.80	0.89	0.98	1.08	1.20	1.34	1.49	11.0%
Sequencing and Library Preparation	0.98	1.10	1.22	1.36	1.52	1.70	1.91	2.14	11.9%
Data Analysis	0.31	0.34	0.37	0.40	0.43	0.47	0.51	0.56	8.9%
Total	**2.01**	**2.23**	**2.47**	**2.74**	**3.04**	**3.37**	**3.76**	**4.20**	**11.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 158 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.83	0.93	1.03	1.15	1.28	1.43	1.61	1.80	11.8%
Whole-genome Sequencing	0.44	0.48	0.52	0.57	0.62	0.68	0.75	0.83	9.5%
Shotgun Sequencing	0.54	0.61	0.68	0.77	0.86	0.97	1.09	1.24	12.7%
Other Types	0.21	0.22	0.24	0.25	0.27	0.29	0.31	0.33	6.9%
Total	**2.01**	**2.23**	**2.47**	**2.74**	**3.04**	**3.37**	**3.76**	**4.20**	**11.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 159 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	0.91	1.01	1.12	1.24	1.38	1.53	1.71	1.92	11.4%
Sequencing by Ligation	0.26	0.29	0.32	0.35	0.39	0.43	0.48	0.53	10.8%
Nanopore Sequencing	0.10	0.12	0.14	0.16	0.19	0.22	0.26	0.30	16.5%
Other Technologies	0.75	0.82	0.90	0.99	1.08	1.19	1.31	1.45	10.0%
Total	**2.01**	**2.23**	**2.47**	**2.74**	**3.04**	**3.37**	**3.76**	**4.20**	**11.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 160 BRAZIL: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.63	0.71	0.80	0.90	1.02	1.15	1.30	1.47	12.9%
Academic and Research Institutes	0.91	1.01	1.11	1.22	1.35	1.49	1.65	1.83	10.5%
Other End Users	0.47	0.51	0.56	0.62	0.67	0.74	0.81	0.89	9.7%
Total	**2.01**	**2.23**	**2.47**	**2.74**	**3.04**	**3.37**	**3.76**	**4.20**	**11.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.5.2 REST OF LATIN AMERICA

The Rest of Latin America mainly comprises Argentina, Colombia, Costa Rica, and Peru. The microbiome sequencing services market in the Rest of Latin America is estimated to reach USD 4.69 million by 2028 from USD 2.79 million in 2023, at a CAGR of 11.0% from 2023 to 2028.

Market growth in this region will be primarily driven by the awareness of and acceptance of personalized medicine, the increasing prevalence of chronic diseases (such as diabetes, cancer, and gastrointestinal disorders), and increasing investment in genomics research by governments and private organizations. However, certain regulatory barriers, such as complex approval processes, may create delays or difficulties in launching new microbiome sequencing services. Moreover, there may be a shortage of qualified professionals with the necessary expertise to perform and interpret microbiome sequencing.

TABLE 161 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	0.82	0.90	1.00	1.10	1.21	1.35	1.49	1.66	10.8%
Sequencing and Library Preparation	1.11	1.24	1.38	1.53	1.71	1.91	2.14	2.40	11.8%
Data Analysis	0.35	0.38	0.41	0.45	0.48	0.53	0.57	0.62	8.5%
Total	**2.28**	**2.52**	**2.79**	**3.08**	**3.41**	**3.78**	**4.20**	**4.69**	**11.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 162 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	0.94	1.04	1.16	1.29	1.44	1.61	1.80	2.02	11.7%
Whole-genome Sequencing	0.50	0.54	0.59	0.64	0.70	0.76	0.84	0.92	9.2%
Shotgun Sequencing	0.61	0.68	0.77	0.86	0.97	1.09	1.23	1.39	12.6%
Other Types	0.23	0.25	0.27	0.28	0.30	0.32	0.34	0.36	6.4%
Total	**2.28**	**2.52**	**2.79**	**3.08**	**3.41**	**3.78**	**4.20**	**4.69**	**11.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 163 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.02	1.13	1.26	1.39	1.54	1.71	1.91	2.13	11.2%
Sequencing by Ligation	0.29	0.32	0.36	0.39	0.44	0.48	0.53	0.60	10.7%
Nanopore Sequencing	0.12	0.14	0.16	0.19	0.22	0.25	0.29	0.34	16.6%
Other Technologies	0.84	0.93	1.01	1.11	1.21	1.33	1.46	1.61	9.8%
Total	**2.28**	**2.52**	**2.79**	**3.08**	**3.41**	**3.78**	**4.20**	**4.69**	**11.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 164 REST OF LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.71	0.80	0.90	1.02	1.14	1.29	1.46	1.66	12.9%
Academic and Research Institutes	1.03	1.14	1.25	1.37	1.51	1.66	1.84	2.04	10.3%
Other End Users	0.53	0.58	0.63	0.69	0.75	0.82	0.90	0.99	9.5%
Total	**2.28**	**2.52**	**2.79**	**3.08**	**3.41**	**3.78**	**4.20**	**4.69**	**11.0%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.5.3 LATIN AMERICA: IMPACT OF RECESSION

A US recession would affect Latin America's real GDP outlook by impacting trade and the overall economy. GDP growth of 3.3% in 2022 is expected to slow by 1.5% in 2023 in Latin America. However, government bodies and pharmaceutical companies are expanding the pharmaceutical and biotechnology industry in countries such as Brazil. In October 2022, Luis Inácio Lula da Silva, one of the candidates in the presidential election and winner of Brazil's first-round elections, pledged to increase healthcare spending by investing in the Unified Health System (SUS) and health infrastructure, thereby expanding the domestic pharmaceutical industry. In November 2022, MedQuímica Indústria Farmacêutica (MedQuímica), a wholly owned subsidiary of Lupin in Brazil, signed a definitive agreement to acquire all rights to nine medicines from BL Indústria Ótica Ltda., a subsidiary of Bausch Health Companies Inc., to enhance its product portfolio in Brazil. Such initiatives are likely to encourage the growth of the pharmaceutical and biotechnology industry in 2023 and 2024.

Inflation has reached an inflection point in most countries in Latin America with tighter monetary conditions, stabilizing prices for energy, health and food commodities, transportation costs, and supply chain disruptions. All these factors would create an impact on the pharmaceutical and biotechnology industry and related technologies, such as metagenomic sequencing. Market growth is expected to slow in 2023-24 and will eventually start to stabilize and recover in 2025-26.

10.6 MIDDLE EAST & AFRICA

The microbiome sequencing services market in the Middle East & Africa is projected to reach USD 5.55 million by 2028 from USD 3.55 million in 2023, at a CAGR of 9.3% from 2023 to 2028.

The increasing awareness and understanding of the importance of the microbiome in human health and diseases are contributing to the growth of this regional market. However, the market growth is likely to be restrained by factors such as the high costs associated with microbiome sequencing services and a lack of skilled professionals in the region. Despite these challenges, there are several opportunities for market growth in MEA, including the potential for increased investment in research and development, partnerships and collaborations between academic and industrial sectors, and the development of novel technologies and analytical tools.

TABLE 165 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	1.06	1.16	1.27	1.38	1.51	1.65	1.80	1.97	9.2%
Sequencing and Library Preparation	1.45	1.60	1.76	1.93	2.13	2.34	2.58	2.85	10.2%
Data Analysis	0.46	0.49	0.53	0.56	0.60	0.64	0.68	0.73	6.8%
Total	**2.97**	**3.25**	**3.55**	**3.88**	**4.23**	**4.63**	**5.06**	**5.55**	**9.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 166 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TYPE, 2021–2028 (USD MILLION)

Type	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Amplicon Sequencing	1.22	1.35	1.48	1.63	1.79	1.97	2.17	2.40	10.1%
Whole-genome Sequencing	0.65	0.70	0.75	0.81	0.87	0.93	1.00	1.08	7.5%
Shotgun Sequencing	0.79	0.88	0.98	1.09	1.21	1.34	1.48	1.65	11.0%
Other Types	0.31	0.32	0.34	0.35	0.37	0.39	0.40	0.42	4.6%
Total	**2.97**	**3.25**	**3.55**	**3.88**	**4.23**	**4.63**	**5.06**	**5.55**	**9.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 167 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY TECHNOLOGY, 2021–2028 (USD MILLION)

Technology	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sequencing by Synthesis	1.33	1.46	1.60	1.75	1.91	2.10	2.30	2.52	9.5%
Sequencing by Ligation	0.38	0.42	0.46	0.50	0.54	0.59	0.65	0.71	9.2%
Nanopore Sequencing	0.15	0.18	0.20	0.24	0.27	0.31	0.36	0.41	15.2%
Other Technologies	1.10	1.19	1.29	1.39	1.50	1.63	1.76	1.91	8.1%
Total	**2.97**	**3.25**	**3.55**	**3.88**	**4.23**	**4.63**	**5.06**	**5.55**	**9.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 168 MIDDLE EAST & AFRICA: MICROBIOME SEQUENCING SERVICES MARKET, BY END USER, 2021–2028 (USD MILLION)

End User	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies	0.93	1.04	1.15	1.28	1.43	1.59	1.77	1.97	11.3%
Academic and Research Institutes	1.35	1.47	1.59	1.73	1.87	2.03	2.21	2.41	8.6%
Other End Users	0.69	0.75	0.81	0.87	0.93	1.01	1.08	1.17	7.7%
Total	**2.97**	**3.25**	**3.55**	**3.88**	**4.23**	**4.63**	**5.06**	**5.55**	**9.3%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

10.6.1 MIDDLE EAST & AFRICA: IMPACT OF RECESSION

The Middle East and Africa (MEA) region has experienced a significant economic slowdown in recent years, which has impacted the healthcare sector. The region has been hit by a combination of factors, including falling oil prices and political instability. The economic recession is having a severe impact on the healthcare industry, with healthcare spending reduced in many countries. Governments have been forced to reduce spending on healthcare due to budget constraints, and private healthcare providers have struggled to maintain profitability. This has led to a reduction in investment in healthcare infrastructure and a decline in the quality of care in some areas. However, there also are opportunities for growth in the MEA healthcare sector, particularly in the areas of digital health and telemedicine. The rising government initiatives to improve healthcare infrastructure and provide better access to advanced diagnostic tools in the region could also drive market growth. For instance, the UAE government has launched several initiatives to promote innovation in healthcare, including supporting research and development activities in the field of microbiome sequencing. These initiatives are expected to create a favorable environment for the growth of the microbiome sequencing services market in the region.

11 COMPETITIVE LANDSCAPE

11.1 OVERVIEW

The competitive landscape includes an analysis of the key growth strategies adopted by major players between October 2020 and April 2023. Players in the microbiome sequencing services market have employed various strategies to expand their footprints and increase their market shares. The key strategies adopted by major players in the microbiome sequencing services market include expansions, service launches, acquisitions, agreements, and collaborations.

Prominent players in this market include Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea).

11.2 STRATEGIES ADOPTED BY KEY PLAYERS

This section includes an in-depth analysis of the key developments undertaken by top players in this market during the study period. These developments are curated based on their overall impact on the market.

FIGURE 28 STRATEGIES ADOPTED BY KEY PLAYERS IN MICROBIOME SEQUENCING SERVICES MARKET, 2020–2023

COMPANY NAME	ORGANIC GROWTH STRATEGIES		INORGANIC GROWTH STRATEGIES
	SERVICE LAUNCHES	EXPANSIONS	AGREEMENTS, PARTNERSHIPS, AND COLLABORATIONS
COSMOSID (US)	The company launched its metabolomics services designed to offer support for multi-omics microbiome research.		Locus Biosciences announced a long-term partnership with CosmosID to gain support of the latter for clinical trial studies.
CHARLES RIVER LABORATORIES (US)			Charles River Laboratories entered into an exclusive partnership with Fios Genomics, a leading provider of bioinformatics data analysis services.
QIAGEN (GERMANY)			QIAGEN announced a partnership with Element Biosciences, Inc. to validate QIAGEN's leading NGS library prep platform and AVITI System associated offerings.
MICROBIOME INSIGHTS (CANADA)		Announced the opening of an office in Dublin, Ireland. This expansion is targeted to serve the company's European clients efficiently.	
CLINICAL-MICROBIOMICS A/S (DENMARK)			Clinical-Microbiomics announced the acquisition of MS-Omics, a metabolomics service provider.

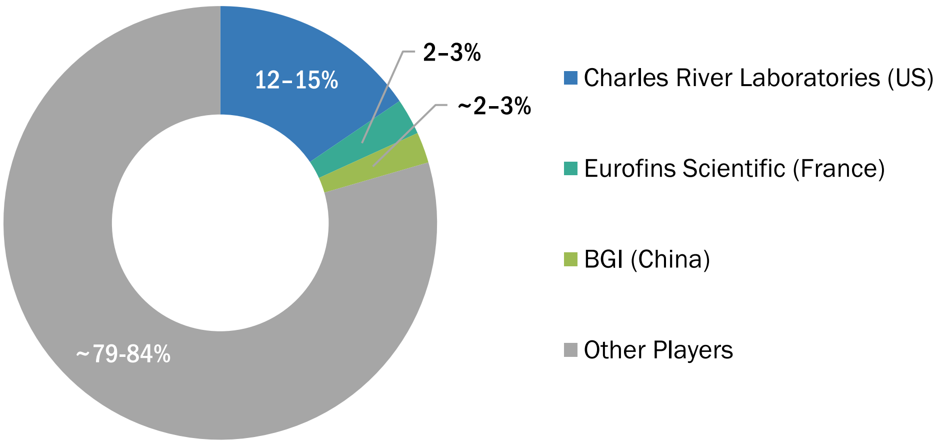
Note: indicates events that will have a significant impact on market growth.

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

11.3 MARKET SHARE ANALYSIS

The global microbiome sequencing services market is a fragmented market, with Charles River Laboratories (US), Eurofins Scientific (France), and BGI (China) together accounting for ~20% of the global market. Players such as CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea) accounted for the remaining 75–80% of the total market.

FIGURE 29 MICROBIOME SEQUENCING SERVICES MARKET SHARE ANALYSIS, BY KEY PLAYER, 2022



Other players include CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea).

Source: Annual Reports, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 169 MICROBIOME SEQUENCING SERVICES MARKET: DEGREE OF COMPETITION

DEGREE OF COMPETITION	FRAGMENTED
Total Market Share of the Top 3 Players	~20%
Other Players	~80%
THE DEGREE OF COMPETITION IS DEFINED AS BELOW:	
Fragmented: When the top 5 players have a total market share of <25%	
Competitive: When the top 5 players have a total market share of 25–50%	
Consolidated: When the top 5 players have a total market share of >50%	

Note: Market shares of the above companies are calculated based on relevant segmental revenues generated by the companies.
Source: Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets

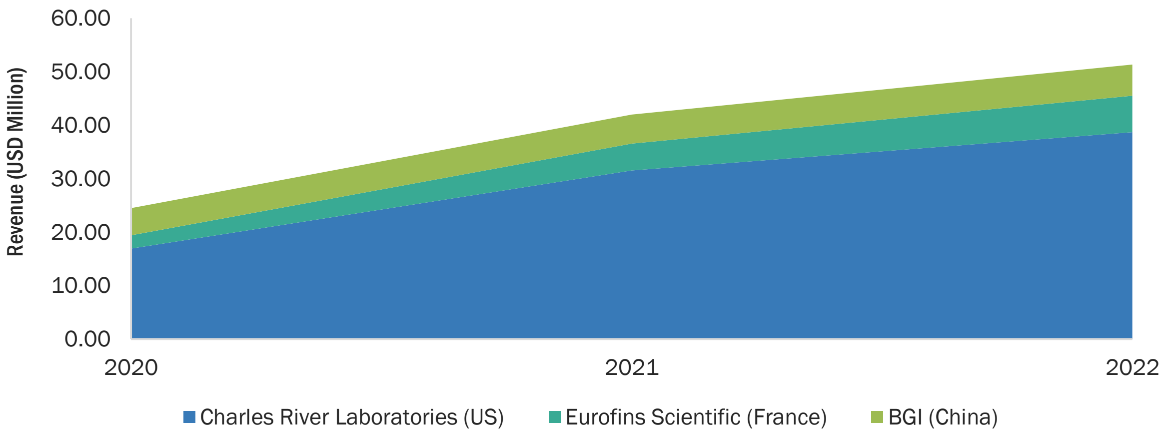
Charles River Laboratories (US) is the leading player in the microbiome sequencing services market. CRL provides superior, flexible, and tailored solutions through its comprehensive portfolio of drug discovery and research models and services. The company has been focusing on organic and inorganic growth through expansions and partnerships. For instance, CRL partnered with Fios Genomics Ltd. (UK) to gain access to its bioinformatics solutions, helping CRL to develop a full spectrum of microbiome profiling/analysis services. In addition, the strong reach of CRL's microbial detection product, Accugenix, is contributing to the company's dominant position in the market.

Eurofins Scientific (France) is a leading player in the global microbiome sequencing services market. The company's strong focus on a broad range of customers, technology-enabled solutions, investments in joint ventures and partnerships, and comprehensive microbiome sequencing solutions and processes are factors contributing to the growth of the company's market share. The company provides drug discovery and development services to more than 220,000 customers, including managed care organizations (MCOs), biopharmaceutical companies, governmental agencies, physicians, hospitals and health systems, food & nutritional companies, and independent clinical laboratories.

11.4 REVENUE SHARE ANALYSIS OF TOP MARKET PLAYERS

The revenue analysis was calculated based on the performance of major players during the past few years and the current years in the microbiome sequencing services market. This gave an estimation of the overall market size, structure, and recent advancements in service technologies. Leading players, such as Charles River Laboratories (US), Eurofins Scientific (France), and BGI (China), influence the microbiome sequencing services market due to their vast services portfolios, recent acquisitions, and wide distribution & sales networks across the globe.

FIGURE 30 REVENUE SHARE ANALYSIS OF TOP PLAYERS IN MICROBIOME SEQUENCING SERVICES MARKET, 2020–2022



Notes: The segmental revenue for the top three players have been considered. The segmental share of microbiome sequencing services may vary significantly from company to company, depending on individual company portfolios.

Source: Press Releases and Company Websites

11.5 COMPANY EVALUATION QUADRANT

The company evaluation quadrant provides information on the top 20 players offering microbiome sequencing services and outlines findings on the performance of each vendor. The industry matrix evaluations are based on two broad categories—Service Offerings and Business Strategies. Each type carries various criteria based on which the companies have been evaluated. The evaluation criteria considered under service offerings include the breadth of offerings, services launched, service technologies, and support. The evaluation criteria considered under the business strategy includes reach (based on market and geographic presence), industry coverage (based on end users catered to), viability, and inorganic growth strategies. The top 20 companies are placed into four categories based on their performance in each criterion—“Stars,” “Emerging Leaders,” “Pervasive Players,” and “Participants.”

We have evaluated 20 companies in this section, which include Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), QIAGEN (Germany), Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), OraSure Technologies (US), MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), Novogene Co., Ltd. (China), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea).

11.5.1 STARS

Players who fall into this category receive high scores for most evaluation criteria. They have a strong and established product portfolio, high market shares, comprehensive application coverage, and a strong geographical footprint. These players provide mature, innovative, and reputable microbiome sequencing services. Charles River Laboratories (US), Eurofins Scientific (France), BGI (China), CosmosID (US), Microba (Australia), and QIAGEN (Germany) are the star players in this market.

11.5.2 EMERGING LEADERS

Emerging leaders have demonstrated substantial product innovations compared to their competitors (resulting in an evolved service portfolio) and significant market presence. However, they do not have robust growth strategies for geographical penetration and overall business growth. Microbiome Insights (Canada), BaseClear (Netherlands), CD Genomics (US), Zymo Research (US), and OraSure Technologies (US) are emerging leaders in the market.

11.5.3 PERVASIVE PLAYERS

Companies falling in this quadrant are established players with very strong business strategies and global distribution/product availability. However, they have a weaker service portfolio and lower market share than the emerging leaders. Pervasive players are Clinical-Microbiomics A/S (Denmark) and Novogene Co., Ltd. (China).

11.5.4 PARTICIPANTS

The companies that fall in this category have niche service offerings. They do not have substantial business strategies as compared to other established vendors. These companies could be new entrants and require more time before getting traction. Participants in the microbiome sequencing services market include MR DNA (US), Eremid Genomic Services (US), Clinical-Microbiomics A/S (Denmark), EzBiome (US), Boster Biological Technology (US), Zifo (India), omics2view.consulting GbR (Germany), and Macrogen, Inc. (South Korea).

FIGURE 31 MICROBIOME SEQUENCING SERVICES MARKET: COMPANY EVALUATION QUADRANT, 2022

Source: MarketsandMarkets Analysis

11.6 COMPETITIVE SCENARIO

TABLE 170 SERVICE LAUNCHES

MONTH & YEAR	COMPANY NAME	SERVICE	DESCRIPTION
March 2023	Microba (Australia)	Metagenomic sequencing and analysis	Microba launched a new human gut microbiome testing service for comprehensive gastrointestinal testing solutions. The solutions are accessible to healthcare professionals in Australia through a different brand name, Co-Biome.
January 2023	CosmosID (US)	CosmosID Metabolomics Service	The company launched its metabolomics services designed to offer support for multi-omics microbiome research. The service consists of a combination of Liquid Chromatography–Mass Spectrometry (LC-MS) and Gas Chromatography-Mass Spectroscopy (GC-MS).
March 2022	OraSure Technologies (US)	Metatranscriptomic sequencing and analysis	Diversigen announced the launch of a new service, metatranscriptomic sequencing and analysis of the gut microbiome. This service will assist research professionals in understanding gene expression while gaining insights into both function and mechanism of microbiomes in samples.

Source: Press Releases and Company Website

TABLE 171 DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
February 2023	Partnership	CosmosID (US)	Locus Biosciences (US)	Locus Biosciences announced a long-term partnership with CosmosID to gain the support of the latter for clinical trial studies. CosmosID's support for microbiome analysis is expected to assist Locus Biosciences in strengthening its precision therapeutics platform.	NA
June 2021	Partnership	Microba (Australia)	Illumina, Inc. (US)	Microba announced a partnership with Illumina to advance research activities in human gut microbiome health and disease. The partnership is expected to combine NGS tools provided by Illumina with Microba's proprietary analysis platform to develop accurate metagenomic databases.	NA

October 2020	Partnership	CRL (US)	Fios Genomics Ltd. (UK)	Charles River Laboratories entered into an exclusive partnership with Fios Genomics, a leading provider of bioinformatics data analysis services. Through this partnership, Charles River's clients will have access to Fios Genomics' expertise in bioinformatics, statistics, and biology, to assist in the sourcing and analysis of the high-dimensional, multi-variant datasets associated with drug development, including microarrays, next-generation sequencing (NGS), proteomics, metabolomics, and epigenetics as well as the associated meta-data.	NA

Source: Press Releases and Company Website

12 COMPANY PROFILES

12.1 KEY PLAYERS

12.1.1 CHARLES RIVER LABORATORIES

12.1.1.1 Business overview

Charles River Laboratories (CRL) is a service and early-stage CRO. The company provides a diverse portfolio of discovery and safety assessment services and offers various products and services to support its clients' manufacturing activities. The company operates through three reporting segments, namely, Research Models and Services (RMS), Drug Discovery and Safety Assessment (DSA), and Manufacturing Support. The company offers microbiome research services (including sequencing services) through all three business segments. CRL offers services to analyze and monitor the health of research models through the RMS segment. The company offers microbiome sequencing services for animal diagnostics.

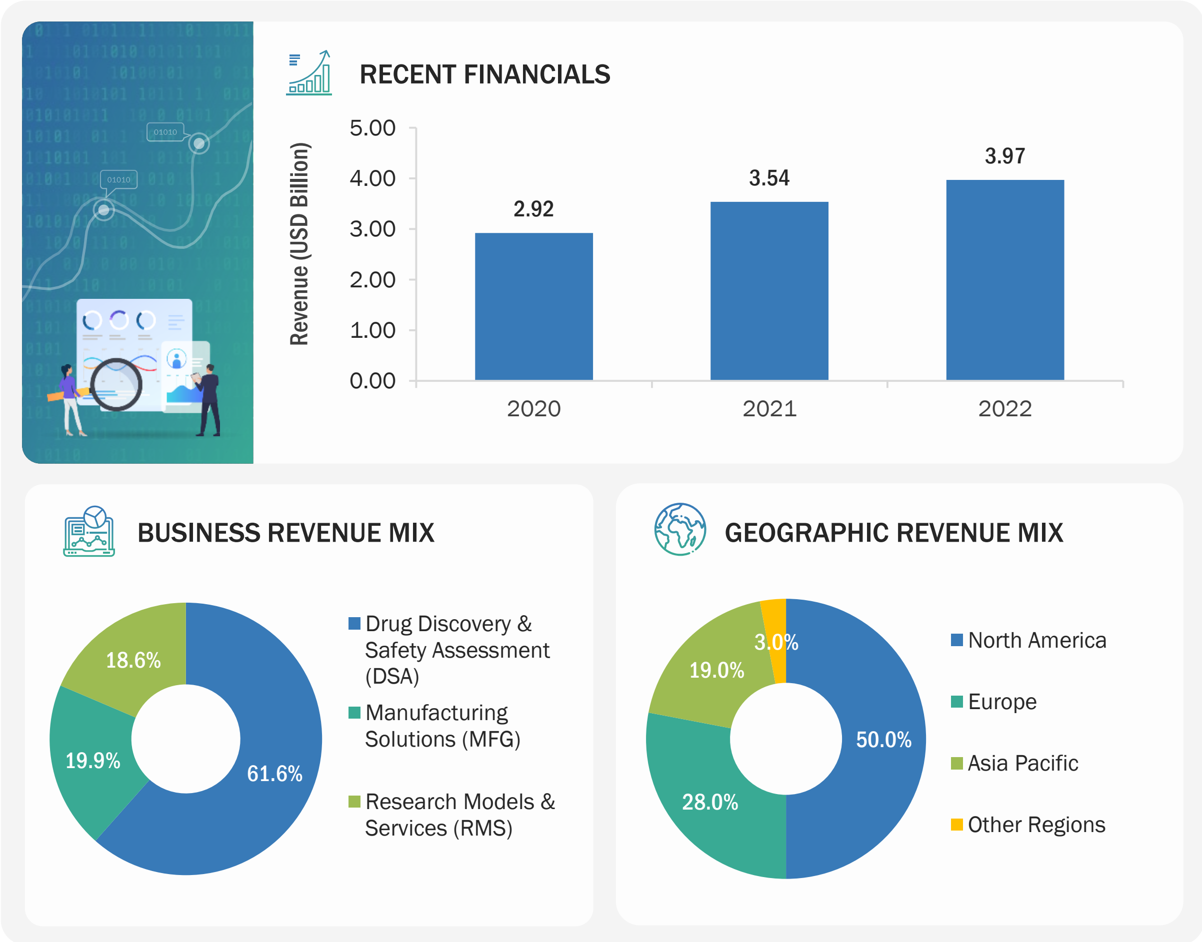
CRL has 100+ facilities in 20+ countries worldwide. It sells products and services principally through its direct sales and business development teams based in North America, Europe, and Asia. Some of the notable subsidiaries of the company are Cognate Bioservices Inc., Vigene Biosciences Inc., MPI Research, Charles River Laboratories Italia Srl, Charles River Laboratories Japan, and Charles River UK Limited.

TABLE 172 CHARLES RIVER LABORATORIES: BUSINESS OVERVIEW

CHARLES RIVER LABORATORIES	
Year Founded	1947
Country	US
City/State	Massachusetts
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 32 CHARLES RIVER LABORATORIES: COMPANY SNAPSHOT (2022)



Note: The company offers microbiome sequencing services through all three business segments (DSA, RMS, and MFG). Hence, the recent financials, business revenue mix, and geographic revenue mix have been provided for the overall company.

Source: Company Website, Annual Reports, and SEC Filings

12.1.1.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
NGS Microbiome Sequencing	16S Next Generation Sequencing	▪ The company offers 16S NGS services for microbiome library preparation and sequencing through three service types, namely, Microbiome Basic, Microbiome Plus, and Advanced Analysis.	Animal health monitoring, germ-free animal contamination testing, and human disease modeling.
Microbial Sequencing ID Services (Strain Typing/ Identification Services)	Whole-genome Sequencing (AccuGENX-ID Sanger Sequencing)	▪ The company offers microbial identification services through its proprietary product line of microbiome products: ▪ Fungal Identification-AccuGENX-ID FunITS ▪ Bacterial Identification-AccuGENX-ID BacSeq ▪ Protein-Coding Gene Sequencing-AccuGENX-ID ProSeq ▪ AccuBLAST Sequencing Data Analysis	Fungal and bacterial identification.

12.1.1.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
October 2020	Partnership	CRL (US)	Fios Genomics Ltd. (UK)	Charles River Laboratories entered into an exclusive partnership with Fios Genomics, a leading provider of bioinformatics data analysis services. Through this partnership, Charles River's clients will have access to Fios Genomics' expertise in bioinformatics, statistics, and biology, to assist in the sourcing and analysis of the high-dimensional, multi-variant datasets associated with drug development, including microarrays, next-generation sequencing (NGS), proteomics, metabolomics, and epigenetics as well as the associated meta-data.	NA

Source: Press Releases

12.1.1.4 MnM view

12.1.1.4.1 Key strengths

CRL is a leading player in the microbiome sequencing services market. Its key strength lies in its strong presence in the adjacent portfolios of research and animal diagnostic services and drug discovery services. The company has been focusing on organic and inorganic growth through expansions and partnerships, as observed in the last three financial years. In October 2020, CRL partnered with Fios Genomics Ltd. (UK) to gain access to its bioinformatics solutions, helping CRL to develop a full spectrum of microbiome profiling/analysis services. In addition, the strong reach of CRL's microbial detection product, Accugenix, has ensured its dominant position in the market.

12.1.1.4.2 Strategic choices

The company focuses on strategic inorganic growth strategies to co-develop advanced technologies and strengthen its research and commercialization capabilities through acquisitions in the microbiome sequencing services market. In August 2012, CRL acquired Accugenix, Inc., a cGMP-compliant contract microbial identification testing provider. This acquisition strengthened CRL's capabilities around state-of-the-art microbial detection services for manufacturing in the biopharmaceutical medical device, nutraceutical, and consumer care industries. In 2023, the company entered into an acquisition agreement with SAMDI Tech for drug discovery research. The acquisition marks the culmination of a partnership between the companies that began in 2018. With its expanding service portfolio, product innovation, and wide geographic presence, the company is expected to grow at a high rate in the microbiome sequencing market during the forecast period.

12.1.1.4.3 Weaknesses and competitive threats

The company faces increased competition from private companies that have dedicated portfolios for microbiome sequencing services across the globe. This may hamper its profitability. The company aims to retain existing customers and obtain new customers and business partners to sustain the competition.

12.1.2 EUROFINS SCIENTIFIC

12.1.2.1 Business overview

Eurofins Scientific is a life science company that provides a range of analytical testing services to its clients across multiple industries. The company operates through only one business segment—Analytical Testing. Different analytical testing services offered under this business segment include Biopharmaceutical Services, Food and Feed Testing, Environment Testing, Clinical Diagnostics, Consumer Product Testing (CPT), and IVD Solutions. The company offers next-generation sequencing services (including microbiome sequencing) through the Biopharmaceutical Services business segment. Eurofins Genomics is part of the Eurofins Scientific Group, which provides NGS, genotyping, gene expression, and custom DNA sequencing, along with oligonucleotides, siRNA, and gene synthesis services.

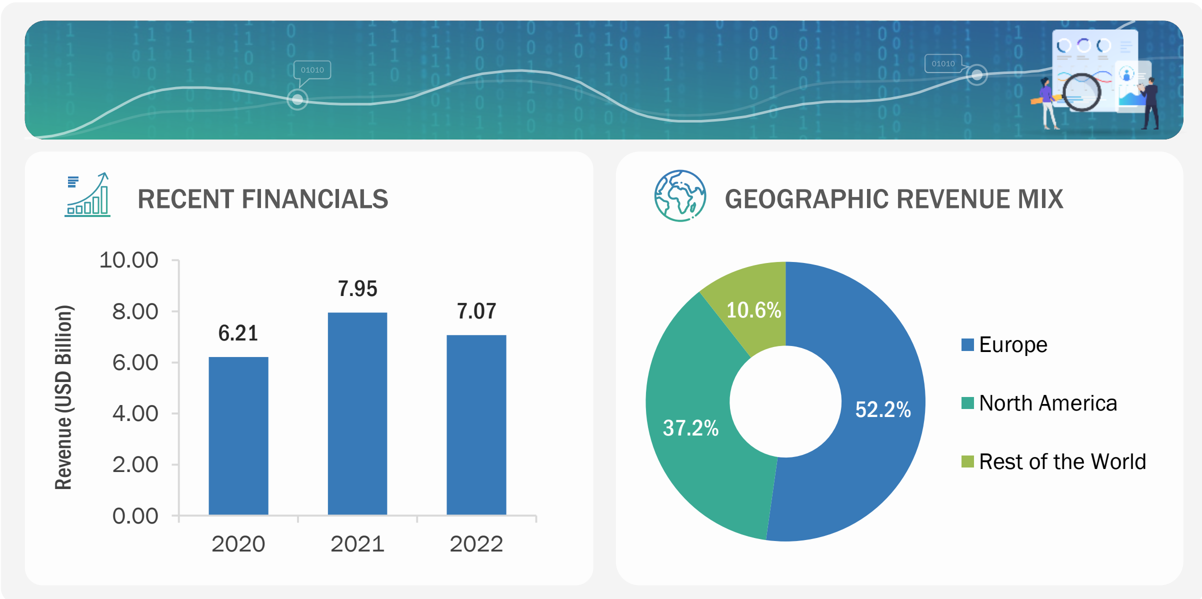
The company has R&D and production sites in the US, Germany, India, and Japan and provides its services to around 200 laboratories across 36 countries in Europe, North America, the Asia Pacific, and Latin America. Some of the company's major subsidiaries are Eurofins BioPharma Product Testing (Shanghai) Co., Ltd. (China), Eurofins Genomics Europe Shared Services GmbH (Germany), Transplant Genomics, Inc. (US), and GATC Biotech SAS (France).

TABLE 173 EUROFINS SCIENTIFIC: BUSINESS OVERVIEW

EUROFINS SCIENTIFIC	
Founded	1987
Country	Luxembourg
City/State	Luxembourg City
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 33 EUROFINS SCIENTIFIC: COMPANY SNAPSHOT (2022)



Note 1: The Business revenue Mix of the company is not available in the public domain. Hence, it is not provided in the company snapshot.

Note 2: The Rest of the World includes countries across South America, Asia, the Middle East, and the Pacific.

Source: Company Website and Annual Reports

12.1.2.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
NGS Microbiome Sequencing/ Profiling	16S Next Generation Sequencing	Eurofins offers microbiome profiling, including sequencing of samples for the below target list: ▪ 18S ▪ plant ITS1 ▪ plant trnL ▪ Different primer pairs for ITS2 ▪ 16S bacteria V4 ▪ 16S bacteria V5-V6 Different primer pairs for: ▪ 16S bacteria V3-V4	Bacterial taxonomic profiling, fungal taxonomic profiling, and archaeal taxonomic profiling.

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

12.1.2.3 MnM view

12.1.2.3.1 Key strengths

Eurofins Scientific is a prominent player in the market. The company's strong focus on a broad range of customers, technology-enabled solutions, investments in joint ventures and partnerships, and comprehensive quality systems and processes contribute to its market growth. The company provides drug discovery and development services to more than 220,000 customers, including managed care organizations (MCOs), biopharmaceutical companies, governmental agencies, physicians, hospitals and health systems, food & nutritional companies, and independent clinical laboratories.

12.1.2.3.2 Strategic choices

The company focuses on organic and inorganic growth strategies such as acquisitions, expansions, and product innovation. The company started integrating Illumina's NGS platforms in 2019 and has since focused on enhancing its service portfolio around microbiome sequencing. Through this integration, the company expanded its service offerings for microbiome or bacterial analysis, which can be used for the monitoring or assessment of skin microbiomes in order to identify diseases.

12.1.2.3.3 Weaknesses and competitive threats

The company faces increased price competition and competes with various specialized microbiome sequencing service providers across the globe. This may affect its profitability or attractiveness. To sustain its market presence, the company must aim to retain existing customers and obtain new customers and business partners.

12.1.3 BGI GROUP

12.1.3.1 Business overview

Founded in 1999 and headquartered in Shenzhen, China, BGI offers products and services for transcriptomics, genomics, epigenomics, proteomics, meta-omics, single-cell sequencing, bioinformatics analysis, and other services. In the field of genomics, the company offers solutions for exome sequencing, whole-genome sequencing, single-cell DNA sequencing, target region sequencing, de novo sequencing, genotyping, and Sanger sequencing. It also offers sequencing and sample preparation systems and kits through its subsidiary, MGI Tech Co., Ltd. (China).

BGI provides its services and solutions in more than 100 countries across the globe, with a customer base of more than 3,000 institutes across Europe, North America, and the Asia Pacific. BGI operates through nine subsidiaries to promote applications in diverse fields. The company's subsidiaries include Complete Genomics, Inc. (US), BGI Diagnosis (Denmark), and BGI Research (China).

TABLE 174 BGI GROUP: BUSINESS OVERVIEW

BGI GROUP	
Year Founded	1999
Country	China
City/State	Shenzhen
Ownership	Public

Note: BGI Group is listed on the Shenzhen Stock Exchange; however, no financial information is available in the public domain. Hence, financial details are not included in this section.

Source: Company Website

12.1.3.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Metagenomic Sequencing	Whole-genome Shotgun Sequencing	The company provides whole-genome shotgun sequencing for DNA samples, which are provided for complex environmental and human samples.	▪ Species composition and abundance analysis ▪ Analysis of functional genes ▪ Gene differences between samples ▪ Metabolic pathways

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

12.1.4 COSMOSID

12.1.4.1 Business overview

CosmosID is a CLIA-certified and GCP-compliant laboratory that offers microbiome sequencing and bioinformatics solutions suitable for research and clinical applications. The company has a developed NGS facility and designs software solutions that enhance the interpretation of metagenomic data. These solutions are designed to provide strain-level resolution to research professionals. The company offers various microbiome sequencing services, including amplicon sequencing, shallow shotgun sequencing, deep shotgun metagenomic sequencing, and metatranscriptomics sequencing.

CosmosID is a private company with operations in the US. It generates its revenue through one reporting segment, Microbiome Solutions.

TABLE 175 COSMOSID: BUSINESS OVERVIEW

COSMOSID	
Year Founded	2008
Country	US
City/State	Germantown, Maryland
Ownership	Private

Note: CosmosID is a privately held company with no financial data available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.4.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Amplicon Sequencing	Bacteria/Archaea - 16S rDNA Sequencing	This sequencing service is designed to support amplicon-based metagenomics projects. The service is suitable for research and commercial applications.	Routine testing, large cohort studies, biobanking, population-based studies, and microbiome-based diagnostics.
Shotgun Sequencing	Shallow Shotgun Sequencing	CosmosID offers shallow shotgun sequencing at 3M reads, which yields consistent species and strain-level resolution of bacteria.	Large cohort studies based on the gut microbiome.
	Deep Shotgun Metagenomic Sequencing	In addition to the sequencing services, the company offers validated DNA extraction protocols suitable for shotgun sequencing.	Clinical metagenomics, infectious disease diagnostics, and biomarker discovery.
Metatranscriptomics Sequencing	NGS Library Preparation/ Sequencing	CosmosID makes use of optimized RNA workflows to offer metatranscriptomic services for microbial isolates and other complex samples.	Nutritional studies, drug discovery and development, and probiotics/live biotherapeutic product (LBP) discovery.

Bacterial Isolate Sequencing	Whole-genome Sequencing (WGS)	The company offers end-to-end services for whole-genome sequencing (WGS) of microbial isolates. The sequencing is carried out in CLIA-certified facilities constructed by the company.	Probiotics and LBP characterization, academic research, and clinical microbiology.

12.1.4.3 Recent developments

SERVICE LAUNCHES

MONTH & YEAR	SEGMENT	SERVICE	DESCRIPTION
January 2023	Metagenomics/ Metabolomics Services	CosmosID Metabolomics Service	The company launched its metabolomics services designed to offer support for multi-omics microbiome research. The service consists of a combination of Liquid Chromatography–Mass Spectrometry (LC-MS) and Gas Chromatography-Mass Spectroscopy (GC-MS).
September 2021	Software & Service Platform	CosmosID-HUB Software Platform	The company announced the addition of a functional analysis module to its online user interface, namely, CosmosID-HUB: Microbiome. This software is combined with microbiome sequencing services offered by CosmosID.

Source: Press Releases

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
February 2023	Partnership	CosmosID (US)	Locus Biosciences (US)	Locus Biosciences announced a long-term partnership with CosmosID to gain the support of the latter for clinical trial studies. CosmosID's support for microbiome analysis is expected to assist Locus Biosciences in strengthening its precision therapeutics platform.	NA
October 2022	Partnership	CosmosID (US)	Microbiome Labs (US)	Microbiome Labs partnered with CosmosID to co-develop gut microbiome tests. This is a stool test designed for microbiome R&D and infectious disease applications.	NA
June 2021	Partnership	CosmosID (US)	Sandwalk BioVentures (Spain)	Sandwalk BioVentures partnered with CosmosID to launch a joint service for Generally-recognized as Safe (GRAS) regulatory genomic reporting. This service combines end-to-end microbial genomics infrastructure and regulatory infrastructure, targeted toward the US probiotics market.	NA

Source: Press Releases

12.1.5 MICROBA

12.1.5.1 Business overview

Microba is engaged in the development of precision microbiome solutions specifically for human gut health. The company has developed an end-to-end analytics platform to measure the human gut microbiome, targeted toward enhancing drug discovery and development. Microba also delivers gut microbiome testing services to clinicians and research professionals.

The company has key operations in Australia and provides microbiome solutions throughout the Asia Pacific region. Microba has three business divisions, namely, Services, Databank, and Therapeutics. The company delivers microbiome sequencing services through the Testing Services segment, which is a part of the Services business division.

TABLE 176 MICROBA: BUSINESS OVERVIEW

MICROBA	
Year Founded	2017
Country	Australia
City/State	Brisbane City, Queensland
Ownership	Public

Note: Financial information for all three years (2020, 2021, and 2022) is not available for Microba in the public domain. Hence, financial details are not included in this section.

Source: Company Website

12.1.5.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Shotgun Sequencing	High-throughput shotgun metagenomic sequencing	Microba uses the Illumina NovaSeq 6000 system to deliver reproducible, high-quality metagenomic sequence data via shotgun metagenomic sequencing.	High-resolution taxonomic and functional data profiling.

12.1.5.3 Recent developments

SERVICE LAUNCHES

MONTH & YEAR	SEGMENT	SERVICE	DESCRIPTION
March 2023	Human Gut Microbiome Testing	Metagenomic Sequencing and Analysis	Microba launched a new human gut microbiome testing service for comprehensive gastrointestinal testing solutions. The solutions are accessible to healthcare professionals in Australia through a different brand name, Co-Biome.

Source: Press Releases

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
June 2021	Partnership	Microba (Australia)	Illumina, Inc. (US)	Microba announced a partnership with Illumina to advance research activities in human gut microbiome health and disease. The partnership is expected to combine NGS tools provided by Illumina with Microba's proprietary analysis platform to develop accurate metagenomic databases.	NA

Source: Press Releases

12.1.6 QIAGEN N.V.

12.1.6.1 Business overview

QIAGEN is a leading global provider of sample-to-insight molecular diagnostics and sample preparation technologies. The company operates through a single business segment and generates its revenue through different product categories, namely, Sample Technologies, Diagnostic Solutions, PCR/Nucleic Acid Amplification, Genomics/NGS, and Others. The company offers a wide range of instruments and assay kits for genetic testing and genomic services through all segments and end-to-end microbiome profiling (including sequencing) services through the Genomics segment.

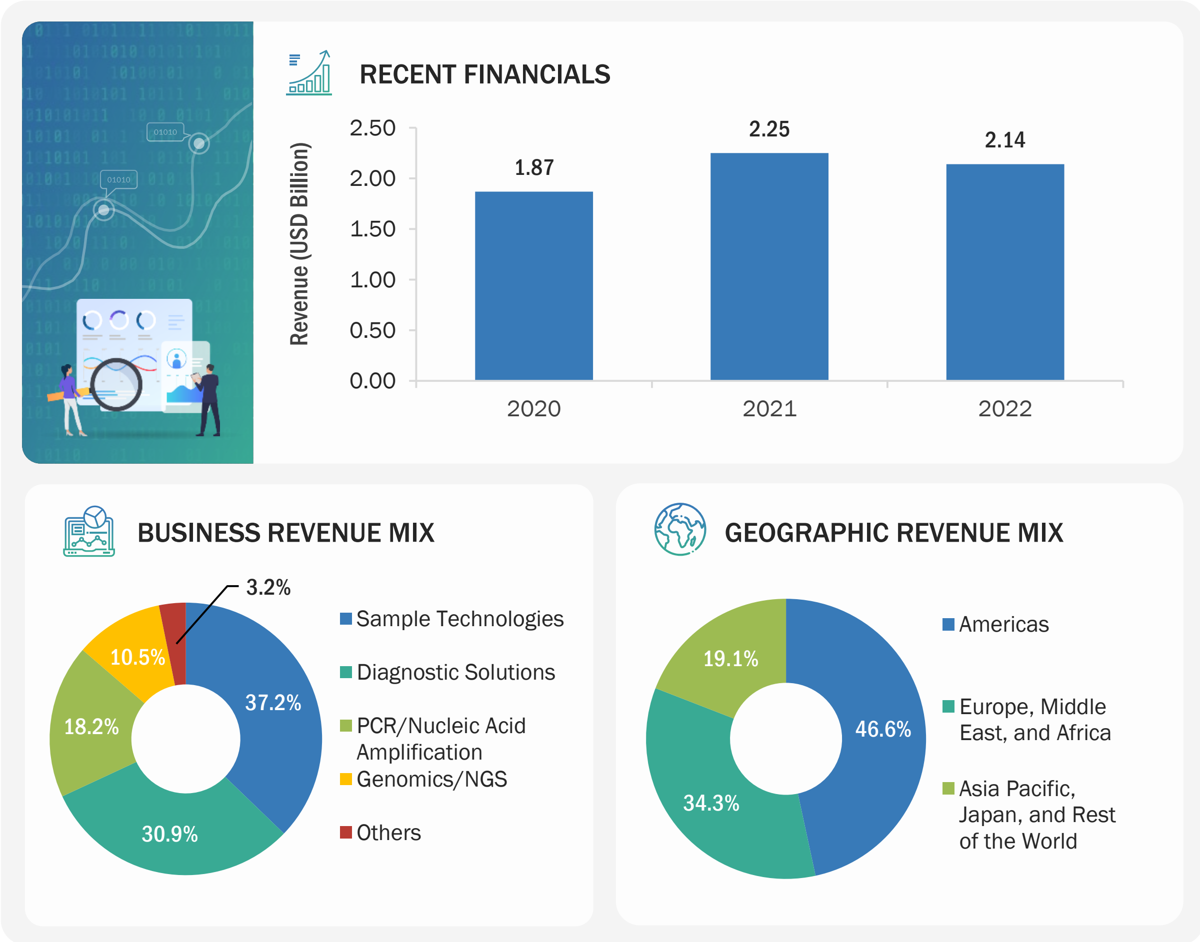
The company operates across Europe, the Americas, the Middle East, Africa, and Asia. Some of the company's major subsidiaries are Amnisure International, LLC (US), Cellestis Pty. Ltd. (Australia), QIAGEN Deutschland Holding GmbH (Germany), QIAGEN Marseille S.A.S. (France), QIAGEN K.K. (Japan), and QIAGEN China (Shanghai) Co. Ltd. (China).

TABLE 177 QIAGEN N.V.: BUSINESS OVERVIEW

QIAGEN N.V.	
Founded	1984
Country	Germany
City/State	Hilden
Ownership	Public

Source: Company Website and Annual Reports

FIGURE 34 QIAGEN N.V.: COMPANY SNAPSHOT (2022)



Note: The company provides microbiome sequencing services through the Genomics/NGS business segment. The information for the Recent Financials, Business Revenue Mix, and Geographic Revenue Mix was not available for the Genomics/NGS. Hence, these sections have been provided for the overall company.

Source: Company Website and Annual Reports

12.1.6.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
NGS Microbiome Sequencing	16S/ITS Microbiome Sequencing	QIAGEN offers a sample-to-insight profile panel, which includes services around sample disruption, sample preparation, NGS library preparation, sequencing, data analysis, and interpretation.	Bacterial and fungal profiling.

12.1.6.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
February 2022	Partnership	QIAGEN (Germany)	Element Biosciences (US)	QIAGEN announced a partnership with Element Biosciences to validate its leading NGS library prep platform and AVITI System-associated offerings. This partnership strengthened AVITI System capabilities through QIAseq target enrichment panels. QIAseq products aid in the detection and characterization of changes in the genome, transcriptome, methylome, and microbiome.	NA

Source: Press Releases

12.1.7 MICROBIOME INSIGHTS

12.1.7.1 Business overview

Microbiome Insights was established through the partnership of the Life Science Institute and the Personalized Medicine Initiative. The company offers end-to-end services for comprehensive microbiome analysis, including different services for microbiome sequencing. Microbiome Insights operates in a single business unit, focused entirely on microbiome analysis/profiling services. These services are offered for different applications, including cosmetics & dermatology, food & beverage, pharma & human health, and agriculture & animal health.

TABLE 178 MICROBIOME INSIGHTS: BUSINESS OVERVIEW

MICROBIOME INSIGHTS	
Year Founded	2015
Country	Canada
City/State	Richmond
Ownership	Private

Note: Microbiome Insights is a privately held company with no financial data available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.7.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Gene Sequencing	16S & ITS2 rRNA Sequencing Service	Microbiome Insights offers 16S & ITS2 rRNA sequencing service with two primer options—16S V4 and 16S V1-V3.	Bacterial taxonomy analysis and fungal taxonomy analysis.
Shotgun Sequencing	Shotgun Metagenomic Sequencing	The company uses sequencing instruments manufactured by Illumina Inc., namely MiSeq, NextSeq, and HiSeq, to offer microbiome shotgun metagenomic sequencing services.	Taxonomic profiling and functional analysis.
Shotgun Sequencing	Shallow Shotgun Metagenome Sequencing (SSMS)	This service combines different samples in a single sequencing run via a modified protocol. This protocol uses a lower volume of reagents for sequencing. library preparation	Taxonomic profiling and functional analysis.

12.1.7.3 Recent developments

OTHER DEVELOPMENTS

MONTH & YEAR	DEVELOPMENT TYPE	COUNTRY/REGION	DESCRIPTION
November 2021	Expansion	Ireland	Microbiome Insights announced the opening of an office in Dublin, Ireland. This expansion was targeted to serve the company's European clients efficiently. This expansion was also expected to support the increasing number of clients across European countries (UK, Germany, France, Ireland, Portugal, Spain, Austria, and Sweden).

Source: Press Releases

12.1.8 BASECLEAR

12.1.8.1 Business overview

BaseClear develops microbial genomic solutions that enhance the understanding and use of microorganisms. The services offered by the company span from product improvement to regulatory approvals. These services include study design consultancy, sampling and logistics, microbiology, genomics, advanced bioinformatics & biostatistics, microbiome analysis, and regulatory affairs. All of the above services are targeted toward microbiome analysis, including sequencing services. Key product lines of the company related to microbiomes include human health, animal health, skin health, and personal care. BaseClear performs in-house DNA research and analysis, followed by data interpretation services.

BaseClear offers services in Europe, which range from sample collection and study design to analysis.

TABLE 179 BASECLEAR: BUSINESS OVERVIEW

BASECLEAR	
Year Founded	1993
Country	Netherlands
City/State	Leiden
Ownership	Private

Note: BaseClear is a privately held company with no financial data available in the public domain. Hence, financial details have not been added in this section.

12.1.8.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Microbial Profiling Service	16S and ITS Sequencing	BaseClear offers complete bioinformatics solutions for microbial profiling analysis. Standard solutions for microbial profiling include 16S and/or ITS amplicon analysis for taxonomic classification and relative frequencies. Illumina MiSeq reads are merged into overlapping pseudo-reads and subsequently aligned against a 16S database.	Bacteria and fungal profiling in samples of food, probiotics, water, microbiome, biofilm, fermentation, and soil.

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023

12.1.9 CD GENOMICS

12.1.9.1 Business overview

CD Genomics is a service-based company providing genomics and bioinformatics service solutions through advanced technologies such as next-generation sequencing, Sanger sequencing, PCR and microarray services. The company provides end-to-end microbiome sequencing services, which include sample collection and preparation, DNA extraction, library preparation and sequencing, and bioinformatics analysis, along with providing detailed reports of microbiome sequencing results. The company provides microbiome sequencing services to end users such as academic research institutes, biotechnology companies, pharmaceuticals and biopharmaceuticals, and clinical diagnostics.

The company provides services across North America, Europe, the Asia Pacific, Latin America, and the Middle East & Africa. Some of the major subsidiaries of CD Genomics are CD Genomics Bioinformatics Technology Co., Ltd. (China), CD Genomics Europe Limited (UK), CD Genomics Inc. (US), CD Genomics (Suzhou) Co., Ltd. (China), CD Genomics (Guangzhou) Co., Ltd. (China), CD Genomics (Hong Kong) Limited (Hong Kong).

TABLE 180 CD GENOMICS: COMPANY OVERVIEW

CD GENOMICS	
Founded	2004
Country	US
City/State	New York
Ownership Type	Private

Note: CD Genomics is a privately held company with no financial data available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.9.2 Services offered

SERVICE TYPE	SERVICE	DESCRIPTION	APPLICATION
Microbial Genomics	16S/18S/ITS Amplicon Sequencing	The company provides DNA extraction, library preparation, sequencing, and bioinformatics analysis for amplicon sequencing.	Microbial identification and phylogeny studies of samples from complicated microbiomes or environments.
Microbial Genomics	Metagenomic Shotgun Sequencing	The company offers DNA extraction, library preparation, sequencing, and bioinformatics analysis for metagenomic shotgun sequencing.	Taxonomic and biological functional characterization of polymicrobial communities.
Microbial Genomics	Viral Metagenomic Sequencing	The company offers DNA extraction, library preparation, sequencing, and bioinformatics analysis for viral metagenomic sequencing.	Analysis of taxonomic composition and structure of viral communities.
Microbial Genomics	Metatranscriptomic Sequencing	The company offers DNA extraction, Library preparation, sequencing and bioinformatics analysis for metatranscriptomic sequencing.	Analysis of active biochemical functions for identifying biomarkers and expression signatures.

Microbial Genomics	Microbial Whole-genome Sequencing	The company offers DNA extraction, Library preparation, sequencing and bioinformatics analysis for microbial whole-genome sequencing.	A comprehensive evaluation of all genetic features of an isolated microorganism.
Microbial Genomics	Absolute Quantitative 16S/18S/ITS Amplicon Sequencing	The company offers DNA extraction, library preparation, sequencing, and bioinformatics analysis for microorganism quantification.	Study of composition and dynamic changes of microbiomes in complex environmental samples.
Microbial Genomics	Microbial Identification	The company offers whole-genome sequencing, Sanger sequencing, and 16S/ITS/18S amplicon sequencing for microbial identification.	Detect, identify, and monitor microbes.
Microbial Identification	Total Microbial Quantification	The company offers qPCR-based microbial quantification services.	PCR quantification is done prior to sequencing a sample of low biomass to determine if the results will have positive amplification.

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

Source: Company Website

12.1.10 ZYMO RESEARCH CORPORATION

12.1.10.1 Business overview

Zymo Research is a biotechnology company that provides products and services for molecular biology applications. The major operating segments of the company are DNA and RNA Purification, Epigenetics, Microbiomics, Next-generation Sequencing, Clinical Diagnostics, and Custom Services. Through the Microbiomics segment, the company provides services and products for sample collection and preservation, microbial standards, DNA and RNA extraction kits, library preparation kits, and microbiome sequencing services.

Zymo Research provides microbiome sequencing services to customers worldwide with locations across the US, Germany, China, Japan, the UK, and Australia.

TABLE 181 ZYMO RESEARCH CORPORATION: COMPANY OVERVIEW

ZYMO RESEARCH CORPORATION	
Founded	1994
Country	US
City/State	California
Ownership Type	Private

Note: Zymo Research Corporation is a privately held company with no financial data available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.10.2 Services offered

SERVICE TYPE	SERVICE	DESCRIPTION	APPLICATION
Microbiome Analysis Services	16S/ITS Amplicon Sequencing Service	The company offers the sequencing of 16S rRNA genes to identify and characterize bacterial and archaeal communities in a sample by providing bioinformatics analysis and raw sequencing data.	Detection and identification of bacteria.
Microbiome Analysis Services	Full-Length 16S Sequencing Service	The company offers full-length sequencing through PacBio HiFi sequencing and bioinformatic analysis.	Bacterial profiling
Microbiome Analysis Services	Shotgun Metagenomic Sequencing Service	The company offers shotgun metagenomic sequencing for the comprehensive profile of microbial communities, including bacteria, archaea, viruses, and fungi, as well as functional information.	Taxonomic identification
Microbiome Analysis Services	Metatranscriptomic Sequencing	The company offers high-throughput sequencing for understanding gene expression patterns of microbial communities in various environments, including soil, water, and human microbiomes.	Analysis, disease diagnosis, biotechnology research, and environmental monitoring.

Microbiome Analysis Services	Custom Solutions for Industry Partners	The company offers customized microbiome sequencing services according to the specific research needs of end users.	Academic research and health & commercial testing.

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

Source: Company Website

12.1.11 ORASURE TECHNOLOGIES

12.1.11.1 Business overview

OraSure Technologies is engaged in the manufacturing and distribution of sample collection devices, sample stabilization devices, and rapid diagnostic tests. Key customers of the company include professionals from research institutes, clinical laboratories, hospitals & clinics, physicians' offices, government agencies, pharmaceutical entities, and direct consumers. In February 2023, OraSure Technologies announced a corporate restructuring that combines its innovation teams with commercial expertise to develop multiple product lines. This restructuring is expected to accelerate innovation, enhance customer experience, and result in operational synergies.

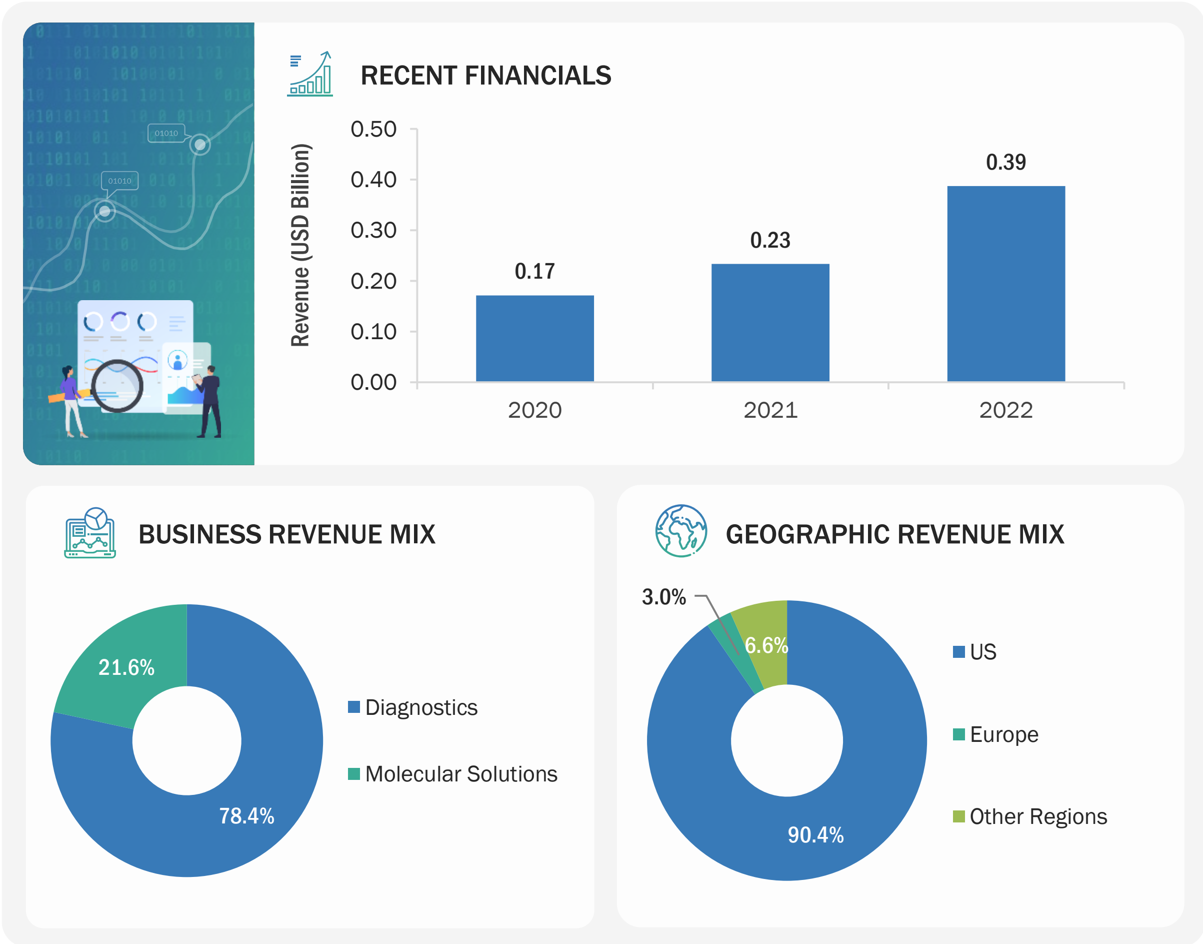
The company has locations in the US, Canada, and Europe and operates through its wholly owned subsidiaries, namely DNA Genotek Inc. (Canada), Diversigen (US), and Novosanis (Romania). Through these subsidiaries, OraSure Technologies offers end-to-end diagnostic and research solutions to its customers, including microbiome sequencing services. OraSure Technologies offers microbiome sequencing services via its subsidiary Diversigen (US). Diversigen offers microbiome sequencing and analysis to customers from academics, research, and direct-to-consumer companies. Microbiome profiling services designed by the company serve researchers in pharmaceutical and agriculture R&D, environmental monitoring, cosmetics development, pet nutrition, direct-to-consumer testing, and other industries.

TABLE 182 ORASURE TECHNOLOGIES: BUSINESS OVERVIEW

ORASURE TECHNOLOGIES	
Year Founded	1987
Country	US
City/State	Pennsylvania
Ownership	Public

Source: Company Website

FIGURE 35 ORASURE TECHNOLOGIES: COMPANY SNAPSHOT (2022)



Source: Company Website and Annual Reports

12.1.11.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Microbial Amplicon Sequencing	BenchMark Amplicon Sequencing	BenchMark is a proprietary technology designed for optimum amplicon sequencing to profile/sequence microbiomes. This service also includes a complimentary Core Analysis Report, which provides fully annotated data tables, interactive visualizations, and all the raw data.	This service is suitable for both high and low-biomass samples. It is especially suitable for low biomass samples, including soil, surface swabs, skin, or biopsy samples.
Shotgun Metagenomic Sequencing	BoosterShot Shallow Shotgun Metagenomic Sequencing	The BoosterShot approach is based on shallow shotgun sequencing that offers full species-level resolution and functional profiling attainable for large-scale studies.	This service offers species-level taxonomic profiling and functional profiling of biological pathways in a particular sample.
	DEEPSEQ Deep Shotgun Metagenomic Sequencing	DeepSeq is based on the deep shotgun sequencing approach, which generates big data and uncovers the signatures in microbiome samples.	This service helps in species-level taxonomic profiling and functional profiling of biological pathways along with strain coverage information.

Whole-genome Sequencing	STRAINVIEW Whole-genome Sequencing of Microbial Isolates	High-throughput method for generating draft-level genomes of hundreds or thousands of cultured isolates in parallel.	Strainview enables the characterization of the entire strain collection comprehensively and efficiently.

Source: Company Website

12.1.11.3 Recent developments

SERVICE LAUNCHES

MONTH & YEAR	SEGMENT	SERVICE	DESCRIPTION
March 2022	Metatranscriptomic Sequencing	Metatranscriptomic Sequencing and Analysis	Diversigen announced the launch of a new service, metatranscriptomic sequencing and analysis of the gut microbiome. This service will assist research professionals in understanding gene expression while gaining insights into both function and mechanism of microbiomes in samples.

Source: Press Releases

12.1.12 MR DNA

12.1.12.1 Business overview

Molecular Research LP (MR DNA) offers next-generation sequencing (NGS) and bioinformatics services through collaborating with instrument/system manufacturers, such as Illumina (US), Applied Biosystems (US), and Pacific Biosciences (US). The company offers a wide array of sequencing services suitable for sequencing small microbial genomes, larger eukaryote genomes, transcriptomes, whole genomes, metagenomes, metatranscriptomes, and exomes. MR DNA is engaged in developing microbiome research solutions such as 16S sequencing services, ITS sequencing, 18S sequencing, Cytochrome c oxidase (COI), or any other type of diversity assays. The company also assists in generating the required data from small or large sequencing projects as a part of an end-to-end service solution.

TABLE 183 MR DNA: BUSINESS OVERVIEW

MR DNA	
Year Founded	2011
Country	US
City/State	Texas
Ownership	Private

Note: MR DNA is a privately held company, and no financial data is available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.12.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Amplicon Sequencing Services	16S Ribosomal Sequencing	MR DNA has an extensive in-house assay collection for 16S sequencing, 18S sequencing, and ITS sequencing.	Taxonomic profiling of bacterial, archaeal, and fungal diversity in samples.
Transcriptome Sequencing	PolyA & RNA Sequencing/ Transcriptome Sequencing	MR DNA offers a number of sequencing services for RNA samples.	This is a comprehensive sequencing service that includes RNA extraction and sequencing.
Whole-genome Sequencing	Whole-genome Sequencing (WGS)	MRDNA has access to a number of NGS platforms, including Illumina NovaSeq 6000, Illumina HiSeq, and the PacBio Sequel IIe, through which the company offers WGS.	Taxonomic profiling of bacterial, archaeal, and fungal diversity in samples.
Metagenome Sequencing	Shotgun Metagenome Sequencing	This service is able to target all genes from a number of different samples found in the environment.	Microbial diversity analysis in environmental samples.

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

12.1.13 EREMID GENOMIC SERVICES

12.1.13.1 Business overview

Eremid Genomic Services is a contract research organization (CRO) engaged in the research and development of products for food & nutrition, agbiotech, and human health. The company's service offerings focus on genomics, leveraging expertise in a range of advanced technologies. Eremid Genomic offers comprehensive services, including scientific consultation, experimental design, sample preparation, sample processing, data analysis, and project reporting.

TABLE 184 EREMID GENOMIC SERVICES: BUSINESS OVERVIEW

EREMID GENOMIC SERVICES	
Year Founded	NA
Country	US
City/State	North Carolina
Ownership	Private

Note: Eremid Genomic Services is a privately held company, and no financial data is available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.13.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Microbiome Profiling/Sequencing	Microbiome Species Profiling (16S/18S/ITS) Microbiome Functional Profiling	This portfolio includes 16S, 18S, and internal transcribed spacer (ITS) sequencing services. The shotgun sequencing service offering helps in microbiome functional profiling.	Helps in taxonomic and molecular phylogenetic studies. The characterization of complex microbial strains, genes, and metabolic pathways.

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

12.1.14 CLINICAL-MICROBIOMICS A/S

12.1.14.1 Business overview

Clinical Microbiomics offers microbiome analysis (including sequencing services) for animal health studies, clinical studies, and preclinical studies. The company's portfolio includes DNA extraction, profiling, sequencing profiling, and assessment of clinical data along with the results. The company's proprietary product line for microbiome sequencing includes 16S rDNA amplicon sequencing and shotgun metagenomic sequencing. Clinical Microbiomics has scientific expertise across bioinformatics, microbiology, antibiotic resistance, machine learning, cancer genomics, and preclinical and clinical studies. The company's customers include academic institutes, pharmaceutical companies, nutrition companies, and biotechnology companies developing microbiome-targeted or microbiome-based products. Clinical Microbiomics offers end-to-end microbiome analysis solutions that span from sample preparation and project planning to the final report and data analysis.

TABLE 185 CLINICAL-MICROBIOMICS A/S: BUSINESS OVERVIEW

CLINICAL-MICROBIOMICS A/S	
Year Founded	2015
Country	Denmark
City/State	Copenhagen
Ownership	Private

Note: Clinical Microbiomics is a privately held company, and no financial data is available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.14.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
Amplicon/16S Microbiome Analysis	16S Microbiome Analysis (Sequencing & data analysis)	The company offers 16S sequencing services using the MiSeq platform (Illumina) coupled with the pipeline of 16S bioinformatics.	▪ Characterization of bacterial populations in clinical and preclinical research. ▪ Taxonomical analysis for research and discovery. ▪ Bacterial species identification for rapid clinical research and infectious disease research.
Custom Microbiome Analysis	Shotgun Metagenomic Sequencing	This service provides tailored microbiome analysis. Assistance is offered throughout the project, from project design to analysis of the output.	▪ Microbiome drug discovery ▪ Pre- and probiotics ▪ Microbiome-drug interactions ▪ Biomarker discovery ▪ Strain tracking ▪ Multi-omics integration ▪ De-novo diversity discovery ▪ Responder and non-responder stratification

12.1.14.3 Recent developments

DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1	COMPANY 2	DESCRIPTION	DEAL SIZE
October 2022	Acquisition	Clinical-Microbiomics (Denmark)	MS-Omics ApS (Denmark)	Clinical-Microbiomics announced the acquisition of MS-Omics, a metabolomics service provider. This acquisition will add value to the expertise of Clinical Microbiomics in profiling microbiomes for clinical applications.	NA

Source: Press Releases

OTHER DEVELOPMENTS

MONTH & YEAR	DEVELOPMENT TYPE	DESCRIPTION
October 2022	Funding	Seventure Partners, a microbiome venture capital, announced an investment of USD 11.05 million in Clinical-Microbiomics A/S.

Source: Press Releases

12.1.15 NOVOGENE CO., LTD.

12.1.15.1 Business overview

Novogene is a provider of genomic services and solutions based on NGS and bioinformatics technologies. The company offers a wide range of NGS-based services for human, plant, animal, and microbial DNA and RNA analysis. Novogene has a presence in 70 countries across the globe, with nearly 6,100 customers. Tianjin Novogene Bioinformatics Technology Co., Ltd. (China), Novogene (HK) Company Limited (China), Novogene Corporation Inc. (US), Novogene (UK) Company Limited (UK), Novogene Netherlands BV (Netherlands), and Novogene Japan K.K. (Japan) are the subsidiaries of the company.

TABLE 186 NOVOGENE CO., LTD.: BUSINESS OVERVIEW

NOVOGENE CO., LTD.	
Year Founded	2011
Country	China
City/State	Beijing
Ownership	Private

Note: Novogene Co., Ltd. is a privately held company, and no financial data is available in the public domain. Hence, financial details have not been added in this section.

Source: Company Website

12.1.15.2 Services offered

SERVICE CATEGORY	SERVICE	DESCRIPTION	APPLICATION
De novo Sequencing	Microbial De Novo Sequencing	The company offers these services using PacBio and Illumina platforms. This service is multifaceted and includes sequencing, survey, draft map, complete map, and fine map of genomes.	▪ Virulence research ▪ Molecular markers ▪ Vaccine development ▪ Drug resistance ▪ Microbial evolution ▪ Epidemiology

Note: No significant developments were reported by the company in the microbiome sequencing services market between October 2020 and April 2023.

12.2 OTHER PLAYERS

12.2.1 EZBIOME

EZBIOME	
Year Founded	2020
Headquarters - Country	US
Headquarters – City/State	Maryland
Ownership	Private
Business Overview	The company provides a comprehensive service portfolio that covers sample preparation services and discovery solutions for microbiome research. The service offerings of the company help in microbial identification, microbial genomics, and cross-disciplinary microbiome research. EzBiome has developed a full suite of sample-to-discovery microbiome sequencing services with EzBioCloud bioinformatics for cross-disciplinary research and studies.
Services	Amplicon Sequencing, Shallow Shotgun Sequencing, and Deep Shotgun Sequencing
Geographical Presence	US

12.2.2 BOSTER BIOLOGICAL TECHNOLOGY

BOSTER BIOLOGICAL TECHNOLOGY	
Year Founded	1993
Headquarters - Country	US
Headquarters – City/State	California
Ownership	Private
Business overview	Boster Biological Technology provides an end-to-end service for microbiome analysis. It has developed optimized workflows for accurate microbiome quantification from sample collection to bioinformatics analysis. The company also provides data analysis and interpretation services with a wide range of microbiome reference standards.
Services	▪ 16S/ITS Amplicon Sequencing ▪ 18S Eukaryote Sequencing ▪ Shotgun Metagenomic Sequencing ▪ Metatranscriptomic Sequencing ▪ Customized Microbiome Analysis
Geographical Presence	US

12.2.3 ZIFO

ZIFO	
Year Founded	2008
Headquarters - Country	India
Headquarters – City/State	Chennai
Ownership	Private
Business overview	Zifo integrates NGS techniques to sequence microbiomes and provide a complete profile of the bacterial and fungal community to its clients. The company has developed a full spectrum of analysis support, including 16S/ITS Metagenomics and Meta-transcriptomics. The service portfolio of the company includes sample preparation, sequencing (dereplication of sequences), and data analysis (OTU picking and quality filtering).
Services	16S/ITS Metagenomics and Meta-transcriptomics, Pre-processing of Raw Data, Dereplication of Sequences and Chimeric Reads Removal, and OTU Picking
Geographical Presence	India

12.2.4 OMICS2VIEW.CONSULTING GBR

OMICS2VIEW.CONSULTING GBR	
Year Founded	2015
Headquarters - Country	Germany
Headquarters – City/State	Hamburg
Ownership	Private
Business overview	omics2view.consulting GbR offers complete service packages with innovative next-generation sequencing technologies. The company accepts samples from customers and proceeds with microbial sequencing with the help of its expertise across sample preparation and sequencing. These services include nucleic acid isolation, library preparation, sequencing, bioinformatics, and biostatistics.
Services	▪ Microbiome Profiling (16S rRNA, 18S rRNA, ITS2, and Functional Marker Genes) ▪ Microbial Genome Sequencing ▪ Microeukaryote Sequencing ▪ Shotgun Metagenomics ▪ Shotgun Metatranscriptomics ▪ RNA Sequencing ▪ Whole-genome Sequencing (WGS) ▪ Targeted Sequencing ▪ Amplicon Sequencing
Geographical Presence	Germany

12.2.5 MACROGEN, INC.

MACROGEN, INC.	
Year Founded	1997
Headquarters - Country	Republic of Korea
Headquarters – City/State	Seoul
Ownership	Private
Business overview	The company offers personalized healthcare solutions by analyzing each person’s gut microbiome based on big data.
Product	Whole-genome Sequencing and Metagenomic Sequencing
Geographical Presence	South Korea

13 APPENDIX

13.1 DISCUSSION GUIDE

Q. 1. **What are your views on the growth prospects of the microbiome sequencing services market? What is the current scenario, and at what rate is the market expected to grow in the future?**

Primary's Viewpoint: ____________________

Q. 2. **What are your views on the major drivers, restraints, opportunities, and challenges for the microbiome sequencing services market (global/country level)?**

Primary's Viewpoint: ____________________

Q. 3. **What is the market size, in terms of value, of the major microbiome sequencing services in 2022 (USD million), and at what rate is the market expected to grow in the coming five years (2023–2028)?**

Primary's Viewpoint: ____________________

Q. 4. **Kindly provide the market shares of the following key players in the microbiome sequencing services market. Kindly validate the key players and let us know if any major players are missing.**

COMPANY NAME	MARKET SHARE (2022)
Charles River Laboratories (US)	
Eurofins Scientific. (France)	
BGI (China)	
Other Players	

Q. 5. **Kindly provide the market shares and projected growth rates of the following service segments in the microbiome sequencing services market. What are the key factors that will drive the fastest-growing market segment in the next five years? Kindly validate the market segmentation and let us know if any segments are missing or overlapping.**

SERVICE	MARKET SHARE (2022)	CAGR (2023–2028)
Sample Preparation		
Sequencing and Library Preparation		
Data Analysis		

Q. 6. **Kindly provide the market shares and projected growth rates of the following type segments in the microbiome sequencing services market. What are the key factors that will drive the fastest-growing market segment in the next five years? Kindly validate the market segmentation and let us know if any segments are missing or overlapping.**

TYPE	MARKET SHARE (2022)	CAGR (2023–2028)
Amplicon Sequencing		
Whole-genome Sequencing		
Shotgun Sequencing		
Other Types (De Novo Sequencing and Metagenomic/Metatranscriptomic Sequencing)		

Q. 7. **Kindly provide the market shares and projected growth rates of the following technology segments in the microbiome sequencing services market. What are the key factors that will drive the fastest-growing market segment in the next five years? Kindly validate the market segmentation and let us know if any segments are missing or overlapping.**

TECHNOLOGY	MARKET SHARE (2022)	CAGR (2023–2028)
Sequencing by Synthesis		
Sequencing by Ligation		
Nanopore Sequencing		
Other Technologies (Semiconductor Sequencing and Sanger Sequencing)		

Q. 8. **Kindly provide the market shares and projected growth rates of the following end-user segments in the microbiome sequencing services market. What are the key factors that will drive the fastest-growing market segments in the next five years? Kindly validate the market segmentation and let us know if any segments are missing or overlapping.**

END USER	MARKET SHARE (2022)	CAGR (2023–2028)
Pharmaceutical and Biotechnology Companies		
Academic and Research Institutes		
Other End Users (Hospitals, Clinics, CROs, and CDMOs)		

Q. 9. Kindly provide the market shares and projected growth rates of the following regions in the microbiome sequencing services market.

REGION	MARKET SHARE (2022)	CAGR (2023–2028)
North America		
Europe		
Asia Pacific		
Latin America		
Middle East and Africa		

Q. 10. Kindly provide the market shares and projected growth rates of the following countries in the microbiome sequencing services market.

NORTH AMERICA	MARKET SHARE (2022)	CAGR (2023–2028)
US		
Canada		

EUROPE	MARKET SHARE (2022)	CAGR (2023–2028)
Germany		
UK		
France		
Italy		
Spain		
Rest of Europe		

ASIA PACIFIC	MARKET SHARE (2022)	CAGR (2023–2028)
China		
India		
Japan		
Rest of Asia Pacific		

LATIN AMERICA	MARKET SHARE (2022)	CAGR (2023–2028)
Brazil		
Rest of Latin America		

Q. 11. Kindly provide additional comments/suggestions, if any.

Primaries Viewpoint: ____

13.2 KNOWLEDGESTORE: MARKETSANDMARKETS' SUBSCRIPTION PORTAL

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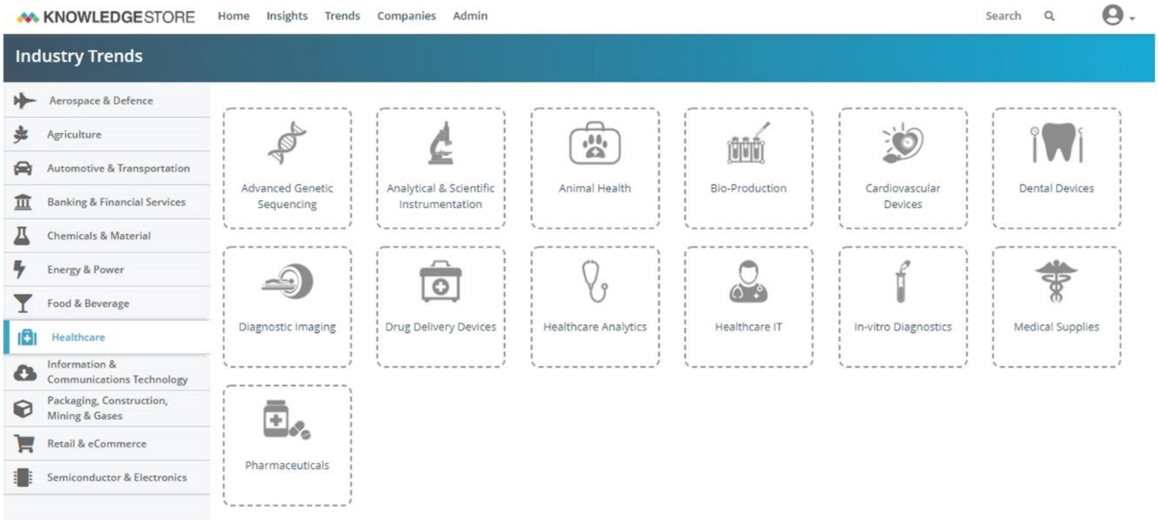

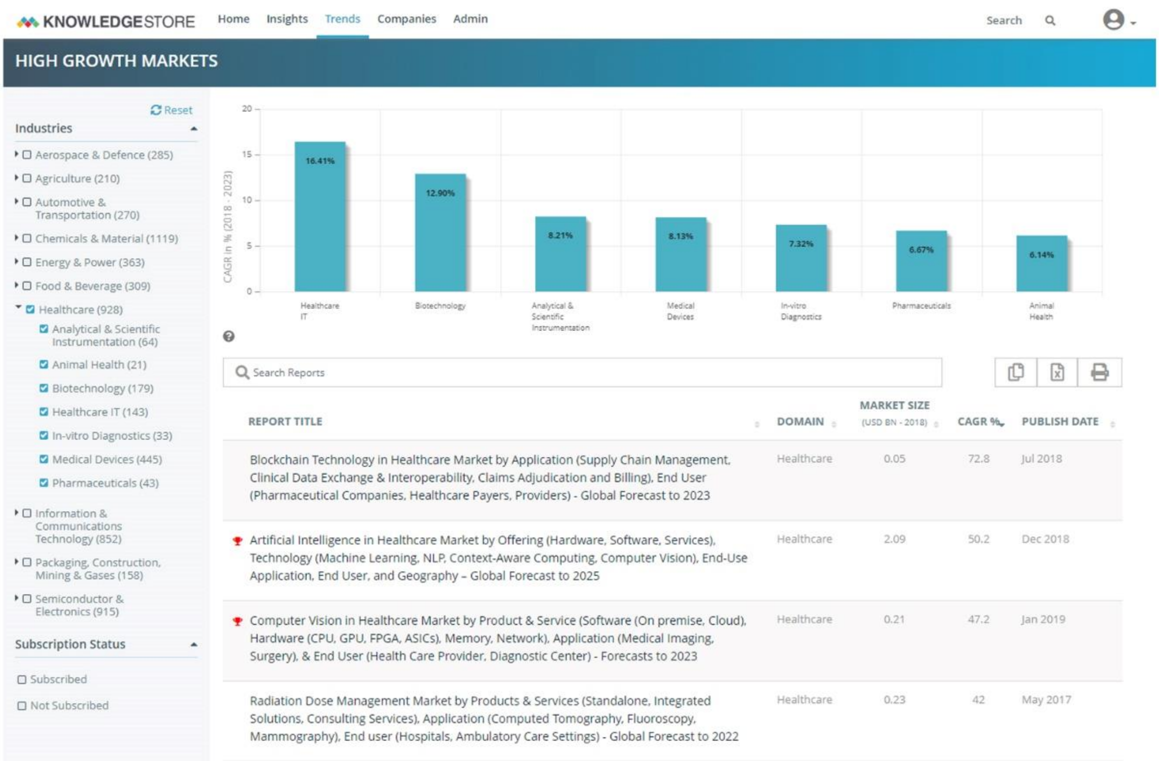

13.3 CUSTOMIZATION OPTIONS

With the given market data, MarketsandMarkets offers customizations as per the company's specific needs. The following customization options are available for the report:

GEOGRAPHICAL ANALYSIS

- Further breakdown of the RoE microbiome sequencing services market, by country
- Further breakdown of the RoAPAC microbiome sequencing services market, by country
- Further breakdown of the RoLATAM and MEA microbiome sequencing services markets, by country

COMPANY INFORMATION

- Detailed analysis and profiling of additional market players (up to five)

13.4 RELATED REPORTS

SR. NO.	REPORT TITLE	PUBLISHED DATE
1	**CONTRACT RESEARCH ORGANIZATION (CROS) SERVICES MARKET - GLOBAL FORECAST TO 2028** By Type (Early phase, Clinical, Laboratory), Therapeutic Area (Oncology, Infectious Disease), Molecule Type (Vaccine, CGT), End User (Pharma, Biopharma, Research Institute) https://www.marketsandmarkets.com/Market-Reports/contract-research-organization-service-market-167410116.html.	February 2023
2	**NEXT-GENERATION SEQUENCING MARKET - GLOBAL FORECAST TO 2027** By Product & Service (Consumables, Platforms, Services), Technology (SBS, Nanopore), Application, End User https://www.marketsandmarkets.com/Market-Reports/next-generation-sequencing-ngs-technologies-market-546.html	January 2023
3	**HUMAN MICROBIOME MARKET - GLOBAL FORECAST TO 2029** By Product (Prebiotics, Probiotics, Food, Diagnostic Tests, Drugs), Application, Disease, Research Technology https://www.marketsandmarkets.com/Market-Reports/human-microbiome-market-37621904.html.	April 2022

13.5 AUTHOR DETAILS

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Rajiv has over 14+ years of experience in market research and consulting in the healthcare vertical. He has worked across diverse areas in the pharma/biotech, medical device, and healthcare IT industries.

He has been instrumental in tracking high-growth and niche markets in the healthcare vertical and has worked on several projects requiring expertise in market estimations, primary & secondary research techniques, strategic & competitive assessment of companies, investment opportunity analysis, and product opportunity evaluations. His current role involves tracking industry developments to develop research areas in collaboration with clients.

Rajiv has diligently worked on addressing clients' problems regarding market and product strategy and formulated detailed solutions in terms of research and consulting outputs. He has completed his master's degree in Pharmacy with a major in Pharmacology from Manipal University and a Postgraduate Diploma in Financial Management.

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